Notes on Varieties of Semiring Solvers

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Contents

1 Preliminaries

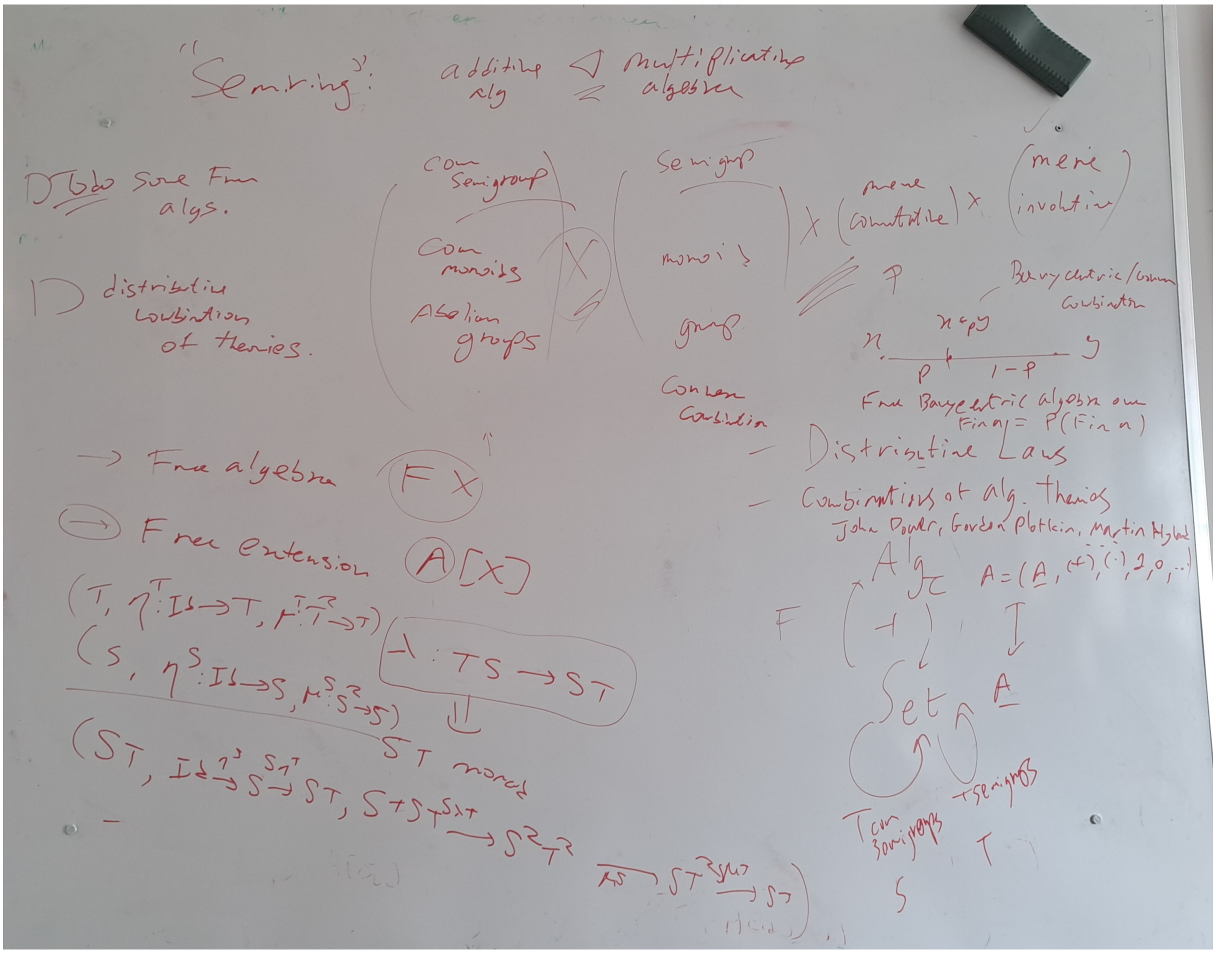

Figure 1: Blackboard at the start of the internship.

For convenience, we recall some definitions we'll be using.

1.1 Algebraic Structures

Definition 1 (Semigroup)

A *semigroup* is a set S equipped with an associative binary operation $\cdot : S \times S \to S$ such that for all $a, b, c \in S$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)

Definition 2 (Monoid)

A *monoid* is a set M equipped with an associative binary operation $\cdot : M \times M \to M$ and an identity element $e \in M$ such that for all $a, b, c \in M$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)

Definition 3 (Group)

A *group* is a set G equipped with an associative binary operation $\cdot : G \times G \to G$, an identity element $e \in G$, and an inverse operation such that for all $a, b, c \in G$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)
- For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 4 (Semiring)

A *semiring* is a set R equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that:

- $(R, +)$ is a commutative monoid with identity element 0.
- $(R, \cdot)$ is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all $a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c,$$
$$(a + b) \cdot c = a \cdot c + b \cdot c.$$

- The additive identity 0 is an absorbing element for multiplication: for all $a \in R$, $0 \cdot a = a \cdot 0 = 0$.

1.2 Categories

Definition 5 (Category)

A *category* $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a class of morphisms $\mathrm{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \mathrm{Ob}(\mathcal{C})$, an identity morphism $\mathrm{id}_a \in \mathrm{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \to b$ and $g : b \to c$, there is a morphism $g \circ f : a \to c$,

satisfying the following axioms:

- Associativity: For morphisms $f : a \to b$, $g : b \to c$, and $h : c \to d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- Identity: For any morphism $f : a \to b$, we have $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$.

Example 6 (Set Category)

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set a is the identity function $\mathrm{id}_a : a \to a$ defined by $\mathrm{id}_a(x) = x$ for all $x \in a$. Composition of morphisms is given by function composition.

Example 7 (Walking Morphism)

The *walking morphism* category $\mathbf{2}$ or I has two objects, say 0 and 1, and a single non-identity morphism $f : 0 \to 1$. The identity morphisms are $\mathrm{id}_0 : 0 \to 0$ and $\mathrm{id}_1 : 1 \to 1$. The composition is defined by the identities and the morphism f.

Definition 8 (Small category)

A category $\mathcal{C}$ is called *small* if both its class of objects $\mathrm{Ob}(\mathcal{C})$ and the class of morphisms $\mathrm{Hom}_{\mathcal{C}}(a,b)$ for all $a, b \in \mathrm{Ob}(\mathcal{C})$ are sets (as opposed to proper classes).

Definition 9 (Product)

Given two objects a and b in a category $\mathcal{C}$, a *product* of a and b is an object $a \times b$ together with two morphisms $\pi_a : a \times b \to a$ and $\pi_b : a \times b \to b$ such that for any object c and morphisms $f : c \to a$ and $g : c \to b$, there exists a unique morphism $\langle f, g\rangle : c \to a \times b$ making the following diagram commute:

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, "g"] & \\ a & a \times b \arrow[l, "\pi_a"] \arrow[r, "\pi_b"'] & b \end{tikzcd}$$

Definition 10 (Coproduct)

Given two objects a and b in a category $\mathcal{C}$, a *coproduct* of a and b is an object $a + b$ together with two morphisms $\iota_a : a \to a + b$ and $\iota_b : b \to a + b$ such that for any object c and morphisms $f : a \to c$ and $g : b \to c$, there exists a unique morphism $[f, g] : a + b \to c$ making the following diagram commute:

$$\begin{tikzcd} a \arrow[r, "\iota_a"] \arrow[dr, "f"'] & a + b \arrow[d, dashed, "{[f,g]}"] & b \arrow[l, "\iota_b"'] \arrow[dl, "g"] \\ & c & \end{tikzcd}$$

Definition 11 (Functor)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a *functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

- A mapping $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ that assigns to each object a in $\mathcal{C}$ an object $F(a)$ in $\mathcal{D}$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a mapping $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(F(a), F(b))$ that assigns to each morphism $f : a \to b$ in $\mathcal{C}$ a morphism $F(f) : F(a) \to F(b)$ in $\mathcal{D}$,

such that for all objects a, b, c in $\mathcal{C}$ and morphisms $f : a \to b$, $g : b \to c$, the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$ (preservation of composition),
- $F(\mathrm{id}_a) = \mathrm{id}_{F(a)}$ (preservation of identity).

For a category $\mathcal{C}$, the *identity functor* $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is defined by $\mathrm{Id}_{\mathcal{C}}(a) = a$ for all objects a and $\mathrm{Id}_{\mathcal{C}}(f) = f$ for all morphisms f.

- A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is injective.
- A functor $F : \mathcal{C} \to \mathcal{D}$ is *full* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is surjective.
- A functor that is both full and faithful is called *fully faithful*.

Notation: We often denote the image of an object a under a functor F as Fa instead of $F(a)$ and the image of a morphism $f : a \to b$ as $Ff : Fa \to Fb$.

Example 12 (Diagonal functor)

Given a category $\mathcal{C}$, the *diagonal functor* $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is defined by $\Delta(a) = (a, a)$ for each object a in $\mathcal{C}$ and $\Delta(f) = (f, f)$ for each morphism $f : a \to b$ in $\mathcal{C}$.

Definition 13 (Product preserving functor)

A functor $F : \mathcal{C} \to \mathcal{D}$ is said to *preserve products* if for every collection of objects $\{a_i\}_{i \in I}$ in $\mathcal{C}$, the natural morphism from $F(\prod_{i \in I} a_i)$ to $\prod_{i \in I} F(a_i)$ is an isomorphism in $\mathcal{D}$. In other words, F preserves the structure of products up to isomorphism.

Definition 14 (Natural Transformation)

Given two functors $F, G : \mathcal{C} \to \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$, a *natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{C})}$ such that for every morphism $f : a \to b$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} Fa \arrow[r, "Ff"] \arrow[d, "\eta_a"'] & Fb \arrow[d, "\eta_b"] \\ Ga \arrow[r, "Gf"'] & Gb \end{tikzcd}$$

Example 15 (Monoidal Category)

A *monoidal category* is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, an object I called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects a, b, c in $\mathcal{C}$, there are natural isomorphisms:

- Associator: $\alpha_{a,b,c} : (a \otimes b) \otimes c \to a \otimes (b \otimes c)$,
- Left unitor: $\lambda_a : I \otimes a \to a$,
- Right unitor: $\rho_a : a \otimes I \to a$,

such that the following diagrams commute:

$$\begin{tikzcd}
((a \otimes b) \otimes c) \otimes d \arrow[r, "\alpha_{a,b,c} \otimes 1_d"] & (a \otimes (b \otimes c)) \otimes d \arrow[r, "\alpha_{a,b\otimes c,d}"] & a \otimes ((b \otimes c) \otimes d) \\
(a \otimes b) \otimes (c \otimes d) \arrow[u, "\alpha_{a,b,c\otimes d}"] & & a \otimes (b \otimes (c \otimes d)) \arrow[u, "1_a \otimes \alpha_{b,c,d}"']
\end{tikzcd}$$

$$(I \otimes a) \otimes b \xrightarrow{\alpha_{I,a,b}} I \otimes (a \otimes b) \xleftarrow{\rho_a} a = a \otimes I \xrightarrow{\lambda_a} a \otimes I \xrightarrow{\alpha_{a,I,I}} (a \otimes I) \otimes I$$

Definition 16 (Opposite category)

Let $\mathcal{C}$ be a category.

The *opposite category* $\mathcal{C}^{\mathrm{op}}$ is defined by:

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) := \mathrm{Ob}(\mathcal{C})$
- $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(x, y) := \mathrm{Hom}_{\mathcal{C}}(y, x)$
- $(g \circ f)^{\mathrm{op}} := f^{\mathrm{op}} \circ g^{\mathrm{op}}$

Definition 17 (Presheaf)

A *presheaf* on a small category $\mathcal{C}$ is a functor

$$F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$$

from the opposite category $\mathcal{C}^{\mathrm{op}}$ of $\mathcal{C}$ to the category $\mathbf{Set}$ of sets.

The category of presheaves on $\mathcal{C}$, usually denoted $\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}$ or $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$, but often abbreviated as $\widehat{\mathcal{C}}$, has:

- functors $F : \mathcal{C}^{\mathrm{op}} \to Set$ as objects
- natural transformations between such functors as morphisms

Definition 18 (Adjunction)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor $F : \mathcal{C} \to \mathcal{D}$ is a *left adjoint* to a functor $G : \mathcal{D} \to \mathcal{C}$ (and G is a *right adjoint* to F) if there is a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{D}}(Fa, b) \cong \mathrm{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects a in $\mathcal{C}$ and b in $\mathcal{D}$. This means that for every morphism $f : Fa \to b$ in $\mathcal{D}$, there is a unique morphism $g : a \to Gb$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{tikzcd} Fa \arrow[r, "f"] & b \arrow[d, "G"] \\ a \arrow[u, "F"] \arrow[r, "g"'] & Gb \end{tikzcd}$$

We often denote this adjunction by $F \dashv G$.

There are actually several different but equivalent ways to define adjunctions. They are linked thanks to the Yoneda lemma.

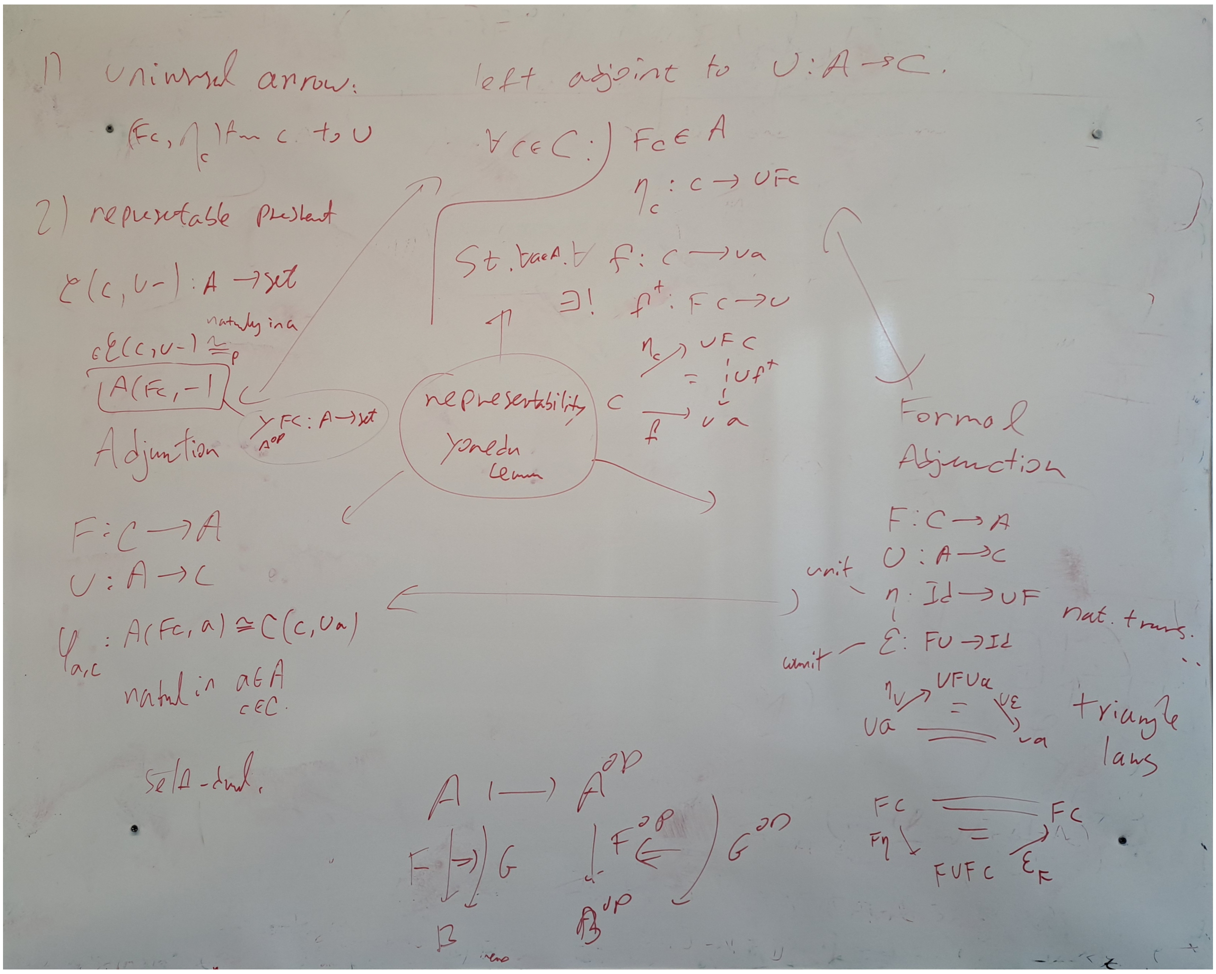

Figure 2: Different equivalent definitions of adjunctions.

One important definition is that of the universal arrow from MacLane's *Categories for the Working Mathematician* [Mac71].

Definition 19 (Universal arrow)

Let $U : A \to C$ be a functor, and let $c \in C$.

An *universal arrow* from c to U is a pair (Fc, η_c) where:

- $Fc \in A$
- $\eta_c : c \to UFc$ is a morphism in C

such that for every $a \in A$, and every $f : c \to Ua$ in C, there exists a unique morphism $f^\dagger : Fc \to a$ in A such that the following diagram commutes:

$$\begin{tikzcd} c \arrow[r, "\eta_c"] \arrow[dr, "f"'] & UFc \arrow[d, dashed, "Uf^\dagger"] \\ & Ua \end{tikzcd} \qquad \begin{tikzcd} Fc \arrow[d, "f^\dagger"] \\ a \end{tikzcd}$$

We can dualize this definition to get the notion of a universal arrow from U to c.

A universal arrow from U to c is a pair (Fc, ε_c) where:

- $Fc \in A$
- $\varepsilon_c : UFc \to c$ is a morphism in C

such that for every $a \in A$, and every $f : Ua \to c$ in C, there exists a unique morphism $f^\dagger : a \to Fc$ in A such that the following diagram commutes:

$$\begin{tikzcd} c & UFc \arrow[l, "\varepsilon_c"'] \\ & Ua \arrow[ul, "f"] \arrow[u, dashed, "Uf^\dagger"'] \end{tikzcd} \qquad \begin{tikzcd} Fc \\ a \arrow[u, "f^\dagger"'] \end{tikzcd}$$

Example 20 (Adjunction from Universal Arrows)

Given a functor $U : A \to C$ such that for every object $c \in C$, there exists a universal arrow from c to U, we can construct an adjunction $F \dashv U$ where:

- The functor $F : C \to A$ is defined by sending each object c in C to the object Fc in A that appears in the universal arrow from c to U, and sending each morphism $f : c \to c'$ in C to the unique morphism $Ff : Fc \to Fc'$ in A that makes the corresponding diagram commute.
- The unit $\eta : \mathrm{Id}_C \to UF$ is given by the family of morphisms $\{\eta_c : c \to UFc\}_{c \in C}$ that are part of the universal arrows.

Universal arrows have more uses than just defining adjunctions. Here are other examples.

Example 21 (Defining the coproduct with an universal arrow)

Let A be a category, and let $a, b \in A$. A *coproduct* of a and b is a universal arrow from (a, b) to the diagonal functor $\Delta : A \to A \times A$, where $\Delta(c) = (c, c)$ for any object $c \in A$.

$$\begin{tikzcd} a \arrow[r, "\iota_a"] \arrow[dr, "f"'] & a + b \arrow[d, dashed, "{[f,g]}"] & b \arrow[l, "\iota_b"'] \arrow[dl, "g"] \\ & c & \end{tikzcd} \sim \begin{tikzcd} (a,b) \arrow[r, "\eta_{(a,b)}"] \arrow[dr, "{(f,g)}"'] & \Delta(a+b) \arrow[d, dashed, "{\Delta[f,g]}"] \\ & \Delta c \end{tikzcd}$$

We have $\eta_{(a,b)} = (\iota_a, \iota_b)$.

Example 22 (Defining the product with an universal arrow)

Let A be a category, and let $a, b \in A$. A *product* of a and b is a universal arrow from the diagonal functor $\Delta : A \to A \times A$ to (a,b).

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\ a & a \times b \arrow[l, "\pi_a"] \arrow[r, "\pi_b"'] & b \end{tikzcd} \quad \sim \quad \begin{tikzcd} \Delta c \arrow[r, "\varepsilon_{(a,b)}"] \arrow[dr, "(f,g)"'] & \Delta(a \times b) \arrow[d, dashed, "(\pi_a, \pi_b)"] \\ & (a,b) \end{tikzcd}$$

We have $\varepsilon_{(a,b)} = \langle f, g \rangle$.

Definition 23 (Equaliser)

Let $f, g : X \to Y$ be two morphisms in a category $\mathcal{C}$.

An *equaliser* of f and g is an object E together with a morphism $e : E \to X$ such that the following diagram commutes:

$$\begin{tikzcd} E \arrow[r, "e"] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \end{tikzcd}$$

Moreover, for any object Z and morphism $z : Z \to X$ such that $f \circ z = g \circ z$, there exists a unique morphism $u : Z \to E$ such that $e \circ u = z$:

$$\begin{tikzcd} Z \arrow[d, dashed, "u"'] \arrow[dr, "z"] & & \\ E \arrow[r, "e"'] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \end{tikzcd}$$

Definition 24 (Pullback of two morphisms)

Let $f : X \to Z$ and $g : Y \to Z$ be two morphisms in a category $\mathcal{C}$. A *pullback* of f and g consists of an object P and two morphisms $p_1 : P \to X$ and $p_2 : P \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

Moreover, for any triple (Q, q_1, q_2) where $q_1 : Q \to X$ and $q_2 : Q \to Y$ are morphisms with $fq_1 = gq_2$, there must exist $u : Q \to P$ such that $p_1 \circ u = q_1$ and $p_2 \circ u = q_2$.
The situation is illustrated in the following commutative diagram:

$$\begin{tikzcd} Q \arrow[dr, dashed, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\ & P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ & X \arrow[r, "f"'] & Z \end{tikzcd}$$

This condition can be reformulated with universal arrows as follows: a pullback of f and g is a universal arrow from the functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ to the object (f, g) in $\mathcal{C} \times \mathcal{C}$.

Example 25 (Pullback in the category of categories)

Let $\mathcal{A}, \mathcal{B}, \mathcal{Z} \in \mathbf{Cat}$ such that we have functors $F : \mathcal{A} \to \mathcal{Z}$ and $G : \mathcal{B} \to \mathcal{Z}$.

The pullback of F and G is the category $\mathcal{P}$ defined as follows:

- The objects of $\mathcal{P}$ are pairs (a, b) where a is an object of $\mathcal{A}$, b is an object of $\mathcal{B}$, and $Fa = Gb$.
- The morphisms of $\mathcal{P}$ from (a, b) to (a', b') are pairs (f, g) where $f : a \to a'$ is a morphism in $\mathcal{A}$, $g : b \to b'$ is a morphism in $\mathcal{B}$, and $Ff = Gg$.
- The composition of morphisms in $\mathcal{P}$ is defined componentwise: if $(f, g) : (a, b) \to (a', b')$ and $(f', g') : (a', b') \to (a'', b'')$ are morphisms in $\mathcal{P}$, then their composition is given by $(f' \circ f, g' \circ g) : (a, b) \to (a'', b'')$.
- The identity morphism on an object (a, b) in $\mathcal{P}$ is given by $(\mathrm{id}_a, \mathrm{id}_b)$, where id_a is the identity morphism on a in $\mathcal{A}$ and id_b is the identity morphism on b in $\mathcal{B}$.

Proof

Consider the following commuting diagram:

$$\begin{tikzcd}
\mathcal{Q} \arrow[dr, dashed, "U"] \arrow[drr, bend left, "Q_2"] \arrow[ddr, bend right, "Q_1"'] & & \\
& \mathcal{P} \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & \mathcal{B} \arrow[d, "G"] \\
& \mathcal{A} \arrow[r, "F"'] & \mathcal{Z}
\end{tikzcd}$$

We need to find U and to prove that it is unique up to isomorphism.

We define U as follows: for any object q in $\mathcal{Q}$, we have $Q_1 q \in \mathcal{A}$ and $Q_2 q \in \mathcal{B}$. Since $FQ_1 q = GQ_2 q$, we can define $Uq = (Q_1 q, Q_2 q)$. For any morphism $m : q \to q'$ in $\mathcal{Q}$, we have $Q_1 m : Q_1 q \to Q_1 q'$ and $Q_2 m : Q_2 q \to Q_2 q'$. Since $FQ_1 m = GQ_2 m$, we can define $Um = (Q_1 m, Q_2 m)$.

It is straightforward to check that U is a functor and that the diagram commutes.

Moreover, if $U' : \mathcal{Q} \to \mathcal{P}$ is another functor such that the diagram commutes, then for any object q in $\mathcal{Q}$, we have $U'q = (Q_1 q, Q_2 q) = Uq$. For any morphism $m : q \to q'$ in $\mathcal{Q}$, we have $U'm = (Q_1 m, Q_2 m) = Um$. Hence, U is unique up to isomorphism. $\square$

Definition 26 (Pushout of two morphisms)

Let $f : Z \to X$ and $g : Z \to Y$ be two morphisms in a category $\mathcal{C}$.

A *pushout* of f and g consists of an object P and two morphisms $p_1 : X \to P$ and $p_2 : Y \to P$ such that the following diagram commutes:

$$\begin{tikzcd} Z \arrow[r, "g"] \arrow[d, "f"'] & Y \arrow[d, "p_2"] \\ X \arrow[r, "p_1"'] & P \end{tikzcd}$$

Moreover, for any triple (Q, q_1, q_2) where $q_1 : X \to Q$ and $q_2 : Y \to Q$ are morphisms with $q_1 f = q_2 g$, there must exist $u : P \to Q$ such that $u \circ p_1 = q_1$ and $u \circ p_2 = q_2$.
The situation is illustrated in the following commutative diagram:

$$\begin{tikzcd} Z \arrow[r, "g"] \arrow[d, "f"'] & Y \arrow[d, "p_2"] \arrow[ddr, bend left, "q_2"] & \\ X \arrow[r, "p_1"'] \arrow[drr, bend right, "q_1"'] & P \arrow[dr, dashed, "u"] & \\ & & Q \end{tikzcd}$$

This condition can be reformulated with universal arrows as follows: a pushout of f and g is a universal arrow from the object (f, g) in $\mathcal{C} \times \mathcal{C}$ to the functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$.

Definition 27 (Coequaliser)

Let $f, g : X \to Y$ be two morphisms in a category $\mathcal{C}$.

A *coequaliser* of f and g consists of an object Q and a morphism $q : Y \to Q$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \arrow[r, "q"] & Q \end{tikzcd}$$

Moreover, for any pair (R, r) where $r : Y \to R$ is a morphism with $r \circ f = r \circ g$, there must exist a unique morphism $u : Q \to R$ such that $u \circ q = r$.
The situation is illustrated in the following commutative diagram:

$$\begin{tikzcd} X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \arrow[r, "q"] \arrow[dr, "r"'] & Q \arrow[d, dashed, "u"] \\ & & R \end{tikzcd}$$

Lemma 28 (Pushout as coproducts and coequalisers)

Let $f : Z \to X$ and $g : Z \to Y$ be two morphisms in a category $\mathcal{C}$. The pushout of f and g can be constructed as the coequaliser of the morphisms $\iota_X f : Z \to X \amalg Y$ and $\iota_Y g : Z \to X \amalg Y$, where $X \amalg Y$ is the coproduct of X and Y, and $\iota_X : X \to X + Y$ and $\iota_Y : Y \to X \amalg Y$ are the canonical injections:

$$\begin{tikzcd}
Z \arrow[rr, "g"] \arrow[dd, "f"'] \arrow[dr, shift left, "\tilde{g}"] \arrow[dr, shift right, "\tilde{f}"'] & & Y \arrow[dl, "\iota_Y"'] \arrow[dd, "\iota_Y \circ q"] \\
& X \amalg Y \arrow[dr, "q"] & \\
X \arrow[ur, "\iota_X"] \arrow[rr, "\iota_X \circ q"'] & & P
\end{tikzcd}$$

ADD REFERENCES

Definition 29 (Diagram)

A *diagram* in a category $\mathcal{C}$ is just a functor

$$D : J \to \mathcal{C}$$

J is called the *index category* or *shape* of the diagram.

Definition 30 (Cone)

Given a diagram $F : J \to \mathcal{C}$, a *cone* to F is:

- an object N of $\mathcal{C}$
- a family ψ of morphisms $\psi_X : N \to FX$ indexed by the objects X of J such that for every morphism $f : X \to Y$ in J,

$$Ff \circ \psi_X = \psi_Y$$

Definition 31 (Cocone)

Given a diagram $F : J \to \mathcal{C}$, a *cocone* to F is:

- an object N of $\mathcal{C}$
- a family ι of morphisms $\iota_X : FX \to N$ indexed by the objects X of J such that for every morphism $f : X \to Y$ in J,

$$\iota_Y \circ Ff = \iota_X$$

Definition 32 (Limit)

A *limit* of a diagram $F : J \to \mathcal{C}$ is a cone (L, φ) to F such that for every cone (N, ψ) to F there is a unique morphism $u : N \to L$ such that $\varphi_X \circ u = \psi_X$ for all X in J:

$$\begin{tikzcd}
& N \arrow[d, dashed, "u"] \arrow[ddl, bend right, "\psi_X"'] \arrow[ddr, bend left, "\psi_Y"] & \\
& L \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\
FX \arrow[rr, "Ff"'] & & FY
\end{tikzcd}$$

A limit is called *small* if J is small.

Definition 33 (Complete category)

A category $\mathcal{C}$ is said to be *complete* if it has all small limits.

Definition 34 (Continuous functor)

A functor G is said to *preserve all limits of shape* J if it preserves the limits of all diagrams $F : J \to \mathcal{C}$.

A *continuous functor* is a functor that preserves all small limits.

Definition 35 (Colimit)

A *colimit* of a diagram $F : J \to \mathcal{C}$ is a cocone (L, φ) to F such that for every cocone (N, ψ) to F there is a unique morphism $u : L \to N$ such that $u \circ \varphi_X = \psi_X$ for all X in J:

$$\begin{tikzcd}
FX \arrow[rr, "Ff"] \arrow[dr, "\varphi_X"] \arrow[ddr, bend right, "\psi_X"'] & & FY \arrow[dl, "\varphi_Y"'] \arrow[ddl, bend left, "\psi_Y"] \\
 & L \arrow[d, dashed, "u"] & \\
 & N &
\end{tikzcd}$$

A colimit is called *small* if J is small.

Definition 36 (Cocomplete category)

A category $\mathcal{C}$ is said to be *cocomplete* if it has all small colimits.

Definition 37 (Cocontinuous functor)

A functor G is said to *preserve all colimits of shape* J if it preserves the colimits of all diagrams $F : J \to \mathcal{C}$.

A *cocontinuous functor* is a functor that preserves all small colimits.

Example 38 (Product as a limit)

The product of two objects A and B in a category $\mathcal{C}$ is the limit of the diagram consisting of the two objects A and B with no arrows between them.

Example 39 (Equaliser as a limit)

The limit of the diagram $X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is the equaliser of f and g.

Example 40 (Pullback as a limit)

The limit of the span $X \xrightarrow{f} Z \xleftarrow{g} Y$ is the pullback $X \times_Z Y$.

Example 41 (Coproduct as a colimit)

The coproduct of two objects A and B in a category $\mathcal{C}$ is the colimit of the diagram consisting of the two objects A and B with no arrows between them.

Example 42 (Coequaliser as a colimit)

The colimit of the diagram $X \underset{g}{\overset{f}{\rightrightarrows}} Y$ is the coequaliser of f and g.

Example 43 (Pushout as a colimit)

The colimit of the cospan $X \xleftarrow{f} Z \xrightarrow{g} Y$ is the pushout $X \sqcup_Z Y$.

Theorem 44 (Preservation of limits by adjoints)
If $F : \mathcal{C} \leftrightarrows \mathcal{D} : G$ with $F \dashv G$, then:

- F preserves colimits
- G preserves limits

In other words, every right adjoint is continuous and every left adjoint is cocontinuous.

1.3 Monads

Definition 45 (Monad)
A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where:

- $T : \mathcal{C} \to \mathcal{C}$ is a functor,
- $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is a natural transformation called the unit,
- $\mu : T^2 \to T$ is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{tikzcd} T \arrow[r, "T\eta"] \arrow[d, "\eta T"'] \arrow[dr, equal] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd} \qquad \begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd}$$

Example 46 (Monad from an Adjunction)
Given an adjunction $F \dashv G$ between categories $\mathcal{C}$ and $\mathcal{D}$, we can construct a monad on $\mathcal{C}$ as follows:

- The functor $T : \mathcal{C} \to \mathcal{C}$ is defined by $T = GF$.
- The unit $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\eta_a : a \to GFa$ corresponding to the identity morphism on Fa under the adjunction.
- The multiplication $\mu : T^2 \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\mu_a : GFGFa \to GFa$ corresponding to the composi-

tion of morphisms in $\mathcal{D}$ under the adjunction.

Definition 47 (Distributive law between monads, [Bec69])

Given two monads (T, η^T, μ^T) and (S, η^S, μ^S) on a category $\mathcal{C}$, a *distributive law* of T over S is a natural transformation $\lambda : TS \to ST$ such that the following diagrams commute:

$$\begin{tikzcd} & T \arrow[dl, "T\eta^S"'] \arrow[dr, "\eta^S T"] & \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd} \qquad \begin{tikzcd} & S \arrow[dl, "\eta^T S"'] \arrow[dr, "S\eta^T"] & \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd}$$

$$\begin{tikzcd} T^2S \arrow[r, "T\lambda"] \arrow[d, "\mu^T S"'] & TST \arrow[r, "\lambda T"] & ST^2 \arrow[d, "S\mu^T"] \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd} \qquad \begin{tikzcd} TS^2 \arrow[r, "\lambda S"] \arrow[d, "T\mu^S"'] & STS \arrow[r, "S\lambda"] & S^2T \arrow[d, "\mu^S T"] \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd}$$

This law induces a composite monad ST on $\mathcal{C}$ with

- unit $\eta^{ST} = \eta^S \circ \eta^T$
- multiplication $STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu^S\mu^T} ST$

Definition 48 (Algebra over a monad)

Let (T, η, μ) be a monad on a category $\mathcal{C}$.

An *algebra over the monad T* (or simply a T-algebra) consists of:

- An object A in $\mathcal{C}$,
- A morphism $a : TA \to A$ in $\mathcal{C}$, called the structure map, such that the following diagrams commute:

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[dr, "\mathrm{id}_A"'] & TA \arrow[d, "a"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} T^2A \arrow[r, "Ta"] \arrow[d, "\mu_A"'] & TA \arrow[d, "a"] \\ TA \arrow[r, "a"'] & A \end{tikzcd}$$

The left one is called the unit property, and the right one is called the action property.

We'll now reformulate the definition of monads in terms of Kleisli categories and extension operators.

Definition 49 (Kleisli category)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *Kleisli category* $\mathcal{C}_T$ has:

- The same objects as $\mathcal{C}$,
- For each pair of objects a, b, the morphisms from a to b in $\mathcal{C}_T$ are given by $\mathrm{Hom}_{\mathcal{C}_T}(a, b) = \mathrm{Hom}_{\mathcal{C}}(a, Tb)$,
- The identity morphism on an object a is given by $\eta_a : a \to Ta$,
- The composition of morphisms $f : a \to Tb$ and $g : b \to Tc$ in $\mathcal{C}_T$ is given by the morphism $g \circ_T f : a \to Tc$ defined as the composition:

$$a \xrightarrow{f} Tb \xrightarrow{Tg} T^2c \xrightarrow{\mu_c} Tc$$

Definition 50 (Extension operator)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *extension operator* $(-)^\sharp$ is defined for each morphism $f : a \to Tb$ in $\mathcal{C}$ as the morphism $f^\sharp : Ta \to Tb$ given by the composition:

$$Ta \xrightarrow{Tf} T^2 b \xrightarrow{\mu_b} Tb$$

In other words, $f^\sharp = \mu_b \circ Tf$.

We have the following properties of the extension operator:

- $\eta_a^\sharp = \mathrm{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- For any morphism $f : a \to Tb$, we have $f^\sharp \circ \eta_a = f$.
- For any morphisms $f : a \to Tb$ and $g : b \to Tc$, we have $(g^\sharp \circ f)^\sharp = g^\sharp \circ f^\sharp$.

Definition 51 (Kleisli triple)

A *Kleisli triple* on a category $\mathcal{C}$ consists of:

- A mapping $T : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{C})$,
- For each object a in $\mathcal{C}$, a morphism $\eta_a : a \to Ta$,
- For each morphism $f : a \to Tb$ in $\mathcal{C}$, a morphism $f^\sharp : Ta \to Tb$,

such that the following conditions hold:

- $\eta_a^\sharp = \mathrm{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- the following diagrams commute

$$\begin{tikzcd} a \arrow[r, "\eta_a"] \arrow[dr, "f"'] & Ta \arrow[d, "f^\sharp"] \\ & Tb \end{tikzcd} \qquad \begin{tikzcd} Ta \arrow[r, "f^\sharp"] \arrow[dr, "(g^\sharp \cdot f)^\sharp"'] & Tb \arrow[d, "g^\sharp"] \\ & Tc \end{tikzcd}$$

Proposition 52

Given a category $\mathcal{C}$, there is a one-to-one correspondence between monads on $\mathcal{C}$ and Kleisli triples on $\mathcal{C}$.

Proof

Given a monad (T, η, μ) on $\mathcal{C}$, we can construct a Kleisli triple by defining the extension operator as $f^\sharp = \mu_b \circ Tf$ for each morphism $f : a \to Tb$.

Conversely, given a Kleisli triple $(T, \eta, (-)^\sharp)$ on $\mathcal{C}$, we can construct a monad by defining the multiplication $\mu_a : T^2 a \to Ta$ as $\mu_a = (\mathrm{id}_{Ta})^\sharp$ for each object a in $\mathcal{C}$.
The conditions of the Kleisli triple ensure that the monad axioms are satisfied, and the construction is reversible, establishing the one-to-one correspondence. $\square$

Remark : Kleisli triples are sometimes referred to as *monads in extensive form*.

Example 53 (Monads in terms of binds and returns)

In the context of programming languages, a monad can be described in terms of a type constructor T and two operations: `return` (or `unit`) and `bind`. The `return` operation corresponds to the unit η of the monad, while the `bind` operation corresponds to the extension operator $(-)^\sharp$. Specifically, for a monad (T, η, μ) on a category $\mathcal{C}$, we can define:

- $\texttt{return}_a = \eta_a : a \to Ta$ for each object a in $\mathcal{C}$,
- $\texttt{bind}_{a,b}(f) = f^\sharp : Ta \to Tb$ for each morphism $f : a \to Tb$ in $\mathcal{C}$.

We use the notion of $\mathbb{S}$-algebra of [MW10] to define the distributive laws for monads in extensive form.

Definition 54 ($\mathbb{S}$-algebra, [MW10])

Given a monad in extensive form $\mathbb{S} = (S, \eta, (-)^{\mathbb{S}})$ on a category $\mathcal{C}$, an $\mathbb{S}$-*algebra* is a pair $\mathbb{B} = (B, (-)^{\mathbb{B}})$ where:

- $B \in \mathcal{C}$
- $(-)^{\mathbb{B}}_X : \mathcal{C}(X, B) \to \mathcal{C}(SX, B)$

such that the following diagrams commute for every $h : X \to B$ and $k : Y \to SX$:

$$\begin{tikzcd} X \arrow[r, "\eta X"] \arrow[dr, "h"'] & SX \arrow[d, "h^{\mathbb{B}}"] \\ & B \end{tikzcd} \qquad \begin{tikzcd} SY \arrow[r, "k^{\mathbb{S}}"] \arrow[dr, "(h^{\mathbb{B}}k)^{\mathbb{B}}"'] & SX \arrow[d, "h^{\mathbb{B}}"] \\ & B \end{tikzcd}$$

A morphism of $\mathbb{S}$-algebras $(B, (-)^{\mathbb{B}})$ to $(A, (-)^{\mathbb{A}})$ can be defined as an arrow $l : B \to A$ in $\mathcal{C}$ such that the following diagram commutes for every $h : X \to B$:

$$\begin{tikzcd} SX \arrow[r, "h^{\mathbb{B}}"] \arrow[dr, "(l \cdot h)^{\mathbb{A}}"'] & B \arrow[d, "l"] \\ & A \end{tikzcd}$$

Theorem 55 (Distributive law between monads in extensive form, [MW10])

Let $\mathbb{T} = (T, \eta_T, (-)^{\mathbb{T}})$ and $\mathbb{S} = (S, \eta_S, (-)^{\mathbb{S}})$ be monads on $\mathcal{C}$. A distributive law of $\mathbb{T}$ over $\mathbb{S}$ can equivalently be given by an $\mathbb{T}$-algebra $(STA, (-)^\lambda)$ for every $A \in \mathcal{C}$ such that

- $(S\eta_T A \cdot \eta_S A)^\lambda = \eta_S TA$
- for every $f : B \to TSA$, $(f^\lambda)^{\mathbb{S}} : (STB, (-)^\lambda) \to (STA, (-)^\lambda)$ is a morphism of $\mathbb{T}$-algebras.

Unfolding the conditions for the $\mathbb{T}$-algebra and for the morphism, the following diagrams must commute for every $h : X \to STA$, $k : Y \to TX$, $f : B \to STA$ and $g : X \to STB$:

$$\begin{tikzcd} X \arrow[r, "\eta_T X"] \arrow[dr, "h"'] & TX \arrow[d, "h^\lambda"] \\ & STA \end{tikzcd} \qquad \begin{tikzcd} TY \arrow[r, "k^{\mathbb{T}}"] \arrow[dr, "(h^\lambda k)^\lambda"'] & TX \arrow[d, "h^\lambda"] \\ & STA \end{tikzcd} \qquad \begin{tikzcd} TB \arrow[r, "g^\lambda"] \arrow[dr, "((f^\lambda)^{\mathbb{S}} \cdot g)^\lambda"'] & STB \arrow[d, "(f^\lambda)^{\mathbb{S}}"] \\ & STA \end{tikzcd}$$

Proposition 56 ([MW10])

Given a distributive law λ of $\mathbb{T}$ over $\mathbb{S}$, the composite monad is given by $\mathbb{S} \circ_\lambda \mathbb{T} = (ST, S\eta_T \cdot \eta_S, \big((-)^\lambda)^{\mathbb{S}}\big)$

Proposition 57 ([MW10])

Let $\mathbb{S}$ and $\mathbb{T}$ be monads on $\mathcal{C}$. There is a bijection between distributive laws of $\mathbb{T}$ over $\mathbb{S}$ and monad structures $(ST, S\eta_T \cdot \eta_S, (-)^{(\mathbb{ST})})$ on ST such that:

- for every $k : Y \to TX, S(k^{\mathbb{T}}) = (\eta_S TX \cdot k)^{(\mathbb{ST})}$
- for every $m : Y \to STX$ and $h : X \to STU$:
  - $h^{(\mathbb{ST})} = (h^{(\mathbb{ST})} \cdot \eta_S TX)^{\mathbb{T}}$
  - the following diagram commutes

$$\begin{tikzcd} SY \arrow[r, "m^{\mathbb{S}}"] \arrow[dr, "(h^{(\mathbb{ST})}\cdot m)^{\mathbb{S}}"'] & STX \arrow[d, "h^{(\mathbb{ST})}"] \\ & STU \end{tikzcd}$$

1.4 2-categories

Definition 58 (2-category)

A *2-category* C consists of:

- Cellular structure:
  - A collection C_0 of 0-cells $a, b, c, \dots$ (objects),
  - For each pair of 0-cells a, b, a collection $C_1(a, b)$ of 1-cells (morphisms) from a to b,
  - For each pair of 1-cells $f, g \in C_1(a, b)$, a collection $C_2(f, g)$ of 2-cells (2-morphisms) from f to g
- Identities:
  - A 1-cell $\mathrm{id}_a : a \to a$ for each 0-cell $a \in C_0$ (identity 1-cell),
  - A 2-cell $\mathrm{id}_f : f \Rightarrow f$ for each 1-cell $f \in C_1(a, b)$ (identity 2-cell)
- Composition:
  - A composition operation on 1-cells $g \circ f$ for every pair of composable 1-cells $f : a \to b$ and $g : b \to c$,
  - A vertical composition operation on 2-cells $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$,
  - A horizontal composition operation on 2-cells $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$ for every pair of composable 2-cells $\alpha : f \Rightarrow f'$ and $\beta : g \Rightarrow g'$

  such that

  - 1-cell compositions are associative and identities are neutral
  - 2-cell vertical compositions are associative and identities are neutral
  - 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
  - the interchange law holds: for any composable 2-cells $\alpha, \alpha', \beta, \beta'$, we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

Example 59 (2-category of Categories)

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,
- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

Definition 60 (Adjunction in a 2-category)

Given a 2-category C, an *adjunction* between 0-cells a and b in C consists of:

- Two 1-cells $f : a \to b$ and $g : b \to a$,
- Two 2-cells $\eta : \mathrm{id}_a \Rightarrow g \circ f$ (unit) and $\varepsilon : f \circ g \Rightarrow \mathrm{id}_b$ (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \mathrm{id}_f \qquad (\eta \bullet g) \circ (g \bullet \varepsilon) = \mathrm{id}_g$$

Definition 61 (2-functor)

Given two 2-categories C and D, a *2-functor* $F : C \to D$ consists of three mappings:

- $F : C_0 \to D_0$ that assigns to each 0-cell a in C a 0-cell Fa in D,
- $F : C_1(a, b) \to D_1(Fa, Fb)$ for each pair of 0-cells a, b in C, that assigns to each 1-cell $f : a \to b$ in C a 1-cell $Ff : Fa \to Fb$ in D,
- $F : C_2(f, g) \to D_2(Ff, Fg)$ for each pair of 1-cells $f, g : a \to b$ in C, that assigns to each 2-cell $\alpha : f \Rightarrow g$ in C a 2-cell $F\alpha : Ff \Rightarrow Fg$ in D,

such that F preserves identities and compositions of 1-cells and 2-cells.

Lemma 62

Let $F : C \to D$ be a 2-functor.

If $\langle f : a \to b, u, \eta, \varepsilon \rangle$ is an adjunction in C, then $\langle Ff : Fa \to Fb, Fu, F\eta, F\varepsilon \rangle$ is an adjunction in D.

Proof

The candidate adjunction has the right type, and a 2-functor preserves pasting diagrams and identities, so the triangle identities are preserved. □

Definition 63 (natural 2-transformation)

Let $F, G : C \to D$ be two 2-functors between 2-categories.

A *(strict) natural 2-transformation* $\begin{tikzcd} C \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow \Xi"] & D \end{tikzcd}$ consists of a 1-cell $\Xi_a : F_0 a \to G_0 a$ for each 0-cell a such that:

- for every 1-cell $f : a \to b$ in C:

$$\begin{tikzcd} F_0 a \arrow[r, "F_1 f"] \arrow[d, "\Xi_a"'] & F_0 b \arrow[d, "\Xi_b"] \\ G_0 a \arrow[r, "G_1 f"'] & G_0 b \end{tikzcd}$$

- for every 2-cell $\begin{tikzcd} a \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow \alpha"] & b \end{tikzcd}$:

$$\begin{tikzcd} & Ga \arrow[dr, bend left, "Gf"] \arrow[dr, bend right, "Gg"'] \arrow[dr, phantom, "\Downarrow G\alpha"] & \\ Fa \arrow[ur, "\Xi_a"] \arrow[dr, "Fg"'] & & Gb \\ & Fb \arrow[ur, "\Xi_b"'] & \end{tikzcd} = \begin{tikzcd} & Ga \arrow[dr, "Gf"] & \\ Fa \arrow[ur, "\Xi_a"] \arrow[dr, bend left, "Ff"] \arrow[dr, bend right, "Fg"'] \arrow[dr, phantom, "\Downarrow F\alpha"] & & Gb \\ & Fb \arrow[ur, "\Xi_b"'] & \end{tikzcd}$$

1.5 Multicategories

Definition 64 (Multicategory)

A multicategory $\mathcal{U}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{U})$,
- For each finite (possibly empty) sequence of objects $(a_1, a_2, ..., a_n)$ and an object b, a set of morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$,
- For each object a, an identity morphism $\mathrm{id}_a : (a) \to a$,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

Example 65 (Category with Finite Products as a Multicategory)

Given a category $\mathcal{C}$ with finite products, we can construct a multicategory $\mathcal{U}$ where:

- The objects of $\mathcal{U}$ are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$ are given by the morphisms in $\mathcal{C}$ from the product $a_1 \times a_2 \times ... \times a_n$ to b,

- The identities rely on the canonical isomorphism

$$\mathrm{id}_a := \prod_{i=1}^{n} a \cong a$$

- The composition is defined using the universal property of products and the composition in $\mathcal{C}$.

Example 66 (Monoidal Category as a Multicategory)

A monoidal category $(\mathcal{C}, \otimes, I)$ can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$ are given by the morphisms in $\mathcal{C}$ from $a_1 \otimes a_2 \otimes ... \otimes a_n$ to b (by nesting the monoidal product)

Definition 67 (Multifunctor)

A *multi-functor* $F : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$ consists of two mappings:

- A mapping $F : \mathrm{Ob}(\mathcal{U}) \to \mathrm{Ob}(\mathcal{V})$ that assigns to each object a in $\mathcal{U}$ an object Fa in $\mathcal{V}$,
- A mapping $F : \mathcal{U}(a_1, a_2, ..., a_n; b) \to \mathcal{V}(Fa_1, Fa_2, ..., Fa_n; Fb)$ that assigns to each morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$ a morphism $Ff : (Fa_1, Fa_2, ..., Fa_n) \to Fb$ in $\mathcal{V}$,

such that F preserves:

- Identities: For each object a in $\mathcal{U}$, $F\mathrm{id}_a = \mathrm{id}_{Fa}$,
- Composition: For morphisms $f : (a_1, a_2, ..., a_n) \to b$ and $g_i : (a_{i1}, a_{i2}, ..., a_{ik_i}) \to a_i$ in $\mathcal{U}$, we have
$$F(f \circ (g_1, g_2, ..., g_n)) = Ff \circ (Fg_1, Fg_2, ..., Fg_n).$$

Definition 68 (Multi-natural transformation)

Given two multi-functors $F, G : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$, a *multi-natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{U})}$ such that for every morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$, the following diagram commutes:

$$\begin{tikzcd}
(Fa_1, Fa_2, ..., Fa_n) \arrow[r, "Ff"] \arrow[d, "{(\eta_{a_1}, \eta_{a_2}, ..., \eta_{a_n})}"'] & Fb \arrow[d, "\eta_b"] \\
(Ga_1, Ga_2, ..., Ga_n) \arrow[r, "Gf"'] & Gb
\end{tikzcd}$$

Example 69 (Category of multicategories)

We have a 2-category **Multicat** where:

- 0-cells are multi-categories
- 1-cells are multi-functors
- 2-cells are multi-natural transformations

with multi-functor composition and identities:

$$\mathrm{Id}_C\, a := a \quad \mathrm{Id}_C\, f := f$$

vertical, horizontal composition of multi-natural transformations, with identities:

$$(\mathrm{id}_F)_a := \mathrm{id}_{Fa} : (a) \to a$$

Example 70 (Ordinary 2-functor)

We have a 2-functor structure $-_o : \mathbf{Multicat} \to \mathbf{Cat}$ sending:

- each multi-category $\mathcal{U}$ to its (**o**rdinary) category of morphisms:
  - $\mathrm{Ob}(\mathcal{U}_o) := \mathrm{Ob}(\mathcal{U})$
  - $\mathcal{U}_o(a,b) := \mathcal{U}(a;b)$, i.e the morphisms in $\mathcal{U}_o$ are the multi-morphisms with a length-1 source
  - identities and composition are given by identity morphisms and composition in the multi-category
- each multi-functor $F : \mathcal{U} \to \mathcal{V}$ to the functor sending:
  - $a \in \mathrm{Ob}(\mathcal{U})$ to $Fa \in \mathrm{Ob}(\mathcal{V})$
  - $f : (a) \to b$ to $F_o f := Ff : (Fa) \to Fb$
- each multi-natural transformation $\begin{tikzcd} \mathcal{U} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & \mathcal{V} \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$ to $(\alpha_o)_a := \alpha_a$

It is 2-faithful.

1.6 Operads

Definition 71 (Operad)

An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n\geq 0}$, where $\mathcal{O}_n$ is the set of n-ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \to \mathcal{O}_{k_1+\cdots+k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: For all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have
$$\gamma(f; g_1, \ldots, g_n) = f(g_1(h_{11}, \ldots, h_{1k_1}), \ldots, g_n(h_{n1}, \ldots, h_{nk_n})).$$
- Identity: For all $f \in \mathcal{O}_n$, we have
$$f(e, \ldots, e) = f.$$

Definition 72 (Algrebraic Theory)

An *algebraic theory* $\mathcal{T}$ consists of:

- A collection of sets $\{\mathcal{T}_n\}_{n\geq 0}$, where $\mathcal{T}_n$ is the set of n-ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

Example 73 (Operad from an Algebraic Theory)

Given an algebraic theory $\mathcal{T}$, we can construct an operad $\mathcal{O}^{\mathcal{T}}$ where:

- $\mathcal{O}^{\mathcal{T}}_n$ consists of the n-ary operations in $\mathcal{T}$,
- The composition function γ is defined by substituting operations according to the equations of the theory,
- The identity element e corresponds to the identity operation in the theory.

Definition 74 (Renaming)

A *renaming* $\rho : n \to m$ is any function from the set of n variables to the set of m variables.

Example 75 (of renaming)

Let $\rho : 3 \to 2$ be the renaming defined by $\rho(x_1) = y_1$, $\rho(x_2) = y_2$, and $\rho(x_3) = y_1$. Then for any term $t \in \mathcal{T}_3$, we have $\rho(t) \in \mathcal{T}_2$ obtained by substituting y_1 for x_1 and x_3, and y_2 for x_2 in t.

Definition 76 (Translation)

Let $\mathcal{O}$ be an operad and $\mathcal{T}$ be a theory.

A *translation* $\mathfrak{T} : \mathcal{O} \to \mathcal{T}$ consists of an assignment $\mathfrak{T}f \in \mathcal{T}_n$ for every $f \in \mathcal{O}_n$ such that:

- the identity $\mathrm{id} : \mathcal{O}_1$ is sent the the unique variable projection: $\mathfrak{T}\mathrm{id} = x \in \mathcal{T}_1$
- the translation respects composition, for all $f \in \mathcal{O}_n$ composable with $g_i \in \mathcal{O}_{m_i}$:

$$\mathfrak{T}(f \circ (g_1, \dots, g_n)) = \mathfrak{T}f[x_i \mapsto \mathfrak{T}g_i]_{i=1}^{n}$$

A translation is *surjective*, written $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{T}$, if for every term $t \in \mathcal{T}_n$, there exists an operator $f \in \mathcal{O}_m$ and a renaming $\rho : m \to n$ such that $t = \mathfrak{T}f[\rho]$.

Definition 77 (Model)

Let $\mathcal{O}$ be an operad, and $\mathcal{U}$ be a multicategory. A *model* of $\mathcal{O}$ in $\mathcal{U}$ is a multi-functor $\mathcal{A} : \mathcal{O} \to \mathcal{U}$ that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

Definition 78

Homomorphism of Models A *homomorphism* $h : \mathcal{A} \to \mathcal{B}$ between $\mathcal{O}$-models in $\mathcal{U}$ consists of a morphism $\underline{h} : (\underline{\mathcal{A}}) \to \underline{\mathcal{B}}$ such that for all $f \in \mathcal{O}_n$,

$$\underline{h} \circ (\mathcal{A}[\![f]\!]) = \mathcal{B}[\![f]\!]\,(\underline{h}, \dots, \underline{h})$$

Example 79 (Category of Models)

Given an operad $\mathcal{O}$ and a multicategory $\mathcal{U}$, we can construct a category $\mathbf{Mod}(\mathcal{O},\mathcal{U})$ where:

- The objects are the models of $\mathcal{O}$ in $\mathcal{U}$,
- The morphisms are the homomorphisms between these models.

Proposition 80

Let $\mathfrak{T} : \mathcal{O} \to \mathcal{T}$ be a translation, and let $\mathcal{C}$ be a category with chosen finite product $\prod$.

There is a carrier-preserving, fully-faithful functor $\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) : \mathbf{Mod}(\mathcal{T}, \mathcal{C}) \to \mathbf{Mod}(\mathcal{O}, \langle \mathcal{C}, \prod \rangle)$ given by:

$$\left(\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle) A\right) [\![f]\!] := A [\![\mathfrak{T} f]\!] \qquad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle) h := h$$

Proof
The compatibility with composition and identities of the operad follows from their preservation by $\mathfrak{T}$. Preservation of identities and composition, and fully-faithfulness are evident.
□

Proposition 81

We define a 2-functor $\mathbf{Mod}(\mathcal{O}, -) : \mathbf{Multicat} \to \mathbf{Cat}$ such that:

- for every multi-category $\mathcal{U}$, $\mathbf{Mod}(\mathcal{O},\mathcal{U})$ is the category of $\mathcal{O}$-models in $\mathcal{U}$.
- for every multi-functor $F : \mathcal{U} \to \mathcal{V}$, the functor $\mathbf{Mod}(\mathcal{O}, F) : \mathbf{Mod}(\mathcal{O},\mathcal{U}) \to \mathbf{Mod}(\mathcal{O},\mathcal{V})$ sends:
  - each $\mathcal{O}$-model $\mathcal{A}$ in $\mathcal{U}$ to the $\mathcal{O}$-model:
    * $\underline{F\mathcal{A}} := F\underline{\mathcal{A}}$
    * $(F\mathcal{A}) [\![f]\!] : (F\underline{\mathcal{A}}, \ldots, F\underline{\mathcal{A}}) \xrightarrow{F(\mathcal{A}[\![f]\!])} F\underline{\mathcal{A}}$
  - each $\mathcal{O}$-homomorphism $h : \mathcal{A} \to \mathcal{B}$ to the $\mathcal{O}$-homomorphism over $F\underline{h} : F\mathcal{A} \to F\mathcal{B}$
- for every multi-natural transformation $\begin{tikzcd} \mathcal{U} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{V} \end{tikzcd}$ we have a natural transformation

$$\begin{tikzcd} \mathbf{Mod}(\mathcal{O},\mathcal{U}) \arrow[r, bend left, "\mathbf{Mod}(\mathcal{O},F)"] \arrow[r, bend right, "\mathbf{Mod}(\mathcal{O},G)"'] \arrow[r, phantom, "\Downarrow\mathbf{Mod}(\mathcal{O},\alpha)"] & \mathbf{Mod}(\mathcal{O},\mathcal{V}) \end{tikzcd}$$ given, for every model $\mathcal{A}$, by

$$\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}} : (F\underline{\mathcal{A}}) \xrightarrow{\alpha_{\underline{\mathcal{A}}}} G\underline{\mathcal{A}}$$

1.7 Free algebras

We'll use the definitions from [All+25].

Definition 82 (Finitary signature)

A *finitary signature* $\Sigma = (\mathrm{op}\,\Sigma, \mathrm{arity})$ consists of a set $\mathrm{op}\,\Sigma$ of *operators*, and an assigment $\mathrm{arity} : \mathrm{op}\,\Sigma \to \mathbb{N}$ associating to each operator in $\mathrm{op}\,\Sigma$ its *arity*.

Definition 83 (Algebra)

An *algebra* $A = (\underline{A}, A[\![-]\!])$ over a finitary signature Σ consists of:

- A set $\underline{A}$ called the *carrier* of A,
- For each operator $f \in \mathrm{op}\,\Sigma$ of arity n, a function $A[\![f]\!] : \underline{A}^n \to \underline{A}$ called the *interpretation* of f in A.

Definition 84 (Terms)

Given a set X of variables and a finitary signature Σ, the set of Σ-*terms over* X, denoted $\mathrm{Term}_\Sigma(X)$, is defined inductively as follows:

- Every variable $x \in X$ is a term,
- If $f \in \mathrm{op}\,\Sigma$ is an operator of arity n and $t_1, t_2, ..., t_n$ are terms, then $f(t_1, t_2, ..., t_n)$ is a term.

Remark : The definition of algebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

Definition 85 (Environment and Interpretation)

An *environment* ρ for a context X in an algebra A is a function $\rho : X \to \underline{A}$ that assigns to each variable in X an element of the carrier of A.

Given a term $t \in \mathrm{Term}_\Sigma(X)$, denoted $X \vdash t$, the algebra A gives an *interpretation* function $A[\![t]\!] : \underline{A}^X \to \underline{A}$. Given an environment $\rho : X \to \underline{A}$, we can evaluate the term t in A under the environment ρ, denoted $A[\![t]\!](\rho)$, by recursively substituting the variables in t with their corresponding values in $\underline{A}$ according to ρ and applying the interpretations of the operators as defined in A.

Definition 86 (Valid equations)

An equation $X \vdash l = r$ between terms in $\mathrm{Term}_\Sigma(X)$ is *valid* in an algebra A if for every environment $\rho : X \to \underline{A}$, we have $A[\![l]\!](\rho) = A[\![r]\!](\rho)$.

Definition 87 (Presentation and Models)

A *presentation* $\mathcal{T} = (\Sigma_\mathcal{T}, \mathcal{T}_{ax})$ consists of a finitary signature $\Sigma_\mathcal{T}$ and a set $\mathcal{T}_{ax}$ of *axioms*, which are equations between terms in $\mathrm{Term}_{\Sigma_\mathcal{T}}(X)$ for some set of variables X.

A $\mathcal{T}$-model A is a $\Sigma_\mathcal{T}$-algebra such that all equations in $\mathcal{T}_{ax}$ are valid in A.

A $\mathcal{T}$-model A over a context X is a $\mathcal{T}$-model A together with an environment $\rho : X \to \underline{A}$.

Definition 88 (Homomorphisms)

Let $A = (\underline{A}, A[\![-]\!])$ and $B = (\underline{B}, B[\![-]\!])$ be algebras over the same signature Σ.

A *homomorphism* of Σ-algebras from A to B is a function $h : \underline{A} \to \underline{B}$ such that for every operator $f \in \mathrm{op}\,\Sigma$ of arity n, we have:

$$B[\![f]\!](h(a_1), \ldots, h(a_n)) = h(A[\![f]\!](a_1, \ldots, a_n))$$

for all $a_1, \ldots, a_n \in \underline{A}$.

A *homomorphism* of $\mathcal{T}$-models is a homomorphism of $\Sigma_\mathcal{T}$-algebras between the underlying algebras of the models.

Definition 89 (Morphism)

A *morphism* $h : A \to B$ of $\mathcal{T}$-models over X is a $\mathcal{T}$-model homomorphism that makes the diagram commute:

$$\begin{tikzcd} X \arrow[r, "\rho_A"] \arrow[dr, "\rho_B"'] & \underline{A} \arrow[d, "h"] \\ & \underline{B} \end{tikzcd}$$

Definition 90 (Free model (or free algebra))

Let $\mathcal{T}$ be a presentation and X be a set of variables.

A *free $\mathcal{T}$-model* over X is a $\mathcal{T}$-model $F(X)$ such that for any $\mathcal{T}$-model A over X, there exists a unique morphism of $\mathcal{T}$-models from $F(X)$ to A.

The existence-and-uniqueness property is called the *universal property* of free $\mathcal{T}$-models over X.

Informaly:

- A signature gives defines the operators and constants (arity 0 operators).
- An algebra gives a concrete interpretation of the signature.
- A presentation defines axioms on top of the signature, for instance the axioms of a group, or of a semiring, etc.
- A model of a presentation is an algebra that satisfies the axioms of the presentation.
- A free algebra is the most general model of a presentation.

We can give an equivalent definition for free algebras using universal arrows.

Proposition 91

Let $\mathcal{T}$ be a presentation, and X be a set. Let $\mathcal{T}$-**Mod** be the category of $\mathcal{T}$-models over X. Let $U : \mathcal{T}\text{-}\mathbf{Mod} \to \mathbf{Set}$ be the forgetful functor that sends a $\mathcal{T}$-model to its underlying set.

Then, a free $\mathcal{T}$-model over X is a universal arrow from X to U.

Proof

Let $F\,X$ be a free $\mathcal{T}$-model over X. We want to show that $(F\,X, \eta_X)$ is a universal arrow from X to U, where $\eta_X : X \to U\,F\,X$ is the function that sends each variable in X to its interpretation in $F\,X$.

Let A be a $\mathcal{T}$-model over X, and let $\rho_A : X \to U\,A$ be the environment of A. Since $F\,X$ is a free $\mathcal{T}$-model over X, there exists a unique morphism of $\mathcal{T}$-models $h : F\,X \to A$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "\rho_A"'] & U\,F\,X \arrow[d, "U\,h"] \\ & U\,A \end{tikzcd}$$

This shows that $(F\,X, \eta_X)$ satisfies the universal property of being a universal arrow from X to U.

Conversely, if $(F\,X, \eta_X)$ is a universal arrow from X to U, then for any $\mathcal{T}$-model A over

X, there exists a unique morphism of $\mathcal{T}$-models from $F\,X$ to A, which means that $F\,X$ is a free $\mathcal{T}$-model over X. Therefore, the two definitions are equivalent. $\square$

1.8 Multi-homomorphisms

This section will be a subset of Ohad's note. In particular, all proofs will be omitted.

In all of this section, we fix an algebraic theory $\mathcal{T}$ and $\langle \mathcal{C}, \prod \rangle$ a category with chosen finite products.

Definition 92 (Strength)

For all $k \in \mathbb{N}^*$ and $X, Y \in \mathcal{C}$, we define the *strength* (for the reader monad X^k) by

$$\mathrm{str} : X^k \times Y \xrightarrow{\langle \pi_i \times \mathrm{id} \rangle_{i=1}^k} (X \times Y)^k$$

Definition 93 (Multi-$\mathcal{T}$-homomorphism)

Let $A_1, A_2, \ldots, A_k$ and B be $\mathcal{T}$-models in $\mathcal{C}$.

A *multi-$\mathcal{T}$-homomorphism* from $A_1, A_2, \ldots, A_k$ to B is a morphism $h : \prod_{i=1}^k \underline{A_i} \to \underline{B}$ such that for every term $t \in \mathcal{T}_k$ and $1 \leqslant i_0 \leqslant n$, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{A}_{i_0}^k \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \arrow[r, "\mathrm{str}"] \arrow[dd, "\mathcal{A}_{i_0}[\![t]\!] \times \mathrm{id}"'] & \left(\mathcal{A}_{i_0} \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \right)^k \arrow[r, "\cong"] & \left(\prod\limits_{i=1}^n \mathcal{A}_i \right)^k \arrow[d, "f^k"] \\
 & & \mathcal{B}^k \arrow[d, "\mathcal{B}[\![t]\!]"] \\
\mathcal{A}_{i_0} \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \arrow[r, "\cong"'] & \prod\limits_{i=1}^n \mathcal{A}_i \arrow[r, "f"'] & \mathcal{B}
\end{tikzcd}$$

▷ We can then define a multi-category structure $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$ over the finite-product multi-category:

- the objects are $\mathcal{T}$-algebra
- the multi-morphisms are the multi-$\mathcal{T}$-homomorphisms
- the identities are the identity morphisms
- composition is inherited from $\langle \mathcal{C}, \prod, \rangle$

▷ We also define a multi-functor $\underline{\;} : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \to \langle \mathcal{C}, \prod \rangle$ which sends an algebra to its carrier and a multi-$\mathcal{T}$-homomorphism to its underlying morphism.

▷ We define another 2-category $\mathbf{Cat}_\times$ consisting of:

- 0-cells are $\langle \mathcal{C}, \prod \rangle$, categories with chosen finite products
- 1-cells are finite product preserving functors, not necessarily preserving the chosen products
- 2-cells are the natural transformations between these functors

Therefore, this is a sub-category of **Cat**.

▷ We define the data of a 2-functor $\mathbf{Multi}(\mathcal{T}, -) : \mathbf{Cat}_\times \to \mathbf{Multicat}$ sending:

- each category with chosen products (0-cells) $\mathcal{C}$ to the multi-category (0-cell) $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$
- each finite-product preserving functor (1-cell) $H : \mathcal{C} \to \mathcal{D}$ to the following multi-functor $\mathbf{Multi}(\mathcal{T}, H) : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \to \mathbf{Multi}(\mathcal{T}, \mathcal{D})$, sending:
  - each $\mathcal{T}$-algebra $\mathcal{A}$ to the $\mathcal{T}$-algebra $\mathbf{Multi}(\mathcal{T}, H) \coloneqq H\mathcal{A}$, given by:
    * its carrier $\underline{H\mathcal{A}}$ is $H\underline{\mathcal{A}}$
    * the interpretation of each $t \in \mathcal{T}_n$ is $H\mathcal{A}[\![t]\!] : (H\underline{\mathcal{A}}^n) \xrightarrow{\cong} H(\underline{\mathcal{A}}^n) \xrightarrow{\mathcal{A}[\![t]\!]} H\underline{\mathcal{A}}$
  - each multi-$\mathcal{T}$-homomorphism $h : (\mathcal{A}_1, \dots, \mathcal{A}_n) \to \mathcal{B}$ to the multi-$\mathcal{T}$-homomorphism $\mathbf{Multi}(\mathcal{T}, H)h : (H\mathcal{A}_1, \dots, H\mathcal{A}_n)$ given by:

$$\mathbf{Multi}(\mathcal{T}, H)h : \prod_{i=1}^{n} H\mathcal{A}_i \xrightarrow{\cong} H\left(\prod_{i=1}^{n} \mathcal{A}_i\right) \xrightarrow{Hh} H\mathcal{B}$$

- each natural transformation $\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$ (2-cell) to the multi-natural transformation

$\begin{tikzcd} \mathbf{Multi}(\mathcal{T},\mathcal{C}) \arrow[r, bend left, "\mathbf{Multi}(\mathcal{T},H)"] \arrow[r, bend right, "\mathbf{Multi}(\mathcal{T},K)"'] \arrow[r, phantom, "\Downarrow\mathbf{Multi}(\mathcal{T},\alpha)"] & \mathbf{Multi}(\mathcal{T},\mathcal{D}) \end{tikzcd}$, given, at an algebra $\mathcal{A} \in \mathbf{Mod}(\mathcal{T},\mathcal{C})$, by the unary multi-$\mathcal{T}$-homomorphism corresponding to the following $\mathcal{T}$-homomorphism:

$$\mathbf{Multi}(\mathcal{T}, \alpha)'_{\mathcal{A}} : H\underline{\mathcal{A}} \xrightarrow{\alpha_{\underline{\mathcal{A}}}} K\underline{\mathcal{A}}$$

▷ We define another 2-functor $\langle - \rangle : \mathbf{Cat}_\times \to \mathbf{Multicat}$, sending:

- 0-cells: categories with specified products $\langle \mathcal{C}, \prod \rangle$ to their associated multi-category $\langle \mathcal{C}, \prod \rangle$ (cf. Example 65)
- 1-cells: finite-product preserving functors $H : \mathcal{C} \to \mathcal{D}$ to the multi-functor sending:
  - objects $a \in \mathrm{Ob}\,(\langle \mathcal{C}, \prod \rangle) = \mathrm{Ob}\,\mathcal{C}$ to $\langle H \rangle\, a \in \mathrm{Ob}\,\mathcal{D}$
  - multi-morphisms $f : (a_1, \dots, a_n) \to b$ to the multi-morphism:

$$\langle H \rangle\, f : \prod_{i=1}^{n} Ha_i \xrightarrow{\cong} H\left(\prod_{i=1}^{n} a_i\right) \xrightarrow{Hf} Hb$$

- 2-cells: natural transformations $\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$ to the multi-natural transformation given for each $a \in \mathrm{Ob}(\mathcal{C})$, by:

$$\langle \alpha \rangle_a : \prod_{i=1}^{1} Ha \cong Ha \xrightarrow{\alpha_a} Ka$$

▷ Finally, we have a natural 2-transformation $\begin{tikzcd} \mathbf{Cat}_\times \arrow[r, bend left, "\mathbf{Multi}(\mathcal{T},-)"] \arrow[r, bend right, "\langle - \rangle"'] \arrow[r, phantom, "\Downarrow\underline{}"] & \mathbf{Multicat} \end{tikzcd}$ whose components, for each category $\langle \mathcal{C}, \prod \rangle$ with chosen finite products, consists of the forgetful multi-functors sending:

- each $\mathcal{T}$-algebra $\mathcal{A}$ to its carrier $\underline{\mathcal{A}}$
- each multi-$\mathcal{T}$-homomorphism $h : (\mathcal{A}_1, \ldots, \mathcal{A}_n) \to \mathcal{B}$ to its underlying morphism $\underline{h} : \prod_{i=1}^{n} \underline{\mathcal{A}_i} \to \underline{\mathcal{B}}$

All of the proofs that the following definitions are indeed of the nature they were described of are in Ohad's notes.

1.9 Yoneda structure

The formalisation uses the work of Street and Walters in [SW78].

In this section, we will use the notations of section 5.2 of Ohad's notes.

Definition 94 (Admissible multi-functor)
A multi-functor $F : \mathcal{U} \to \mathcal{V}$ is *admissible* when, for all $a_1, \ldots, a_n$ in $\mathcal{U}$ and b in $\mathcal{V}$, the multi-homset $\mathcal{V}(Fa_1, \ldots, Fa_n; b) \in \mathbf{Set}$ is small.

Definition 95 (Admissible multi-category)
A multi-category is *admissible* when its identity multi-functor is admissible, i.e when it is locally small.

Lemma 96

The admissible multi-functors are closed under precomposition, i.e, for $\mathcal{U} \xrightarrow{F} \mathcal{V} \xrightarrow{G} \mathcal{W}$, if G is admissible, then $G \circ F$ is also admissible.

Definition 97 (Opposite multi-category)

Let $\mathcal{U}$ be a multi-category. We define a multi-category $\mathcal{U}^{\mathrm{op}}$ as follows:

- $\mathrm{Ob}(\mathcal{U}^{\mathrm{op}}) \coloneqq \mathrm{List}\,\mathrm{Ob}(\mathcal{U})$ are finite sequences $\mathbf{a} = \langle \mathbf{a}_1 \ldots, \mathbf{a}_n\rangle$ of objects of $\mathcal{U}$.
- Morphisms $f \in \mathcal{U}^{\mathrm{op}}(\mathbf{a}, \mathbf{b})$, which we will write as "$f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$", consists of:
  - an ordered partition $\pi_f \coloneqq \langle \pi_1, \ldots, \pi_n\rangle \in \mathrm{List}^2\,\mathrm{Ob}(\mathcal{U})$, i.e, $\mathbf{b} = \mathbin{+\!\!+}_{i=1}^{n} \pi_i$
  - for each $1 \leqslant i \leqslant n$, a multi-morphism $f_i : \mathbf{a}_i \leftarrow (\mathbf{b}_i)$.

$$
\begin{tikzcd}[row sep=small]
 & & \pi_{11} & = & \mathbf{b}_{\langle 1,1\rangle} \\
 & & \vdots & & \\
\mathbf{a}_1 & f_1 \arrow[l] \arrow[uur, no head] \arrow[r, no head] & \pi_{1|\mathbf{b}_1|} & = & \mathbf{b}_{\langle 1,|\mathbf{b}_1|\rangle} \\
\vdots & & & & \\
\mathbf{a}_n & f_n \arrow[l] \arrow[r, no head] \arrow[ddr, no head] & \pi_{n1} & = & \mathbf{b}_{\langle n,1\rangle} \\
 & & \vdots & & \\
 & & \pi_{n|\mathbf{b}_n|} & = & \mathbf{b}_{\langle n,|\mathbf{b}_n|\rangle}
\end{tikzcd}
$$

  - The identity $\mathrm{id}_{\mathbf{a}}^{[\mathcal{U}^{\mathrm{op}}]}$ consists of:
    - $\pi_{\mathrm{id}_{\mathbf{a}}} \coloneqq \langle \langle a_1 \rangle, \ldots, \langle a_n \rangle\rangle$, the partition into singletons, and
    - $(\mathrm{id}_{\mathbf{a}})_i \coloneqq \mathrm{id}_{\mathbf{a}_i} : \mathbf{a}_i \leftarrow (\mathbf{a}_i)$
  - To compose morphisms $\mathbf{a} \overset{f}{\leftarrow} \mathbf{b} \overset{g}{\leftarrow} \mathbf{c}$, use the multiplication of the Tree monad to get an iterated partition of depth 3, and then flatten all the depth 1 iterated lists into lists. Then, compose the multi-morphisms to get a morphism $f \circ g : \mathbf{a} \leftarrow \mathbf{c}$ (note the order of arguments in this notation). Letting $|\mathbf{a}| \eqqcolon n$ and $|\pi_{f,i}| \eqqcolon m_i$:
    - $\pi_{f\circ g} \coloneqq \left\langle \left\langle \pi_{g_{\langle 1,1\rangle}} \mathbin{+\!\!+} \ldots \mathbin{+\!\!+} \pi_{g_{\langle 1,m_1\rangle}} \right\rangle, \ldots, \left\langle \pi_{g_{\langle n,1\rangle}} \mathbin{+\!\!+} \ldots \mathbin{+\!\!+} \pi_{g_{\langle n,m_n\rangle}} \right\rangle \right\rangle$
    - $(f \circ g)_i \coloneqq f_i \circ (g_{\langle i,1\rangle}, \cdots, g_{\langle i,m_i\rangle})$

$$
\begin{tikzcd}[row sep=small]
 & & & & \mathbf{c}_{\langle 1,1,1\rangle} \\
 & & & & \vdots \\
 & & \mathbf{b}_{\langle 1,1\rangle} & g_{\langle 1,1\rangle} \arrow[l] \arrow[uur, no head] \arrow[r, no head] & \mathbf{c}_{\langle 1,1,k_{11}\rangle} \\
 & & \vdots & \vdots & \\
\mathbf{a}_1 & f_1 \arrow[l] \arrow[uur, no head] \arrow[r, no head] & \mathbf{b}_{\langle 1,m_1\rangle} & g_{\langle 1,m_1\rangle} \arrow[l] \arrow[r, no head] \arrow[ddr, no head] & \mathbf{c}_{\langle 1,m_1,1\rangle} \\
 & & & & \vdots \\
 & & & & \mathbf{c}_{\langle 1,m_1,k_{1m_1}\rangle} \\
\vdots & & & & \\
 & & & & \mathbf{c}_{\langle n,1,1\rangle} \\
 & & & & \vdots \\
\mathbf{a}_n & f_n \arrow[l] \arrow[r, no head] \arrow[ddr, no head] & \mathbf{b}_{\langle n,1\rangle} & g_{\langle n,1\rangle} \arrow[l] \arrow[uur, no head] \arrow[r, no head] & \mathbf{c}_{\langle n,1,k_{n1}\rangle} \\
 & & \vdots & \vdots & \\
 & & \mathbf{b}_{\langle n,m_n\rangle} & g_{\langle n,m_n\rangle} \arrow[l] \arrow[r, no head] \arrow[ddr, no head] & \mathbf{c}_{\langle n,m_n,1\rangle} \\
 & & & & \vdots \\
 & & & & \mathbf{c}_{\langle n,m_n,k_{nm_n}\rangle}
\end{tikzcd}
$$

1.10 Free extensions

Free extensions where introduced in [YGK18]. They are a unifying theory for specifying (generalized) partially-static data-structures by recourse to universal algebra.

In this section, let $\mathcal{T}$ be a presentation, X be a set of variables and A be a $\mathcal{T}$-algebra.

Definition 98 (Extension)

An *extension* of A by X is a triple $\langle B, h, e\rangle$ where:

- B is a $\mathcal{T}$-algebra
- $h : A \to B$ is a $\mathcal{T}$-homomorphism
- $e : X \to B$ is a function

Definition 99 (Morphism of extensions)

Let $b_1 = \langle B_1, h_1, e_1\rangle, b_2 = \langle B_2, h_2, e_2\rangle$ be two extensions of A by X.

A *morphism of extensions* $h : b_1 \to b_2$ is a $\mathcal{T}$-homomorphism $h : B_1 \to B_2$ that makes the following diagram commute:

$$\begin{tikzcd} & B_1 \arrow[dd, "h"] & \\ X \arrow[ur, "e_1"] \arrow[dr, "e_2"'] & & A \arrow[ul, "h_1"'] \arrow[dl, "h_2"] \\ & B_2 & \end{tikzcd}$$

Definition 100 (Free extension)

A *free extension* of A by X is an extension for which there exists a unique morphism of extensions to every other extension of A by X.

1.11 Distributive combination of theories

Definition 101 (Coproduct of signatures)

Given two signatures Σ_1 and Σ_2, we define their coproduct $\Sigma_1 + \Sigma_2$ as the signature with operators the disjoint union of the operators, but retaining their arity:

$$\Sigma_1 + \Sigma_2 = \{\iota_i f : n \mid (f : n) \in \Sigma_i, i = 1, 2\}$$

Given a Σ_i-term-in-context $n \vdash_{\Sigma_i} t$, we denote by $n \vdash_{\Sigma_1+\Sigma_2} \iota_i[t]$ the $\Sigma_1+\Sigma_2$-term-in-context obtained by applying the injection ι_i to all the operators of t.

Similarly, given a Σ_i-equation in context $eq := (n \vdash_{\Sigma_i} t = t')$, we denote by $\iota_i[eq]$ the $\Sigma_1 + \Sigma_2$-equation in context $n \vdash_{\Sigma_1+\Sigma_2} \iota_i[t] = \iota_i[t']$.

Definition 102 (Distributive combination of theories)

Let $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ be a translation from an operad $\mathcal{O}$ to a theory $\mathcal{M}$ and $\mathcal{A}$ be any theory.

We define the *distributive combination of* $\mathfrak{T}$ *over* $\mathcal{A}$ to be the theory $\mathfrak{T} \lhd \mathcal{A}$ given by

$$\begin{aligned}
\Sigma_{\mathfrak{T}\lhd A} &:= \Sigma_M + \Sigma_A \\
\mathrm{Ax}_{\mathfrak{T}\lhd A} &:= \iota_1[\mathrm{Ax}_{\mathcal{M}}] \cup \iota_2[\mathrm{Ax}_{\mathcal{A}}] \\
&\cup \left\{ \begin{array}{c} \langle x_i \rangle_{\substack{i=1 \\ i\neq i_0}}^{n}, \mathbf{y} \mid \mathfrak{T}f(x_1,\dots,x_{i_0-1}, s\mathbf{y}, x_{i_0+1},\dots,x_n) \\ = s\,\langle \mathfrak{T}f(x_1,\dots,x_{i_0-1},\mathbf{y}_j,x_{i_0+1},\dots,x_n)\rangle_{j=1}^{|\mathbf{y}|} \end{array} \middle| \begin{array}{c} (f:n)\in O_n \\ (s:|\mathbf{y}|)\in\Sigma_{\mathcal{A}} \end{array} \right\}
\end{aligned}$$

2 Constructing free algebras

2.1 Semigroups

2.1.1 Free commutative semigroups

Let's try to construct the free algebra for the presentation of commutative semigroups. We cannot rely on the order of the terms, nor the order of applying the operations.

Let X be a set. We consider the theory of commutative semigroups **CSG**, with a signature Σ with a single operator $(+)$ of arity 2, and with the two axioms

$$a, b \vdash a + b = b + a$$
$$a, b, c \vdash a + (b + c) = (a + b) + c$$

We pose

$$SX = \{f : X \to \mathbb{N} \mid f \text{ has finite non-empty support}\}$$

For $f, g \in SX$, we pose

$$f + g = \lambda x.f\ x + g\ x \in SX$$

We define $\eta_{\mathbf{CSG}} : X \to SX$ for all $x \in X$ by

$$\eta_{\mathbf{CSG}}(x) = \lambda y. \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases}$$

We pose $F_{\mathbf{CSG}}X = (SX, (+))$ where $+$ is the operator defined above.

Proposition 103

$F_{\mathbf{CSG}}X$ is a free **CSG**-model over X.

Proof

- ▷ Let's first check that $F_{\mathbf{CSG}}X$ is indeed a **CSG**-model. We need to check that the axioms of **CSG** are valid in $F_{\mathbf{CSG}}X$. This is straightforward and comes down to the commutativity and associativity of addition in $\mathbb{N}$.
- ▷ Let A be a **CSG**-model over X with environment $\varphi_A : X \to U\,A$. We want to show that there exists a unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A such that

the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{CSG}}"] \arrow[dr, "\varphi_A"'] & SX \arrow[d, "h"] \\ & UA \end{tikzcd}$$

with U being the forgetful functor from **CSG**-models to **Set**.

We define $\Sigma\varphi_A : SX \to \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f\ x \cdot \varphi_A(x)$$

We want to show that $\Sigma\varphi_A$ is the unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operator $+$, and that the diagram above commutes.

- Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f+g) &= \Sigma\varphi_A(\lambda x. f\ x + g\ x) \\ &= \sum_{x\in X} (f\ x + g\ x) \cdot \varphi_A(x) \\ &= \sum_{x\in X} f\ x \cdot \varphi_A(x) + \sum_{x \in X} g\ x \cdot \varphi_A(x) \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{CSG}}(x)) &= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases}\right) \\ &= \sum_{y \in X} \begin{cases} 1 \cdot \varphi_A(x) & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \\ &= \varphi_A(x) \end{aligned}$$

  Therefore, the diagram commutes.

- Uniqueness: let $h : F_{\mathbf{CSG}}X \to A$ be a morphism of **CSG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We have:

$$\begin{aligned} h(f) &= h\left(\sum_{x\in X} f\ x \cdot \eta_{\mathbf{CSG}}(x)\right) \\ &= \sum_{x\in X} f\ x \cdot h(\eta_{\mathbf{CSG}}(x)) \\ &= \sum_{x\in X} f\ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(f) \end{aligned}$$

$\boxed{\text{Therefore, we proved that } F_{\mathbf{CSG}}X \text{ is a free } \mathbf{CSG}\text{-model over } X.}$

□

Finally we'll define a multiplication $\mu_{\mathbf{CSG}}$ so that we have a monad structure $(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ on **Set**. For all $f \in SSX$, we pose

$$\mu_{\mathbf{CSG}}(f) = \lambda x. \sum_{g \in SX} f\ g \cdot g\ x$$

Proposition 104

$(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ is a monad on **Set**.

Proof

We need to check the monad axioms:

- Left unit: let $f \in SX$. We have:

$$\begin{aligned}
\mu_{\mathbf{CSG}}(\eta_{\mathbf{CSG}} S(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} 1 \text{ if } g = f \\ 0 \text{ otherwise} \end{cases}\right) \\
&= \lambda x. \sum_{g \in SX} \begin{cases} 1 \cdot g\ x & \text{if } g = f \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x. f\ x \\
&= f
\end{aligned}$$

- Right unit: let $f \in SX$. We have:

$$\begin{aligned}
\mu_{\mathbf{CSG}}(S\eta_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} f\ x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases}\right) \\
&= \lambda x. \sum_{g \in SX} \begin{cases} f\ x \cdot g\ x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x. f(x) \cdot \eta_{\mathbf{CSG}}(x)(x) \\
&= \lambda x. f\ x \\
&= f
\end{aligned}$$

- Associativity: let $f \in SSSX$. We have:

$$\begin{aligned}
\mu_{\mathbf{CSG}}(S\mu_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \sum_{h \in SSX} f\ h \cdot h\ g\right) \\
&= \lambda x. \sum_{g \in SX} \left(\sum_{h \in SSX} f\ h \cdot h\ g\right) \cdot g\ x \\
&= \lambda x. \sum_{h \in SSX} f\ h \cdot \left(\sum_{g \in SX} h\ g \cdot g\ x\right) \\
&= \lambda x. \sum_{h \in SSX} f\ h \cdot \mu_{\mathbf{CSG}}(h) \\
&= \mu_{\mathbf{CSG}}(\mu_{\mathbf{CSG}} S(f))
\end{aligned}$$

□

2.1.2 Free semigroups

Compared to the free commutative semigroup, we can rely on the order of the terms. We'll use lists to represent the terms of the free semigroup.

Let X be a set. We consider the theory of semigroups **SG**, with a signature Σ with a single operator $(\cdot)$ of arity 2, and with the axiom

$$a, b, c \vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

We pose
$$TX = \{l \in X^* \mid l \text{ is finite and non-empty}\}$$
For $f, g \in TX$, we pose
$$f \cdot g = f +\!\!+\, g$$
where $+\!\!+$ is the concatenation of lists.

We define $\eta_{\mathbf{SG}} : X \to TX$ for all $x \in X$ by
$$\eta_{\mathbf{SG}}(x) = [x]$$

We pose $F_{\mathbf{SG}}X = (TX, (\cdot))$ where $\cdot$ is the operator defined above.

Proposition 105

$F_{\mathbf{SG}}X$ is a free **SG**-model over X.

Proof

- ▷ $F_{\mathbf{SG}}$ is a **SG**-model: the associativity of $\cdot$ in $F_{\mathbf{SG}}$ comes from the associativity of list concatenation.

- ▷ Let A be a **SG**-model over X with environment $\varphi_A : X \to U\,A$.

  We define $\Sigma\varphi_A : TX \to \underline{A}$ for all $l \in TX$ by
  $$\Sigma\varphi_A(l) = \prod_{x \in l} \varphi_A(x)$$
  where the product[a] is taken in the order of the list l, which is well defined because the list is finite, non-empty, and the product is associative.

  Let's show that $\Sigma\varphi_A$ is the unique morphism h of **SG**-models from $F_{\mathbf{SG}}X$ to A such that the following diagram commutes:
  $$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}}"] \arrow[dr, "\varphi_A"'] & TX \arrow[d, "h"] \\ & U\,A \end{tikzcd}$$

  - Preservation of the operator $\cdot$: let $f, g \in TX$. We have:
  $$\begin{aligned} \Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(f +\!\!+\, g) \\ &= \prod_{x \in f +\!\!+ g} \varphi_A(x) \\ &= \prod_{x \in f} \varphi_A(x) \cdot \prod_{x \in g} \varphi_A(x) && \text{by associativity of the product} \\ &= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g) \end{aligned}$$

  - Commutation of the diagram: let $x \in X$. We have:
  $$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{SG}}(x)) &= \Sigma\varphi_A([x]) \\ &= \varphi_A(x) \end{aligned}$$

  - Uniqueness: let $h : F_{\mathbf{SG}}X \to A$ be a morphism of **SG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $l \in TX$. We have:

$$\begin{aligned} h(l) &= h\left(\prod_{x\in l} \eta_{\mathbf{SG}}(x)\right) \\ &= \prod_{x\in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\ &= \prod_{x\in l} \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(l) \end{aligned}$$

Therefore we proved that $F_{\mathbf{SG}}X$ is a free **SG**-model over X.

□

[a] product with the operator ·

We'll also define a multiplication $\mu_{\mathbf{SG}}$ so that we have a monad structure $(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ on **Set**. For all $l \in TTX$, we pose

$$\mu_{\mathbf{SG}}(l) = +\!\!+_{x\in l} x$$

where the concatenation is taken in the order of the list l, which is well defined because the list l is finite, non-empty, and the concatenation is associative.

Proposition 106

$(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ is a monad on **Set**.

Proof

We need to check the monad axioms:

- Left unit: let $l \in TX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(\eta_{\mathbf{SG}}T(l)) &= \mu_{\mathbf{SG}}([l]) \\ &= +\!\!+_{x\in[l]} x \\ &= l \end{aligned}$$

- Right unit: let $l \in TX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(T\eta_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}\left([[x_i] \mid x_i \in l]\right) \\ &= +\!\!+_{x\in[[x_i]|x_i\in l]} x \\ &= +\!\!+_{x_i\in l}[x_i] \\ &= l \end{aligned}$$

- Associativity: let $l \in TTTX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(T\mu_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}\left([+\!\!+_{x\in l_i} x \mid l_i \in l]\right) \\ &= +\!\!+_{x\in[+\!\!+_{x\in l_i} x|l_i\in l]} x \\ &= +\!\!+_{l_i\in l} +\!\!+_{x\in l_i} x \\ &= +\!\!+_{x\in +\!\!+_{l_j\in l} l_j} x \\ &= +\!\!+_{x\in l} x \\ &= \mu_{\mathbf{SG}}(\mu_{\mathbf{SG}}T(l)) \end{aligned}$$

□

2.2 Combining semigroups and commutative semigroups

We will call $\mathbf{SG} \triangleleft \mathbf{CSG}$ the distributive combination of $\mathbf{SG}$ and $\mathbf{CSG}$.

The signature of $\mathbf{SG} \triangleleft \mathbf{CSG}$ has two operators: $(\cdot)$ of arity 2, and $(+)$ of arity 2, and the axioms are the union of the axioms of $\mathbf{SG}$ and $\mathbf{CSG}$:

$$\begin{aligned} a, b, c &\vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c \\ a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We also have the distributivity axioms:

$$\begin{aligned} a, b, c &\vdash a \cdot (b + c) = a \cdot b + a \cdot c && \text{left distributivity} \\ a, b, c &\vdash (a + b) \cdot c = a \cdot c + b \cdot c && \text{right distributivity} \end{aligned}$$

We'll try to combine the two previous examples to hopefully get the free algebra for the presentation.

The carrier set of our candidate for the free $\mathbf{SG} \triangleleft \mathbf{CSG}$-model over X is STX.

We will keep the same definition for the operator $+$ as in the free commutative semigroup, but we'll need to change the definition of the operator $\cdot$ to take into account the fact that we are now working with STX instead of TX.
If we write $(p : n) \in f$ when $f(p) = n$, then what we want is

$$f \cdot g = \sum_{(p:n) \in f} \sum_{(q:m) \in g} (p \cdot_{\mathbf{SG}} q : n \cdot m)$$

Formaly, this can be defined for all $f, g \in STX$ as

$$f \cdot g = \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2)$$

We define $\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}} : X \to STX = \eta_{\mathbf{CSG}} \circ \eta_{\mathbf{SG}}$. Let's first prove that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}}"] \arrow[dr, "\eta_{\mathbf{CSG}}"'] & TX \arrow[r, "\eta_{\mathbf{CSG}} T"] & STX \\ & SX \arrow[ur, "S\eta_{\mathbf{SG}}"'] & \end{tikzcd}$$

Let $x \in X$. We have:

$$\begin{aligned} (S\eta_{\mathbf{SG}})(\eta_{\mathbf{CSG}}(x)) &= S\eta_{\mathbf{SG}} \left(\lambda y. \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases} \right) \\ &= \lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\ &= \eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x)) \end{aligned}$$

We pose $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X = (STX, (\cdot), (+))$, where $(\cdot)$ and $(+)$ are defined above.

Proposition 107

$F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$ is a free $\mathbf{SG} \triangleleft \mathbf{CSG}$-model over X.

Proof

- ▷ Let's first check that $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ is indeed a $\mathbf{SG}\lhd\mathbf{CSG}$-model.

  The associativity and commutativity of $(+)$ was already proven for $F_{\mathbf{CSG}}X$.

  For the associativity of $(\cdot)$: let $f,g,h \in STX$. We have:

$$\begin{aligned}
(f\cdot g)\cdot h &= \lambda l : TX.\ \sum_{l=l_1\cdot_{\mathbf{SG}}l_2} (f\cdot g)(l_1)\cdot h(l_2)\\
&= \lambda l : TX.\ \sum_{l=l_1\cdot_{\mathbf{SG}}l_2}\ \sum_{l_1=l_{11}\cdot_{\mathbf{SG}}l_{12}} f(l_{11})\cdot g(l_{12})\cdot h(l_2)\\
&= \lambda l : TX.\ \sum_{l=l_{11}\cdot_{\mathbf{SG}}(l_{12}\cdot_{\mathbf{SG}}l_2)} f(l_{11})\cdot g(l_{12})\cdot h(l_2) \qquad \text{by associativity of } \cdot_{\mathbf{SG}}\\
&= \lambda l : TX.\ \sum_{l=l_{11}\cdot_{\mathbf{SG}}l_3}\ \sum_{l_3=l_{12}\cdot_{\mathbf{SG}}l_2} f(l_{11})\cdot g(l_{12})\cdot h(l_2)\\
&= \lambda l : TX.\ \sum_{l=l_{11}\cdot_{\mathbf{SG}}l_3} f(l_{11})\cdot (g\cdot h)(l_3)\\
&= f\cdot(g\cdot h)
\end{aligned}$$

  The distributivity of $(\cdot)$ over $(+)$ comes from the distributivity of the product in $\mathbb{N}$ over addition, and from the definition of $(+)$.

  Therefore, $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ is a $\mathbf{SG}\lhd\mathbf{CSG}$-model.

- ▷ Let A be a $\mathbf{SG}\lhd\mathbf{CSG}$-model over X with environment $\varphi_A : X \to UA$. We want to show that there exists a unique morphism h of $\mathbf{SG}\lhd\mathbf{CSG}$-models from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}\lhd\mathbf{CSG}}"] \arrow[dr, "\varphi_A"'] & STX \arrow[d, "h"] \\ & UA \end{tikzcd}$$

  We define $\Sigma\varphi_A : STX \to \underline{A}$ for all $f \in STX$ by

$$\Sigma\varphi_A(f) = \sum_{l\in TX} f\ l \cdot \prod_{x\in l}\varphi_A(x)$$

  Let's show that $\Sigma\varphi_A$ is the unique morphism h of $\mathbf{SG}\lhd\mathbf{CSG}$-models from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operators $(\cdot)$ and $(+)$, and that the diagram above commutes.

  - Preservation of the operator $(\cdot)$: let $f,g \in STX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f\cdot g) &= \Sigma\varphi_A(\lambda l : TX.\ \sum_{l=l_1\cdot_{\mathbf{SG}}l_2} f(l_1)\cdot g(l_2))\\
&= \sum_{l\in TX}\left(\sum_{l=l_1\cdot_{\mathbf{SG}}l_2} f(l_1)\cdot g(l_2)\right)\cdot\prod_{x\in l}\varphi_A(x)\\
&= \sum_{l_1,l_2\in TX} f(l_1)\cdot g(l_2)\cdot \prod_{x\in l_1\cdot_{\mathbf{SG}}l_2}\varphi_A(x)\\
&= \sum_{l_1,l_2\in TX} f(l_1)\cdot g(l_2)\cdot \prod_{x\in l_1}\varphi_A(x)\cdot\prod_{x\in l_2}\varphi_A(x) \qquad \text{by associativity of the product}\\
&= \sum_{l_1\in TX} f(l_1)\cdot\prod_{x\in l_1}\varphi_A(x)\cdot\sum_{l_2\in TX} g(l_2)\cdot\prod_{x\in l_2}\varphi_A(x)\\
&= \Sigma\varphi_A(f)\cdot\Sigma\varphi_A(g)
\end{aligned}$$

- Preservation of the operator $(+)$: let $f, g \in STX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f+g) &= \Sigma\varphi_A(\lambda l : TX. f\ l + g\ l)\\
&= \sum_{l \in TX} (f\ l + g\ l) \cdot \prod_{x \in l} \varphi_A(x)\\
&= \sum_{l \in TX} f\ l \cdot \prod_{x \in l} \varphi_A(x) + \sum_{l \in TX} g\ l \cdot \prod_{x \in l} \varphi_A(x)\\
&= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)
\end{aligned}$$

- Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned}
(\Sigma\varphi_A)(\eta_{\mathbf{SG} \lhd \mathbf{CSG}}(x)) &= \Sigma\varphi_A(\eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x)))\\
&= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases}\right)\\
&= \sum_{y \in TX} \begin{cases} 1 \cdot \prod_{x' \in y} \varphi_A(x') & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases}\\
&= \prod_{x' \in \eta_{\mathbf{SG}}(x)} \varphi_A(x')\\
&= \varphi_A(x)
\end{aligned}$$

- Uniqueness: let $h : F_{\mathbf{SG} \lhd \mathbf{CSG}} X \to A$ be a morphism of $\mathbf{SG} \lhd \mathbf{CSG}$-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in STX$. We have:

$$\begin{aligned}
h(f) &= h\left(\sum_{l \in TX} f\ l \cdot \prod_{x \in l} \eta_{\mathbf{SG}}(x)\right)\\
&= \sum_{l \in TX} f\ l \cdot \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism}\\
&= \sum_{l \in TX} f\ l \cdot \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes}\\
&= \Sigma\varphi_A(f)
\end{aligned}$$

$\boxed{\text{Therefore } F_{\mathbf{SG} \lhd \mathbf{CSG}} X \text{ is a free } \mathbf{SG} \lhd \mathbf{CSG}\text{-model over } X.}$

□

We also define a distributive law $\lambda : TSX \to STX$ for all $l \in TSX$ by

$$\lambda([f_1, \ldots, f_n]) = \sum_{(x_1 : n_1) \in f_1} \cdots \sum_{(x_n : n_n) \in f_n} \left([x_1, \cdots, x_n] : \prod_{i=1}^{n} n_i\right)$$

Formaly, this can be defined for all $l \in TSX$ as

$$\lambda(l) = \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f) \in x * l} f(y) & \text{otherwise} \end{cases}$$

where $x * l$ is the elementwise pairing of the list x and the list l, which is well defined because x and l have the same length.

Proposition 108

λ is a distributive law of the monad T over the monad S.

Proof

We need to check the following axioms:

- $\lambda \circ T\eta_S = \eta_S T$: let $l \in TX$. We have:

$$\begin{aligned}
(\lambda \circ T\eta_S)(l) &= \lambda(T\eta_S(l)) \\
&= \lambda\left([\eta_S(x) \mid x \in l]\right) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f) \in x * [\eta_S(x) | x \in l]} f(y) & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,z) \in x * l} \eta_S(z)(y) & \text{otherwise} \end{cases} \\
&= \eta_S T(l)
\end{aligned}$$

- $\lambda \circ \eta_T S = S\eta_T$: let $f \in SX$. We have:

$$\begin{aligned}
(\lambda \circ \eta_T S)(f) &= \lambda(\eta_T S(f)) \\
&= \lambda([f]) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq 1 \\ \prod_{(y,g) \in x * [f]} g(y) & \text{otherwise} \end{cases} \\
&= S\eta_T(f)
\end{aligned}$$

- $\lambda \circ T\mu_S = \mu_S T \circ S\lambda \circ \lambda S$: let $l = [f_1, \ldots, f_n] \in TSSX$. We have:

$$\begin{aligned}
(\lambda \circ T\mu_S)(l) &= \lambda(T\mu_S(l)) \\
&= \lambda\left([\mu_S(f_i) \mid i = 1, \ldots, n]\right) \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \mu_S(f_i)(x_i) & x = [x_1, \ldots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \sum_{g \in SX} f_i(g) \cdot g(x_i) & x = [x_1, \ldots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \sum_{g_1, \ldots, g_n \in SX} \prod_{i=1}^{n} f_i(g_i) \cdot g_i(x_i) & x = [x_1, \ldots, x_m] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$
\begin{aligned}
(\mu_S T \circ S\lambda \circ \lambda S)(l) &= \mu_S T(S\lambda(\lambda S(l))) \\
&= \mu_S T\left(S\lambda \left(\lambda y : TSX. \begin{cases} \prod_{i=1}^{n} \sum_{g \in SX} f_i(g) \cdot g(y_i) & y = [y_1, \dots, y_n] \\ 0 & \text{otherwise} \end{cases} \right)\right) \\
&= \mu_S T\left(\lambda h : STX. \sum_{g_1, \dots, g_n \in SX \wedge \lambda([g_1, \dots, g_n]) = h} \prod_{i=1}^{n} f_i(g_i) \cdot g_i(x_i) \right) \\
&= \lambda x : TX. \sum_{g_1, \dots, g_n \in SX} \left(\prod_{i=1}^{n} f_i(g_i) \right) \cdot \lambda([g_1, \dots, g_n])(x) \\
&= \lambda x : TX. \begin{cases} \sum_{g_1, \dots, g_n \in SX} \left(\prod_{i=1}^{n} f_i(g_i) \right) \cdot \left(\prod_{i=1}^{n} g_i(x_i) \right) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

Therefore, we have $(\lambda \circ T\mu_S)(l) = (\mu_S T \circ S\lambda \circ \lambda S)(l)$.

- $\lambda \circ \mu_T S = S\mu_T \circ \lambda T \circ T\lambda$: let $l = [[f_{11}, \dots, f_{1m_1}], \dots, [f_{n1}, \dots, f_{nm_n}]] \in TTSX$. We have:

$$
\begin{aligned}
(\lambda \circ \mu_T S)(l) &= \lambda(\mu_T S(l)) \\
&= \lambda(+\!\!+_{i=1}^{n} [f_{i1}, \dots, f_{im_i}]) \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = +\!\!+_{i=1}^{n} [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

$$
\begin{aligned}
&(S\mu_T \circ \lambda T \circ T\lambda)(l) \\
&= S\mu_T(\lambda T(T\lambda(l))) \\
&= S\mu_T(\lambda T([\lambda([f_{11}, \dots, f_{1m_1}]), \dots, \lambda([f_{n1}, \dots, f_{nm_n}])])) \\
&= S\mu_T\left(\lambda T\left(\left[\lambda y : TX. \begin{cases} \prod_{j=1}^{m_1} f_{1j}(y_j) & y = [y_1, \dots, y_{m_1}] \\ 0 & \text{otherwise} \end{cases}, \dots, \lambda y : TX. \begin{cases} \prod_{j=1}^{m_n} f_{nj}(y_j) & y = [y_1, \dots, y_{m_n}] \\ 0 & \text{otherwise} \end{cases} \right] \right)\right) \\
&= S\mu_T\left(\lambda h : STX. \sum_{i=1}^{n} \sum_{y_1, \dots, y_{m_i} \in TX} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & h = \lambda y : TX. \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y = [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\ 0 & \text{otherwise} \end{cases} \right) \\
&= \lambda x : TX. \begin{cases} \sum_{i=1}^{n} \sum_{y_1, \dots, y_{m_i} \in TX} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y_i = [y_{i1}, \dots, y_{im_i}] \\ 0 & \text{otherwise} \end{cases} & x = +\!\!+_{i=1}^{n} [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = +\!\!+_{i=1}^{n} [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

Therefore, we have $(\lambda \circ \mu_T S)(l) = (S\mu_T \circ \lambda T \circ T\lambda)(l)$.

□

Corollary 109

$(ST, \eta_{\mathbf{SG}\lhd\mathbf{CSG}}, \mu_{\mathbf{SG}\lhd\mathbf{CSG}})$ is a monad on **Set**, where

$$\mu_{\mathbf{SG}\lhd\mathbf{CSG}} : STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu_S\mu_T} ST$$

Another approach to get a monad structure on ST is to use monads in extension form.

[MW10] shows[1] that we can define an **SG**-algebra $(STX, (-)^\lambda)$ where for all $f : X \to STY$, we have $f^\lambda : TX \to STY$ defined by $f^\lambda([x_1, \dots, x_n]) = \lambda([f(x_1), \dots, f(x_n)])$. It is then showed that this **SG**-algebra gives a distributive law of the monad T over the monad S, and therefore a monad structure on ST.

In extensive form, the monad is $(ST, \eta_{\mathbf{SG}\lhd\mathbf{CSG}}, (-)^{(ST)})$ where for all $f : X \to STY$, we have $f^{(ST)} : STX \to STY$ defined by $f^{(ST)} = (f^\lambda)^T$.
We also have that $\lambda = (S\eta_T)^\lambda$ as expected.

We now may want to see whether the monad we got from the distributive law is the same as the one we get from the free $\mathbf{SG} \lhd \mathbf{CSG}$-model construction.
Informally, we want to check whether the following diagram commutes (also see Definition 102):

$$\begin{tikzcd}
\mathbf{SG}, \mathbf{CSG} \arrow[r, "\text{combining theory}"] \arrow[d, "\text{adding monad structure}"'] \arrow[dr, phantom, "\overset{?}{=}"] & \mathbf{SG} \lhd \mathbf{CSG} \arrow[d, "\text{adding monad structure}"] \\
\mathbf{Mon}(\mathbf{SG}), \mathbf{Mon}(\mathbf{CSG}) \arrow[r, "\text{distributive law}"'] & \mathbf{Mon}(\mathbf{SG} \lhd \mathbf{CSG})
\end{tikzcd}$$

The classical monadic structure on $\mathbf{SG} \lhd \mathbf{CSG}$ comes from the adjunction between **Set** and $\mathbf{SG} \lhd \mathbf{CSG}$-models given by the free $\mathbf{SG} \lhd \mathbf{CSG}$-model construction.

Lemma 110

$F_{\mathbf{SG}\lhd\mathbf{CSG}} \dashv U$ is an adjunction, where $U : \mathbf{SG} \lhd \mathbf{CSG}\text{-}\mathbf{Mod} \to \mathbf{Set}$ is the forgetful functor.

Proof

We need to show that for all $X \in \mathbf{Set}$ and all $\mathbf{SG} \lhd \mathbf{CSG}$-model A, there is a natural bijection between $\mathbf{SG} \lhd \mathbf{CSG}$-model morphisms from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A and functions from X to $U\,A$.

Let $X \in \mathbf{Set}$ and let $A = (\underline{A}, \cdot_A, +_A)$ be a $\mathbf{SG} \lhd \mathbf{CSG}$-model. We have already shown that there is a unique morphism of $\mathbf{SG} \lhd \mathbf{CSG}$-models from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A such that the diagram below commutes:

$$\begin{tikzcd}
X \arrow[r, "\eta_{\mathbf{SG}\lhd\mathbf{CSG}}"] \arrow[dr, "\varphi_A"'] & STX \arrow[d, "\Sigma\varphi_A"] \\
& \underline{A}
\end{tikzcd}$$

Therefore, we have a function $\Phi_{X,A}$ from $\mathbf{SG} \lhd \mathbf{CSG}$-model morphisms from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A to functions from X to $U\,A$ defined by $\Phi_{X,A}(h) = h \circ \eta_{\mathbf{SG}\lhd\mathbf{CSG}}$.

Conversely, we have a function $\Psi_{X,A}$ from functions from X to $U\,A$ to $\mathbf{SG} \lhd \mathbf{CSG}$-model morphisms from $F_{\mathbf{SG}\lhd\mathbf{CSG}}X$ to A defined by $\Psi_{X,A}(\varphi_A) = \Sigma\varphi_A$.

We need to show that $\Phi_{X,A}$ and $\Psi_{X,A}$ are inverses of each other.

[1] example 8.5

Let $h : F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X \to A$ be a morphism of $\mathbf{SG} \triangleleft \mathbf{CSG}$-models. We have:

$$\begin{aligned}(\Psi_{X,A} \circ \Phi_{X,A})(h) &= \Psi_{X,A}(h \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}) \\ &= \Sigma(h \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}) \\ &= h\end{aligned}$$

Let $\varphi_A : X \to U\,A$ be a function. We have:

$$\begin{aligned}(\Phi_{X,A} \circ \Psi_{X,A})(\varphi_A) &= \Phi_{X,A}(\Sigma\varphi_A) \\ &= \Sigma\varphi_A \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}} \\ &= \varphi_A\end{aligned}$$

Therefore, $\Phi_{X,A}$ and $\Psi_{X,A}$ are inverses of each other, and we have a natural bijection between $\mathbf{SG} \triangleleft \mathbf{CSG}$-model morphisms from $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$ to A and functions from X to $U\,A$. □

This adjunction gives us a monad structure on ST.

Theorem 111
The monad structure on ST given by the adjunction $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} \dashv U$ is the same as the one given by the distributive law λ.

Proof
We'll consider the extensive form of the monads.

▷ **Carrier sets.**

Let $X \in \mathbf{Set}$. $UF_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$ is the carrier set of the free $\mathbf{SG} \triangleleft \mathbf{CSG}$-model over X, which is STX. The carrier set of the monad given by the distributive law is also STX. Therefore, the carrier sets are the same.

▷ **Unit.**

The unit is the same for both monads, sending $x \in X$ to the sum of the singleton list $[x]$ in STX.

▷ **Extension.**

Let $f : X \to STY$. Since $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} Y$ is a model of $\mathbf{SG} \triangleleft \mathbf{CSG}$, freeness of $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$ gives us a unique morphism of $\mathbf{SG} \triangleleft \mathbf{CSG}$-models from $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$ to $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} Y$ such that the diagram below commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}"] \arrow[dr, "f"'] & STX \arrow[d, "\Sigma f"] \\ & STY \end{tikzcd}$$

Therefore, the extension of f in the monad given by the adjunction is Σf. The extension of f in the monad given by the distributive law is $f^{(ST)} = (f^\lambda)^T$. We need to show that $\Sigma f = (f^\lambda)^T$.

Let $f : X \to STY$. Suppose that

$$f(x_k) = \sum_j u_{k,j}$$

for some $u_{k,j} \in TY$. Then

$$\Sigma f([x_1, \ldots, x_n]) = \sum_{j_1,\ldots,j_n} u_{1,j_1} \cdots u_{n,j_n}$$

On the other hand, we have:

$$\begin{aligned} f^\lambda([x_1, \ldots, x_n]) &= \lambda([f(x_1), \ldots, f(x_n)]) \\ &= \lambda([\sum_{j_1} u_{1,j_1}, \ldots, \sum_{j_n} u_{n,j_n}]) \\ &= \sum_{j_1,\ldots,j_n} u_{1,j_1} \cdots u_{n,j_n} \end{aligned}$$

Therefore, we have $f^\lambda([x_1, \ldots, x_n]) = \Sigma f([x_1, \ldots, x_n])$ for all $[x_1, \ldots, x_n] \in TX$.

We can apply $(-)^T$ to both sides of the equation $f^\lambda = \Sigma f$ to get $(f^\lambda)^T = (\Sigma f)^T$. We have $(\Sigma f)^T = \Sigma f$ because Σf is a morphism of $\mathbf{SG} \triangleleft \mathbf{CSG}$-models, and therefore preserves the operators of $\mathbf{SG} \triangleleft \mathbf{CSG}$. Therefore, we have $f^{(ST)} = (f^\lambda)^T = \Sigma f$.

This shows that the monad structure on ST given by the adjunction $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} \dashv U$ is the same as the one given by the distributive law λ. □

2.3 Monoids

Now that we have the free semigroup, we can easily construct the free monoid by adding a new constant (arity 0 operator) to the signature, and adding the corresponding axioms.

2.3.1 Free commutative monoids

Let X be a set. We consider the theory of commutative monoids **CM**, with a signature Σ with two operators: $(+)$ of arity 2, and 0 of arity 0, and with the three axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \\ a &\vdash a + 0 = a = 0 + a \end{aligned}$$

We pose[2]

$$SX = \{f : X \to \mathbb{N} \mid f \text{ has finite support}\}$$

There is no non-empty support condition as in the free commutative semigroup, which means that the function f can be the constant 0 function. This corresponds to the new constant 0 in the signature.

We pose

$$\mathbf{0} = \lambda x.0 \in SX$$

.

We keep the same definition for the operator + and for $\eta_{\mathbf{CM}}$ as in the free commutative semigroup.

We pose $F_{\mathbf{CM}}X = (SX, (+), \mathbf{0})$ where + and $\eta_{\mathbf{CM}}$ are defined as above.
We want to show that $F_{\mathbf{CM}}X$ is a free **CM**-model over X.

[2]we may want to find a different name, as it may be confused with the one for the commutative semigroup

▷ $F_{\mathbf{CM}}X$ is a **CM**-model: the axioms of **CM** are valid in $F_{\mathbf{CM}}X$ because of the commutativity and associativity of addition in $\mathbb{N}$, and because of the existence of the constant function $\mathbf{0}$ which acts as the identity for $+$.

▷ Let $A = (\underline{A}, +_A, 0_A)$ be a **CM**-model over X with environment $\varphi_A : X \to U\,A$. We define $\Sigma\varphi_A : SX \to \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f\ x \cdot \varphi_A(x)$$

We have $\Sigma\varphi_A(\mathbf{0}) = 0_A$.

For the sake of doing the proof at least once, we'll prove that $\Sigma\varphi_A$ is well defined on $SX \setminus \{\mathbf{0}\}$.

Proposition 112

For all $f \in SX$, the sum $\sum_{x \in X} f\ x \cdot \varphi_A(x)$ is well defined.

Proof

Let $f \in SX \setminus \{\mathbf{0}\}$. We note $\operatorname{Supp} f = \{x \in X \mid f(x) \neq 0\}$.

Since f has finite support, $\operatorname{Supp} f$ is a finite set. Moreover, since $f \neq \mathbf{0}$, $\operatorname{Supp} f$ is non-empty. Therefore, we can write $\operatorname{Supp} f = \{x_1, x_2, \dots, x_n\}$ for some $n \geq 1$ and we have a bijection σ between $\{1, 2, \dots, n\}$ and $\operatorname{Supp} f$ given by $i \mapsto x_i$.

Therefore, we can rewrite the sum as follows:

$$\begin{aligned}\sum_{x \in X} f\ x \cdot \varphi_A(x) &= \sum_{x \in \operatorname{Supp} f} f\ x \cdot \varphi_A(x) \\ &= \sum_{i=1}^{n} f\ \sigma(i) \cdot \varphi_A(\sigma(i))\end{aligned}$$

The definition is valid because $+_A$ is associative.

Let σ' be another bijection between $\{1, 2, \dots, n\}$ and $\operatorname{Supp} f$. Let's show that we get the same value for the sum if we use σ' instead of σ. This is true because σ and σ' are bijections, so there exists a permutation π of $\{1, 2, \dots, n\}$ such that $\sigma' = \sigma \circ \pi$. Therefore, we have:

$$\begin{aligned}\sum_{i=1}^{n} f\ \sigma'(i) \cdot \varphi_A(\sigma'(i)) &= \sum_{i=1}^{n} f\ \sigma(\pi(i)) \cdot \varphi_A(\sigma(\pi(i))) \\ &= \sum_{i=1}^{n} f\ \sigma(i) \cdot \varphi_A(\sigma(i)) && \text{by commutativity of } +_A\end{aligned}$$

Therefore, the definition of $\Sigma\varphi_A(f)$ does not depend on the choice of the bijection between $\{1, 2, \dots, n\}$ and $\operatorname{Supp} f$, and is well defined.

Now if $f = \mathbf{0}$, we have $\operatorname{Supp} f = \varnothing$, and the sum is the empty sum, which is defined to be 0_A. □

We can now prove that $\Sigma\varphi_A$ is the unique morphism h of **CM**-models from $F_{\mathbf{CM}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{CM}}"] \arrow[dr, "\varphi_A"'] & SX \arrow[d, "h"] \\ & U\,A \end{tikzcd}$$

- Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f+g) &= \Sigma\varphi_A(\lambda x.f\ x + g\ x) \\
&= \sum_{x\in X}(f\ x + g\ x)\cdot\varphi_A(x) && \text{by definition} \\
&= \sum_{x\in X} f\ x\cdot\varphi_A(x) + \sum_{x\in X} g\ x\cdot\varphi_A(x) && \text{since both } f \text{ and } g \text{ have finite support} \\
&= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)
\end{aligned}$$

- Commutation of the diagram: $\eta_{\mathbf{CM}} = \eta_{\mathbf{CSG}}$, so the proof is the same as for the free commutative semigroup.
- Uniqueness: let $h : F_{\mathbf{CM}}X \to A$ be a morphism of **CM**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We'll show this lemma first:

Lemma 113
For all $f \in SX$, we have $f = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.

Proof
Let $f \in SX$. We have $\operatorname{Supp} f = \{x \in X \mid f(x) \neq 0\}$. Since f has finite support, $\operatorname{Supp} f$ is a finite set. Moreover, since f is not necessarily non-zero, $\operatorname{Supp} f$ can be empty.
If $\operatorname{Supp} f$ is empty, then $f = \mathbf{0}$ and we have:

$$\sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x) = \sum_{x\in X} 0\cdot\eta_{\mathbf{CM}}(x) = 0 = f$$

If $\operatorname{Supp} f$ is non-empty, we can write $\operatorname{Supp} f = \{x_1, x_2, \ldots, x_n\}$ for some $n \geq 1$. We'll prove by induction on n that $f = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.

* If $n = 1$, then $\operatorname{Supp} f = \{x_1\}$ and we have $f = f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.
* If $n > 1$, we have $\operatorname{Supp} f = \{x_1, x_2, \ldots, x_n\}$.
  We can write $f = f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + f'$ where $f' = \lambda x.\begin{cases} 0 & \text{if } x = x_1 \\ f\ x & \text{otherwise}\end{cases}$
  We have $\operatorname{Supp} f' = \{x_2, \ldots, x_n\}$, so by the induction hypothesis, we have $f' = \sum_{x\in X} f'\ x\cdot\eta_{\mathbf{CM}}(x)$. Therefore, we have:

$$\begin{aligned}
f &= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + f' \\
&= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + \sum_{x\in X} f'\ x\cdot\eta_{\mathbf{CM}}(x) \\
&= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x) - f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) && \text{by definition of } f' \\
&= \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)
\end{aligned}$$

□

We then have:

$$\begin{aligned}
h(f) &= h\left(\sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)\right) \\
&= \sum_{x\in X} f\ x\cdot h(\eta_{\mathbf{CM}}(x)) \\
&= \sum_{x\in X} f\ x\cdot\varphi_A(x) && \text{since the diagram commutes} \\
&= \Sigma\varphi_A(f)
\end{aligned}$$

Therefore we proved that $F_{\mathbf{CM}}X$ is a free **CM**-model over X.

2.4 Distributive combination of semigroups and commutative semigroups

Let's first start with a case of study. We'll take $\mathcal{A}$ to be commutative semigroups and $\mathcal{M}$ to be semigroups. We denote by $(+)$ the operator on $\mathcal{A}$, and $(\cdot)$ the operator on $\mathcal{M}$.

We define $\mathcal{O}$ by $\mathcal{O}_0 = \varnothing$ and $\mathcal{O}_n = \{x_1, \ldots, x_n \vdash x_1 \cdots x_n\}$ for $n \geqslant 1$. We will simply write $\mathcal{O}_n = \{\mu_n\}$.

Proposition 114

$\mathcal{O}$ is the smallest operad containing[a] $(\cdot)$.

[a] we may say that $\mathcal{O}$ is the operad spanned by $(\cdot)$

Proof

▷ **$\mathcal{O}$ is an operad.**

associativity comes from the associativity of $(\cdot)$ and the neutrality of the identity is trivial.

▷ **minimality condition.**

Let $\mathcal{O}'$ be an operad containing $(\cdot)$. We want to show that $\mathcal{O}$ is a sub-operad of $\mathcal{O}'$. We have $\mu_2 \in \mathcal{O}'_2$ since $\mathcal{O}'$ contains $(\cdot)$. We can then construct μ_n for all $n \geqslant 1$ by composition of μ_2 with itself, and using the identity for the case $n = 1$. Therefore, we have $\mu_n \in \mathcal{O}'_n$ for all $n \geqslant 1$, which means that $\mathcal{O}$ is a sub-operad of $\mathcal{O}'$.

□

We then define a translation $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ by interpreting each operator as the n-ary multiplication, i.e $\mathfrak{T}\mu_n(x_1, \ldots, x_n) = x_1 \cdots x_n$.

Proposition 115

$\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ is a surjective translation.

Proof

▷ **$\mathfrak{T}$ is a translation.**

- the identity condition is trivialy true.
- the translation respects composition thanks to the associativity of $(\cdot)$.

▷ **$\mathfrak{T}$ is surjective.**

Let $t \in \mathcal{M}_n$. Since $\mathcal{M}$ is the algebraic theory of semigroups, t is represented by a nonempty word in the variables $x_1, \ldots, x_n$. Thus we may write

$$t = x_{i_1} x_{i_2} \cdots x_{i_m}$$

for some $m \geqslant 1$ and indices $i_j \in \{1, \ldots, n\}$.

Consider the unique m-ary operation $\mu_m \in \mathcal{O}_m$. By definition of the translation,

$$\mathfrak{T}(\mu_m) = x_1 x_2 \cdots x_m.$$

We define a renaming $\rho : m \to n$ by

$$\rho(j) = i_j.$$

Applying this renaming to $\mathfrak{T}(\mu_m)$ gives

$$\mathfrak{T}(\mu_m)[\rho] = (x_1 x_2 \cdots x_m)[\rho] = x_{\rho(1)} x_{\rho(2)} \cdots x_{\rho(m)} = x_{i_1} x_{i_2} \cdots x_{i_m} = t.$$

Therefore every term $t \in \mathcal{M}_n$ is of the form $t = \mathfrak{T}\mu_m[\rho]$ for some renaming $\rho : m \to n$. Hence the translation $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ is surjective.

□

Let's give an explicit description of the theory $\mathfrak{T} \lhd \mathcal{A}$.

- **Signature.** The signature $\Sigma_{\mathfrak{T} \lhd \mathcal{A}}$ contains the operators of $\mathcal{M}$ and the operators of $\mathcal{A}$. Therefore, it contains the operator $(\cdot)$ of arity 2 and the operator $(+)$ of arity 2.
- **Axioms.**

$$\begin{gathered}
\mathcal{M} \text{ axioms} \\
x, y, z \vdash x \cdot (y \cdot z) = (x \cdot y) \cdot z \\
\mathcal{A} \text{ axioms} \\
x, y \vdash x + y = y + x \\
x, y, z \vdash x + (y + z) = (x + y) + z \\
\text{distributivity axioms} \\
x, y, z \vdash x \cdot (y + z) = x \cdot y + x \cdot z \\
x, y, z \vdash (x + y) \cdot z = x \cdot z + y \cdot z
\end{gathered}$$

We define the multi-functor $\underline{-} : \mathbf{Multi}(\mathcal{T}, \mathcal{U}) \to \langle \mathcal{U}, \prod \rangle$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathbf{Multi}(\mathcal{T},\mathcal{U}) \arrow[r, "\underline{-}"] \arrow[d, "-_o"'] & \langle \mathcal{U}, \prod \rangle \arrow[d, "-_o"] \\
\mathbf{Mod}(\mathcal{T},\mathcal{U}) \arrow[r, "U"'] & \mathcal{U}
\end{tikzcd}$$

This is the anaology in **Multicat** of the forgetful functor in **Cat**, i.e the forgetful multi-functor.

Proposition 116

Let $\mathcal{T}$ be a theory. $\underline{-} : \mathbf{Multi}(\mathcal{T}, \mathbf{Set}) \to \langle \mathbf{Set}, \prod \rangle$ has a left adjoint if and only if $\mathcal{T}$ is commutative.

Proof

See section 6 of Ohad's notes. □

Now we'll try to give an explicite description of each element of the following pullback square:

$$\begin{tikzcd}
\mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set}) \arrow[r, "\underline{-}_{\mathcal{O} \lhd \mathcal{A}}"] \arrow[d, "\underline{-}_{\mathcal{M}}"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \arrow[d, "\mathbf{Mod}(\mathcal{O}, \underline{-})"] \\
\mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \arrow[r, "\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})"'] & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)
\end{tikzcd}$$

For vertices:

- $\mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$ is the category of $\mathfrak{T} \lhd \mathcal{A}$-models in $\mathbf{Set}$ (i.e with carrier sets in $\mathbf{Set}$), i.e the category of sem-rgs.
- $\mathbf{Mod}(\mathcal{M}, \mathbf{Set})$ is the category of $\mathcal{M}$-models in $\mathbf{Set}$, i.e the category of semigroups.
- $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$ is the category of $\mathcal{O}$-models in $\langle \mathbf{Set}, \prod \rangle$. This category coincides with the category of semigroups in $\mathbf{Set}$ (cf. example 9 of Ohad's notes).
- $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ is the category of $\mathcal{O}$-models in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$.

  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ is the multicategory where:

  - objects are $\mathcal{A}$-models in $\mathbf{Set}$, i.e commutative semigroups.
  - multi-morphisms are multi-$\mathcal{A}$-homomorphisms.

  Lemma 117
  Multi-$\mathcal{A}$-homomorphisms are multi-linear.

  Proof
  Let $f : \prod_{i=1}^n \underline{\mathcal{A}_i} \to \underline{\mathcal{B}}$ be a multi-$\mathcal{A}$-homomorphism. Consider the term $x_1, \ldots, x_n \vdash t := \sum_{i=1}^n x_i$. The multi-$\mathcal{A}$-homomorphic property of f implies that

$$\begin{tikzcd}
\langle v_1, \ldots, v_{i-1}, \langle u_1, \ldots, u_n \rangle, v_{i+1}, \ldots, v_n \rangle \arrow[r, mapsto, "\mathrm{str}"] \arrow[dddd, mapsto, "\mathcal{A}_i[\![t]\!] \times \mathrm{id}"'] & \langle \langle v_1, \ldots, v_{i-1}, u_j, v_{i+1}, \ldots, v_n \rangle \rangle_{j=1}^n \arrow[r, mapsto, "\cong"] & \langle \langle v_1, \ldots, v_{i-1}, u_j, v_{i+1}, \ldots, v_n \rangle \rangle_{j=1}^n \arrow[d, mapsto, "f^n"] \\
 & & \langle f(v_1, \ldots, v_{i-1}, u_j, v_{i+1}, \ldots, v_n) \rangle_{j=1}^n \arrow[d, mapsto, "\mathcal{B}[\![t]\!]"] \\
 & & \sum_{j=1}^n f(v_1, \ldots, v_{i-1}, u_j, v_{i+1}, \ldots, v_n) \arrow[d, equal] \\
 & & \\
\langle v_1, \ldots, v_{i-1}, \sum_{j=1}^n u_j, v_{i+1}, \ldots, v_n \rangle \arrow[r, mapsto, "\cong"'] & \langle v_1, \ldots, v_{i-1}, \sum_{j=1}^n u_j, v_{i+1}, \ldots, v_n \rangle \arrow[r, mapsto, "f"'] & f\left(v_1, \ldots, v_{i-1}, \sum_{j=1}^n u_j, v_{i+1} \ldots, v_n\right)
\end{tikzcd}$$

  and therefore f is multi-linear. □

  Similarly to the previous case, $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ coincides with the category of sem-rgs in $\mathbf{Set}$. The the distributivity axioms hold in $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ thanks to the multi-linearity of the multi-morphisms in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$.

For arrows:

- $\underline{-}_{\mathcal{O} \lhd \mathcal{A}}$ is such that for $A = (UA, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$, $\underline{A}_{\mathcal{O} \lhd \mathcal{A}}$ consists of:
  - $\underline{A} = (UA, (+))$ the $\mathcal{A}$-model given by the interpretation of the operators of $\mathcal{A}$ in A.
  - $\underline{A}_{\mathcal{O} \lhd \mathcal{A}} [\![\mu_n]\!] (x_1, \ldots, x_n) = x_1 \cdot \cdots \cdot x_n \in \underline{A}$
- $\underline{-}_{\mathcal{M} \lhd}$ is the functor sending a $\mathfrak{T} \lhd \mathcal{A}$-model M to the $\mathcal{M}$-model in $\mathbf{Set}$ given by the interpretation of the operators of $\mathcal{M}$ in M, i.e the semigroup structure of M.

  For $A = (\underline{A}, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$, we have $\underline{A}_{\mathcal{M} \lhd} = (\underline{A}, (\cdot))$. It is easy to check that this is indeed a $\mathcal{M}$-model in $\mathbf{Set}$ because the axioms of $\mathcal{M}$ are included in the axioms of $\mathfrak{T} \lhd \mathcal{A}$.
- $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$ is defined in Proposition 80. We have

$$\left(\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle) A\right) [\![f]\!] := A [\![\mathfrak{T} f]\!] \qquad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle) h := h$$

- $\mathbf{Mod}(\mathcal{O}, \underline{-})$ is defined in Proposition 81. It forgets about the model structure for $\mathcal{A}$, keeping only its carrier set, therefore we are left with a $\mathcal{O}$-model in $\mathbf{Set}$.

Remark : $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$ is therefore an isomorphism of categories.

Proposition 118

The square above is a pullback square.

Proof

Informaly, this is true because a model of $\mathfrak{T} \lhd \mathcal{A}$ consists of a model of $\mathcal{M}$ and a model of $\mathcal{A}$ such that the operations of $\mathcal{O}$ distribute over the operations of $\mathcal{A}$. Conversely, we can reconstruct a model for the combination with both those models.

We'll give a constructive proof anyway.

▷ **commutativity.**

Let $A \in \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$.

We clearly have $\underline{\mathbf{Mod}(\mathfrak{T},\mathbf{Set})(\underline{A}_{\mathcal{M}\lhd})} = \underline{\mathbf{Mod}(\mathcal{O},\underline{\ })(\underline{A}_{\mathcal{O}\lhd\mathcal{A}})}$, since both sides are equal to $\underline{A}$.

Moreover, for $f \in \mathcal{O}_n$, we have:

$$\begin{aligned}
\mathbf{Mod}(\mathfrak{T},\mathbf{Set})(\underline{A}_{\mathcal{M}\lhd})\,[\![f]\!] &= \underline{A}_{\mathcal{M}\lhd}\,[\![\mathfrak{T}f]\!] && \text{by definition of } \mathbf{Mod}(\mathfrak{T},\mathbf{Set}) \\
&= \underline{A}_{\mathcal{M}\lhd}\,[\![\mu_n]\!] && \text{since } \mathcal{O} \text{ is spanned by } \mu_n \\
&= \underline{A}_{\mathcal{O}\lhd\mathcal{A}}\,[\![\mu_n]\!] && \text{by definition of } \underline{\ }_{\mathcal{O}\lhd\mathcal{A}} \\
&= \mathbf{Mod}(\mathcal{O},\underline{\ })(\underline{A}_{\mathcal{O}\lhd\mathcal{A}})\,[\![f]\!] && \text{by definition of } \mathbf{Mod}(\mathcal{O},\underline{\ })
\end{aligned}$$

▷ **pullback.**

Let $\mathcal{C}$ be a category, $F : \mathcal{C} \to \mathbf{Mod}(\mathcal{M},\mathbf{Set})$ and $G : \mathcal{C} \to \mathbf{Mod}(\mathcal{O},\mathbf{Multi}(\mathcal{A},\mathbf{Set}))$ be two functors such that $\mathbf{Mod}(\mathfrak{T},\mathbf{Set}) \circ F = \mathbf{Mod}(\mathcal{O},\underline{\ }) \circ G$. We want to show that there exists a unique functor $H : \mathcal{C} \to \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C} \arrow[dr, dashed, "H"] \arrow[drr, bend left, "G"] \arrow[ddr, bend right, "F"'] & & \\
& \mathbf{Mod}(\mathfrak{T}\lhd\mathcal{A},\mathbf{Set}) \arrow[r, "\underline{\ }_{\mathcal{O}\lhd\mathcal{A}}"] \arrow[d, "\underline{\ }_{\mathcal{M}}"'] & \mathbf{Mod}(\mathcal{O},\mathbf{Multi}(\mathcal{A},\mathbf{Set})) \arrow[d, "\mathbf{Mod}(\mathcal{O},\underline{\ })"] \\
& \mathbf{Mod}(\mathcal{M},\mathbf{Set}) \arrow[r, "\mathbf{Mod}(\mathfrak{T},\mathbf{Set})"'] & \mathbf{Mod}(\mathcal{O},\langle\mathbf{Set},\prod\rangle)
\end{tikzcd}$$

We define $H : \mathcal{C} \to \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$ as follows:

- for $c \in \mathcal{C}$, we have $Hc = (UFc, (+_c), (\cdot_c))$ where:
  - UFc is the underlying set of the $\mathcal{M}$-model Fc.
  - $(+_c)$ is the binary operation on UFc given by the interpretation of the operator $(+)$ of $\mathcal{A}$ in Gc, i.e $(+_c) = Gc[\![+]\!]$.
  - $(\cdot_c)$ is the binary operation on UFc given by the interpretation of the operator $(\cdot)$ of $\mathcal{M}$ in Fc, i.e $(\cdot_c) = Fc[\![\cdot]\!]$.

  We then have to check that Hc is indeed a $\mathfrak{T} \lhd \mathcal{A}$-model, i.e that the axioms of $\mathfrak{T} \lhd \mathcal{A}$ hold in Hc. This is true because:

  - the axioms of $\mathcal{M}$ hold in Hc since Fc is a $\mathcal{M}$-model and the interpretation of the operators of $\mathcal{M}$ in Hc is the same as the interpretation of the operators of $\mathcal{M}$ in Fc.

  * the axioms of $\mathcal{A}$ hold in Hc since Gc is a $\mathcal{O}$-model in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ and the interpretation of the operators of $\mathcal{A}$ in Hc is the same as the interpretation of the operators of $\mathcal{A}$ in Gc.
  * the distributivity axioms hold in Hc since Gc is a $\mathcal{O}$-model in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ and the interpretation of the operators of $\mathcal{M}$ in Hc is the same as the interpretation of the operators of $\mathcal{M}$ in Fc, and the interpretation of the operators of $\mathcal{A}$ in Hc is the same as the interpretation of the operators of $\mathcal{A}$ in Gc, and the distributivity axioms hold in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$.

- for $m : c \to c'$ in $\mathcal{C}$, define Hm to be the underlying function of Fm and Gm (these agree because the two composites to the lower-right corner agree).

  This is indeed a $\mathfrak{T} \lhd \mathcal{A}$-homomorphism:

  * it is an $\mathcal{M}$-homomorphism because Fm is
  * it is an $\mathcal{A}$-homomorphism because Gm is

□

We can construct another pullback using the generic construction of Example 25. We can deduce an isomorphism

$$\mathbf{Mod}(\mathfrak{T} \lhd \mathcal{T}, \mathbf{Set}) \cong \langle \mathbf{Mod}(\mathcal{M}, \mathbf{Set}), \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \rangle$$

As in [HP06], we can obtain an universal characterisation of $\mathfrak{T} \lhd \mathcal{A}$ as a pushout of theories:

$$\begin{tikzcd}
\mathfrak{T} \lhd \mathcal{A} & \mathcal{O} * \mathcal{A} \arrow[l] \\
\mathcal{M} \arrow[u] \arrow[ur, phantom, "\lrcorner" very near end] & F\mathcal{O} \arrow[l] \arrow[u]
\end{tikzcd}$$

2.5 Free algebras for the combination

We continue with the example of the previous section and we now try to construct the free algebra for $\mathfrak{T} \lhd \mathcal{A}$ using the free algebras for $\mathcal{M}$ and $\mathcal{A}$, similarly to Section 2.2, but in a more abstract way. We fix a set X.

We'll use the results of Section 2.1, and we write $F_{\mathcal{M}} = F_{\mathbf{SG}}$ and $F_{\mathcal{A}} = F_{\mathbf{CSG}}$.

Suppose that for each $X \in \mathbf{Set}$ we know how to construct the free algebra $F_{\mathcal{A}}X \in \mathbf{Mod}(\mathcal{T}, \mathbf{Set})$. We can extend this to a multi-functor $F_{\mathcal{A}} : \langle \mathbf{Set}, \prod \rangle \to \mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ the following way[3]:

- The definition is the same on objects: $F_{\mathcal{A}}X$
- Let $f : X_1 \times \cdots \times X_n \to Y$ be a multi-morphism on $\langle \mathbf{Set}, \prod \rangle$.

  We denote by η the unit $\eta_X : X \to \underline{F_{\mathcal{A}}X}$.

  We recall that the free algebra construction gives us an homomorphism

  $$\rangle\!\rangle\!\!\models g : F_{\mathcal{A}}X \to A$$

  for every function $g : X \to \underline{A}$.

  We define a bijection $\psi : \langle \mathbf{Set}, \prod \rangle\, (\mathbf{X}; \underline{A}) \xrightarrow{\cong} \mathbf{Multi}(\mathcal{A}, \mathbf{Set})(F_{\mathcal{A}}[\mathbf{X}]; A)$ such that

  $$\underline{\psi(f)} \circ (\eta_{X_1}, \ldots, \eta_{X_n}) = f$$

  by[4]:

[3]this is the translation of proof 24/28 of Ohad's notes
[4]This is the currified version of ψ.

- $\psi_{X_1,\dots,X_n}(f)(a_1,\dots,a_n) := \gg\!\!= \Big(x_1 \mapsto \psi_{X_2,\dots,X_n}\big((x_2,\dots,x_n) \mapsto f(x_1,x_2,\dots,x_n)\big)(a_2,\dots,a_n)\Big)(a_1)$
- for x a constant, $\psi_\varnothing(x)() := x$

Then, we define

$$F_\mathcal{A} f := \psi(\eta_Y \circ (f)) : (F_\mathcal{A} X_1,\dots,F_\mathcal{A} X_n) \to F_\mathcal{A} Y$$

Indeed, we have

$$(X_1,\dots,X_n) \xrightarrow{f} Y \xrightarrow{\eta_Y} \underline{F_\mathcal{A} Y}$$

so

$$\psi(\eta_Y \circ f)(F_\mathcal{A} X_1,\dots,F_\mathcal{A} X_n) \longrightarrow F_\mathcal{A} Y$$

Proposition 23 of Ohad's notes show that this is indeed a multi-morphism in $\mathbf{Multi}(\mathcal{A},\mathbf{Set})$.

This multi-functor is the left-adjoint of the forgetful multi-functor

$$\underline{} : \mathbf{Multi}(\mathcal{A},\mathbf{Set}) \to \left\langle \mathbf{Set}, \prod \right\rangle$$

Step 1.
Let $A_0 := F_\mathcal{M} X \vDash \mathcal{M}$ be the free $\mathcal{M}$-algebra over X.

$$\underline{A_0} = TX \qquad A_0 [\![(\cdot)]\!] = (+\!\!+) \qquad \eta^{A_0}(x) = [x]$$

Step 2.
We apply the functor

$$\underline{}_\mathfrak{T} := \mathbf{Mod}(\mathfrak{T},\mathbf{Set}) : \mathbf{Mod}(\mathcal{M},\mathbf{Set}) \to \mathbf{Mod}(\mathcal{O}, \left\langle \mathbf{Set}, \prod \right\rangle)$$

to A_0. This gives $A_1 := \underline{A_0}_\mathfrak{T} \vDash \mathcal{O}$.

$$\underline{A_1} = \underline{A_0} = TX \qquad A_1 [\![\mu_n]\!] (x_1,\dots,x_n) = x_1 +\!\!+ \cdots +\!\!+ x_n \qquad \eta^{A_1}_{A_0} = \mathrm{id}$$

Step 3.
By Proposition 116, we have an adjunction

$$F_\mathcal{A} \dashv \underline{} : \mathbf{Multi}(\mathcal{A},\mathbf{Set}) \to \left\langle \mathbf{Set}, \prod \right\rangle$$

in **Multicat** since $\mathcal{A}$ is commutative. Applying the 2-functor $\mathbf{Mod}(\mathcal{O},-) : \mathbf{Multicat} \to \mathbf{Cat}$ gives a left adjoint

$$F_{\mathcal{O}\to\mathcal{A}} := \mathbf{Mod}(\mathcal{O},F_\mathcal{A}) : \mathbf{Mod}(\mathcal{O}, \left\langle \mathbf{Set}, \prod \right\rangle) \to \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A},\mathbf{Set}))$$

We pose $A_2 := F_{\mathcal{O}\to\mathcal{A}} A_1 \vDash \mathcal{O} * \mathcal{A}$.

$$\underline{A_2} = STX \qquad \eta^{A_2}_{A_1}(x) = \sum_{w \in TX} \delta^w_x \cdot w$$

$F_\mathcal{A}$ extends the morphisms in $\langle \mathbf{Set}, \prod \rangle$ multi-linearly. Therefore if $a_i = \sum_{w\in TX} n^{(i)}_w \cdot w \in STX$, then

$$\begin{aligned} A_2 [\![\mu_n]\!] (a_1,\dots,a_n) &= \sum_{w_1,\dots,w_n \in TX} (n^{(1)}_{w_1} \cdots n^{(n)}_{w_n}) \cdot (A_1 [\![\mu_n]\!] (w_1,\dots,w_n)) \\ &= \sum_{w_1,\dots,w_n \in TX} (n^{(1)}_{w_1} \cdots n^{(n)}_{w_n}) \cdot (w_1 +\!\!+ \cdots +\!\!+ w_n) \end{aligned}$$

Lemma 33 of Ohad's notes then is just saying that any $x \in STX$ can be written as

$$x = \sum_{i=1}^n w_i$$

for some $(w_i) \in (TX)^n$, which is no surprise.

Step 4.
Let $A_3 := \mathbf{Mod}(\mathcal{O}, \underline{\ })A_2 \vDash O$ be the underlying operadic algebra.

$$\underline{A_3} = \underline{A_2} = STX \qquad A_3 [\![\mu_n]\!] = A_2 [\![\mu_n]\!] \qquad \eta_{A_2}^{A_3} = \mathrm{id}$$

For each $t \in \mathcal{M}_n$, we use surjectivity to get some $f \in \mathcal{O}_m$ and $\theta : m \to n$ such that $t = \mathfrak{T} f[\theta]$. Then $A_{\mathcal{M}} [\![t]\!] e := A_3 [\![f]\!] e \circ \theta$.

Lemma 119 (Lemma 34 of Ohad's notes, false)

With the preceding notation:

- the interpretation $A_{\mathcal{M}} [\![t]\!] e$ is independent of the decomposition $t = \mathfrak{T} f[\theta]$
- the resulting algebra validates $\mathcal{M}$, i.e $A_{\mathcal{M}} \vDash \mathcal{M}$

Even though the lemma is false, we can still define $A_{\mathcal{M}}$ that way and we do indeed get an $\mathcal{M}$-model, thanks to Lemma 122.
At this point we managed to go from a model of $\mathcal{M}$ with carrier set TX to a model of $\mathcal{M}$ with carrier set STX.

$$\underline{A_{\mathcal{M}}} = \underline{A_3} = \underline{A_2} = STX \qquad A_{\mathcal{M}} [\![m \vdash x_{i_1} \cdots x_{i_n}]\!] e = A_3 [\![\mu_n]\!] (e \circ (j \mapsto i_j)) \qquad \eta_{A_2}^{A_{\mathcal{M}}} = \mathrm{id}$$

Step 5.
We have $A_{\mathcal{M}} \vDash \mathcal{M}$ and $A_2 \vDash \mathcal{O} * \mathcal{A}$. We can then use the pullback square to get a model $A := A_{\mathcal{M}} \lhd A_2$ of $\mathfrak{T} \lhd \mathcal{A}$:

- $\underline{A} = \underline{A_{\mathcal{M}}} = \underline{A_2} = STX$
- $A [\![(\cdot)]\!] (x, y) = A_{\mathcal{M}} [\![(\cdot)]\!] (x, y) = A_2 [\![\mu_2]\!] (x, y)$
- $A [\![(+)]\!] (x, y) = A_2 [\![(+)]\!] (x, y)$

All axioms hold since they hold either in $A_{\mathcal{M}}$ or in A_2.

We define $\eta : X \xrightarrow{\eta^{A_0}} \underline{A_0} = \underline{A_1} \xrightarrow{\eta_{A_1}^{A_2}} \underline{A_2} = STX = \underline{A}$.

Proposition 120

A is the free $\mathfrak{T} \lhd \mathcal{A}$-algebra over X.

Proof

We need to show that (A, η) is an universal arrow from X to the forgetful functor $\mathcal{A} \to \mathbf{Set}$. Since the definitions for $(+)$ and $(\cdot)$ in A are the same as we defined in Section 2.2, the same proof as in Section 107 works here. $\square$

Here is a little diagram that summarizes the steps:

$$\begin{tikzcd}
\text{carrier} = TX & A_0 := F_{\mathcal{M}} X \vDash \mathcal{M} \arrow[r, mapsto, "-_{\mathfrak{T}}"'] & A_1 \vDash \mathcal{O} \arrow[rd, mapsto, bend left, "\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}})"] & \\
\text{carrier} = STX & A_{\mathcal{M}} \vDash \mathcal{M} & A_3 \vDash \mathcal{O} \arrow[l, mapsto, "\text{surjectivity}"'] & A_2 \vDash \mathcal{O} * \mathcal{A} \arrow[l, mapsto, "\mathbf{Mod}(\mathcal{O}, \underline{\ })"'] \\
& & A \vDash \mathfrak{T} \lhd \mathcal{A} \arrow[ul, "\text{combination}"] \arrow[ur] &
\end{tikzcd}$$

2.6 Counter example of Lemma 119

working point

Let $\mathcal{M} = \mathbf{BA}$ be the theory of barycentric algebras, $\mathcal{A} = \mathbf{SL}$ be the theory of join-semilattices, $\mathcal{O}$ be the simplex operad and $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ be the usual translation sending a weight vector to the corresponding convex-combination term.

Formaly:

- **BA** consists of a binary operator $\oplus_p$ for every $p \in [0,1]$ and the axioms

$$\begin{aligned} x &\vdash x \oplus_p x = x && \text{idempotence} \\ x, y &\vdash x \oplus_p y = y \oplus_{1-p} x && \text{skew commutativity} \\ x, y, z &\vdash x \oplus_p (y \oplus_r z) = \left(x \oplus_{\frac{p}{p+(1-p)r}} y\right) \oplus_{p+(1-p)r} z && \text{skew associativity} \end{aligned}$$

- **SL** consists of a binary operator $\vee$ and a constant $\perp$, along with the axioms

$$\begin{aligned} x &\vdash x \vee \perp = x && \text{unit} \\ x, y, z &\vdash (x \vee y) \vee z = x \vee (y \vee z) && \text{associativity} \\ x, y &\vdash x \vee y = y \vee x && \text{commutativity} \\ x &\vdash x \vee x = x && \text{idempotence} \end{aligned}$$

- $\mathcal{O}$ is the operad where

$$\mathcal{O}_n = \left\{ \lambda = (\lambda_1, \dots, \lambda_n) \in [0,1]^n \mid \sum_{i=1}^n \lambda_i = 1 \right\}$$

  Composition is substitution of convex combinations, and the identity is $(1) \in \mathcal{O}_1$.

- $\mathfrak{T} : \mathcal{O} \to \mathbf{BA}$ sends a weighted vector $\lambda \in \mathcal{O}_n$ to the corresponding barycentric term $\sum_i \lambda_i x_i$.

The free semilattice on a set X is $F_{\mathbf{SL}} X = \mathcal{P}_f^+(X)$ the set of non-empty[5] finite subsets of X, with union as join. to prove

The free barycentric algebra on X is $F_{\mathbf{BA}} X = \mathcal{D}_f(X)$ the set of finitely-supported probability distributions on X. to prove

Let $X = \{x, y\}$. Following the pipeline of the previous section, let

$$\begin{aligned} A_0 &= F_{\mathbf{BA}}(X) = \mathcal{D}_f(X) \\ A_1 &= \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) A_0 \\ A_2 &= \mathbf{Mod}(\mathcal{O}, F_{\mathbf{SL}}) A_1 \end{aligned}$$

The carrier set of A_2 is

$$\underline{A_2} = \mathcal{P}_f^+(\mathcal{D}_f(X))$$

the set of non-empty finite subsets of finitely supported probability distributions on X.

For $p \in]0,1[$, let $f_p \in \mathcal{O}_2$ be the binary convex-combination operation of weight p.
The lifted operation on A_2 is

$$A_2 [\![f_p]\!] \langle S, T \rangle = \{ p\mu + (1-p)\nu \mid \mu \in S,\ \nu \in T \}$$

Considering the barycentric axiom $x \oplus_p x = x$, it is of the form

$$1 \vdash_{\mathbf{BA}} \mathfrak{T} f_p[\theta] = \mathfrak{T}\mathrm{id}$$

where

[5] we may actually want to also allow empty sets

- $\theta : 2 \to 1$ is the constant map
- $\mathrm{id} \in \mathcal{O}_1$ is the unary identity operation

Take $e := \{\delta_x,\ \delta_y\} \in \mathcal{P}_f^+(\mathcal{D}_f(X))$.

$$A_2 \llbracket f_p \rrbracket \langle e, e\rangle = \{\delta_x,\ p\delta_x + (1-p)\delta_y,\ p\delta_y + (1-p)\delta_x,\ \delta_y\}$$

whereas

$$A_2 \llbracket \mathrm{id} \rrbracket e = e = \{\delta_x,\ \delta_y\}$$

Since $0 < p < 1$, $p\delta_x + (1-p)\delta_y$ and $p\delta_y + (1-p)\delta_x$ do not belong to e. Therefore

$$A_2 \llbracket f_p \rrbracket \langle e, e\rangle \neq A_2 \llbracket \mathrm{id} \rrbracket e$$

and Lemma 119 does not hold.

It follows from this that the construction presented above does not yield the free algebra for the distributive combination of **BA** over **SL**.

2.7 General case

Steps 1 to 4 of the previous section can be adapted to the general case.

η^{A_0} comes from the adjunction $F_{\mathcal{M}} \dashv U : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \to \mathbf{Set}$ and $\eta^{A_2}_{A_1}$ comes from the adjunction $\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) \dashv \mathbf{Mod}(\mathcal{O}, \_) : \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \to \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$.

We encounter a problem when trying to generalize Lemma 119: it doesn't hold anymore (counter example of Section 2.6). We need to replace it with another weaker lemma which limits which theories we can apply the construction to.

Definition 121 ($\mathfrak{T}$-affine presentation)

Let $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ be a surjective translation.
A presentation of $\mathcal{M}$ over $\mathfrak{T}$ is $\mathfrak{T}$-*affine* if:

- for every operation symbol $g : n$ of the chosen presentation of $\mathcal{M}$, there is a chosen operation $\widetilde{g} \in \mathcal{O}_n$ such that
$$n \vdash_{\mathcal{M}} \mathfrak{T}\widetilde{g} = g(x_1, \ldots, x_n),$$
- the chosen operations $\widetilde{g}$ generate $\mathcal{O}$ under identities and operadic composition,
- every axiom of the presentation is of the form
$$x_1, \ldots, x_n \vdash_{\mathcal{M}} \mathfrak{T} f_1[\theta_1] = \mathfrak{T} f_2[\theta_2],$$
for some $f_1 \in \mathcal{O}_{n_1}, f_2 \in \mathcal{O}_{n_2}$, and injective renamings
$$\theta_1 : [n_1] \hookrightarrow [m] \qquad \theta_2 : [n_2] \hookrightarrow [m]$$

Lemma 122 (Step 4 in the $\mathfrak{T}$-affine case)
With the preceding notation, assume that $\mathcal{M}$ admits a $\mathfrak{T}$-affine presentation. Let

$$A_3 := \mathbf{Mod}(\mathcal{O}, -)(A_2).$$

For each operation symbol $g : n$ of the chosen presentation of $\mathcal{M}$, define

$$A_{\mathcal{M}} [\![g]\!] := A_3 [\![\widetilde{g}]\!].$$

Then:

- for every $f \in \mathcal{O}_n$, the interpretation in $A_{\mathcal{M}}$ of the $\mathcal{M}$-term $\mathfrak{T}f$ is equal to $A_3 [\![f]\!]$;
- $A_{\mathcal{M}} \vDash \mathcal{M}$.

Proof
Let $X := \underline{A_1}$ and let $\eta_X : X \to \underline{F_{\mathcal{A}} X}$ be the unit of the adjunction $F_{\mathcal{A}} \dashv U_{\mathcal{A}}$.
By construction of

$$A_2 = \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}})(A_1),$$

for every $f \in \mathcal{O}_n$, the operation $A_2 [\![f]\!]$ is the unique multi-A-homomorphism extending

$$X^n \xrightarrow{A_1[\![f]\!]} X \xrightarrow{\eta_X} \underline{F_{\mathcal{A}} X}.$$

We first prove that, for every $f \in \mathcal{O}_n$, the interpretation in $A_{\mathcal{M}}$ of the term $\mathfrak{T}f$ is equal to $A_3 [\![f]\!]$. Since the chosen operations $\widetilde{g}$ generate $\mathcal{O}$, we may prove this by induction on the operadic expression of f.

- If $f = \mathrm{id}$, the claim is immediate.
- If $f = \widetilde{g}$ is one of the chosen generators, the claim is exactly the definition of $A_{\mathcal{M}} [\![g]\!]$.
- If $f = h \circ (f_1, \dots, f_k)$, then, using preservation of operadic composition by $\mathfrak{T}$, the induction hypothesis, and the fact that A_3 is an $\mathcal{O}$-model, we get

$$A_{\mathcal{M}} [\![\mathfrak{T}f]\!] = A_{\mathcal{M}} [\![\mathfrak{T}h[\mathfrak{T}f_1, \dots, \mathfrak{T}f_k]]\!] = A_3 [\![h]\!] \circ (A_3 [\![f_1]\!], \dots, A_3 [\![f_k]\!]) = A_3 [\![f]\!].$$

Now let

$$x_1, \dots, x_m \vdash_{\mathcal{M}} \mathfrak{T}f_1[\theta_1] = \mathfrak{T}f_2[\theta_2]$$

be an axiom of the chosen $\mathfrak{T}$-affine presentation. For each $i = 1, 2$, define $\Phi_i : X^m \to \underline{F_A X}$ by

$$\Phi_i(e) := \eta_X\big(A_1 [\![f_i]\!] (e \circ \theta_i)\big).$$

Since $A_0 \vDash \mathcal{M}$, these two maps are equal: $\Phi_1 = \Phi_2$.

For each $i = 1, 2$, define $\overline{\Phi_i} : \underline{F_A X^m} \to \underline{F_A X}$ by

$$\overline{\Phi_i}(e) := A_2 [\![f_i]\!] (e \circ \theta_i).$$

Because θ_i is injective, each coordinate of e is used in at most one argument of $A_2 [\![f_i]\!]$, and therefore $\overline{\Phi_i}$ is a multi-$\mathcal{A}$-homomorphism in each variable. By construction, it extends Φ_i. By uniqueness of multi-$\mathcal{A}$-homomorphic extension from generators, we conclude that

$$\overline{\Phi_1} = \overline{\Phi_2}.$$

Hence, for every environment $e : [m] \to \underline{A_2}$,

$$A_2 [\![f_1]\!] (e \circ \theta_1) = A_2 [\![f_2]\!] (e \circ \theta_2).$$

Forgetting from A_2 to A_3 preserves this equality, so the two sides of the axiom have the same interpretation in $A_{\mathcal{M}}$. Hence all axioms of the chosen presentation hold in $A_{\mathcal{M}}$, and therefore $A_{\mathcal{M}} \vDash \mathcal{M}$. □

Step 5.

We define $A \vDash \mathfrak{T} \lhd \mathcal{A}$ the same way as in the example of Section 2.5, i.e as the pullback of $A_{\mathcal{M}}$ and A_2. Let's prove that it is free.

Let B be any sem-rg algebra and $f : X \to \underline{B}$ be a function.

A_0 is the free $\mathcal{M}$-algebra over X, so we have a unique morphism $h_0 : A_0 \to B_{\mathcal{M}}$ of $\mathcal{M}$-algebras such that $h_0 \circ \eta^{A_0} = f$.

$$\begin{tikzcd} X \arrow[r, "\eta^{A_0}"] \arrow[dr, "f"'] & A_0 \arrow[d, "h_0"] \\ & B_{\mathcal{M}} \end{tikzcd}$$

There is also an homomorphism $h_1 : A_1 \to B_{\mathcal{O}}$ of $\mathcal{O}$-algebras such that $h_1 \circ \eta^{A_1}_{A_0} = h_0$, and $h_1 = h_0$ since $\eta^{A_1}_{A_0} = \mathrm{id}$.

$$\begin{tikzcd} X \arrow[r, "\mathrm{id}\circ\eta^{A_0}"] \arrow[dr, "f"'] & A_1 \arrow[d, "h_1=h_0"] \\ & B_{\mathcal{O}} \end{tikzcd}$$

B has an $\mathcal{A}$-structure that distributes over $\mathcal{O}$, therefore thanks to the adjunction $\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) \dashv \mathbf{Mod}(\mathcal{O}, \underline{\ })$, we have a unique morphism $h_2 : A_2 \to B$ of $\mathcal{O} * \mathcal{A}$-algebras such that $h_2 \circ \eta^{A_2}_{A_1} = h_1$.

$$\begin{tikzcd} A_1 \arrow[r, "\eta^{A_2}_{A_1}"] \arrow[dr, "h_1"'] & A_2 \arrow[d, "h_2"] \\ & B_{\mathcal{O}*\mathcal{A}} \end{tikzcd}$$

Apply the 2-functor $\mathbf{Mod}(\mathcal{O}, \underline{\ })$ to the morphism $h_2 : A_2 \to B$ gives a morphism $h_3 : A_3 \to B_{\mathcal{O}}$ of $\mathcal{O}$-algebras such that $h_3 \circ \eta^{A_3}_{A_2} = h_2$ and $h_3 = h_2$ since $\eta^{A_3}_{A_2} = \mathrm{id}$.

By the surjectivity of $\mathfrak{T}$ and the independance of the decomposition given by Lemma 122, we now have a $\mathcal{M}$-morphism $h_4 = h_3 : A_{\mathcal{M}} \to B_{\mathcal{M}}$ such that $h_4 \circ \eta^{A_{\mathcal{M}}}_{A_3} = h_3$.

Incomplete: rewrite the way h_4 is defined now that Lemma 119 is fixed, and then prove that it is indeed an homomorphism

Proof

Let $t \in \mathcal{M}_n$ and $e : [n] \to \underline{A_{\mathcal{M}}}$. By surjectivity, there is some $f \in \mathcal{O}_m$ and $\theta : m \to n$ such that $t = \mathfrak{T} f[\theta]$. We define h_4 as

$$h_4 (A_{\mathcal{M}} [\![t]\!] e) = h_3 (A_3 [\![f]\!] e \circ \theta)$$

This is well-defined by Lemma 122. We then check that h_4 is a morphism of $\mathcal{M}$-algebras. Since h_3 is a morphism of $\mathcal{O}$-algebras,

$$\begin{aligned} h_3 (A_3 [\![f]\!] e \circ \theta) &= B_{\mathcal{O}} [\![f]\!] (h_3 \circ e \circ \theta, \ldots, h_3 \circ e \circ \theta) \\ &= B_{\mathcal{M}} [\![f]\!] (h_4 \circ e, \ldots, h_4 \circ e) \end{aligned}$$

This proves that $h_4 \left(A_{\mathcal{M}} [\![t]\!] \, e\right) = B_{\mathcal{M}} [\![t]\!] \left(h_4 \circ e\right)$, so h_4 is a morphism of $\mathcal{M}$-algebras[a]. $\square$

[a] I should really work on notations before writing the report as the current ones are really hard to read and don't really make sense sometimes.

In the end we have the following diagram:

$$\begin{tikzcd}
X \arrow[r, "\eta^{A_0}"] \arrow[dr, "f"'] & \underline{A_0} \arrow[r, "\mathrm{id}"] \arrow[d, "h_0"] & \underline{A_1} \arrow[dl, "h_1"'] \arrow[d, "\eta^{A_2}_{A_1}"] \\
 & \underline{B} & \underline{A_2} \arrow[l, "h_2"'] \arrow[d, "\mathrm{id}"] \\
 & \underline{A_4} \arrow[u, "h_4"] & \underline{A_3} \arrow[ul, "h_3"] \arrow[l, "\mathrm{id}"]
\end{tikzcd}$$

We then consider the walking morphism $I : \{0 \xrightarrow{u} 1\}$ and the two functors $F_1 : I \to \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ and $F_2 : I \to \mathbf{Mod}(\mathcal{M}, \mathbf{Set})$ defined by $F_1\, 0 = A_2$, $F_1\, 1 = B_{\mathcal{O} * \mathcal{A}}$, $F_1\, u = h_2$ and $F_2\, 0 = A_{\mathcal{M}}$, $F_2\, 1 = B_{\mathcal{M}}$, $F_2\, u = h_4$. By the construction, we can then use the pullback square to get a unique $H : I \to \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A})$ such that the following diagram commutes:

$$\begin{tikzcd}
I \arrow[dr, dashed, "H"] \arrow[drr, bend left, "F_1"] \arrow[ddr, bend right, "F_2"'] & & \\
 & \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set}) \arrow[r, "__{\mathcal{O} \lhd \mathcal{A}}"] \arrow[d, "__{\mathcal{M}}"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \arrow[d, "\mathbf{Mod}(\mathcal{O}, _)"] \\
 & \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \arrow[r, "\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})"'] & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)
\end{tikzcd}$$

This gives us a unique morphism $h = H\, u : A \to B$ of $\mathfrak{T} \lhd \mathcal{A}$-algebras.

We need to show that h extends f along η. $h_2 \circ \eta = h_4 \circ \eta = f$, therefore $h \circ \eta = f$ since η comes from the pullback square.

This morphism is unique for a given h_2 and h_4, but these are also uniquely determined by h_0 and h_1, which are themselves uniquely determined by f. Therefore h is the unique morphism of $\mathfrak{T} \lhd \mathcal{A}$-algebras such that the following diagram commutes:

$$\begin{tikzcd}
X \arrow[r, "\eta"] \arrow[dr, "f"'] & A \arrow[d, "h"] \\
 & B
\end{tikzcd}$$

$\boxed{\text{This proves that } A \text{ is the free } \mathfrak{T} \lhd \mathcal{A}\text{-algebra over } X.}$

Now that we proved that A is the free $\mathfrak{T} \lhd \mathcal{A}$-algebra over X, we can give an explicit description of the unique morphism $h : A \to B$ of $\mathfrak{T} \lhd \mathcal{A}$-algebras such that $h \circ \eta = f$.

Since we cannot access the internal structure anymore, we have the following lemma:

Lemma 123

For every $e \in \underline{A_2}$, there is some $s \in \mathcal{A}_k$ and $\mathbf{e} \in \underline{A_1}^k$ such that

$$e = A_2 [\![s]\!] \left\langle \eta^{A_2}_{A_1} \mathbf{e}_i \right\rangle_{i=1}^{k}$$

Proof

This is Lemma 33 of Ohad's notes. $\square$

We can actually prove a stronger statement:

Lemma 124
For all $e \in \underline{A_2}$, there exists $s \in \mathcal{A}_k$ such that for all $i \in [k]$, there exists $t_i \in \mathcal{M}_{k_i}$ and $\mathbf{e}_{\langle i,j\rangle} \in X$ such that:

$$e = A_2 \llbracket s \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j\rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^{k}$$

Proof
Similar to the proof of Lemma 123.

You first prove that for every $e \in \underline{A_1}$, there is some $t \in \mathcal{M}_k$ and $\mathbf{e} \in X^k$ such that

$$e = A_1 \llbracket t \rrbracket \left\langle \eta^{A_1} \mathbf{e}_i \right\rangle_{i=1}^{k}$$

You can then substitue this result in Lemma **??** to get the final result. □

Let $e \in \underline{A} = \underline{A_2}$. By Lemma 124, we can write

$$e = A_2 \llbracket s \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j\rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^{k}$$

We can then define $h : X \to \underline{B}$ by

$$h\, e = B \llbracket s \rrbracket \left\langle B \llbracket t_i \rrbracket \left\langle f\, \mathbf{e}_{\langle i,j\rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^{k}$$

Lemma 125
h is well-defined, i.e. it does not depend on the choice of s, t_i and $\mathbf{e}_{\langle i,j\rangle}$.

Proof
Suppose

$$e = A_2 \llbracket s_1 \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{1,k} \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle 1,k,j\rangle} \right\rangle_{j=1}^{k_{1,k}} \right\rangle_{k=1}^{n_1}$$

and

$$e = A_2 \llbracket s_2 \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{2,k} \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle 2,k,j\rangle} \right\rangle_{j=1}^{k_{2,k}} \right\rangle_{k=1}^{n_2}$$

Equalities in A are exactly generated by the equations of $\mathfrak{T} \lhd \mathcal{A}$ because A is the free $\mathfrak{T} \lhd \mathcal{A}$-algebra.
Since B is a model of $\mathfrak{T} \lhd \mathcal{A}$, we have that B validates the same equations, and therefore

$$\begin{aligned} & B \llbracket s_1 \rrbracket \left\langle B \llbracket t_{1,k} \rrbracket \left\langle f\, \mathbf{e}_{\langle 1,k,j\rangle} \right\rangle_{j=1}^{k_{1,k}} \right\rangle_{k=1}^{n_1} \\ = & B \llbracket s_2 \rrbracket \left\langle B \llbracket t_{2,k} \rrbracket \left\langle f\, \mathbf{e}_{\langle 2,k,j\rangle} \right\rangle_{j=1}^{k_{2,k}} \right\rangle_{k=1}^{n_2} \end{aligned}$$

□

We now prove that this is indeed a $\mathfrak{T} \lhd \mathcal{A}$-homomorphism and that it is the unique one such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta"] \arrow[dr, "f"'] & \underline{A} \arrow[d, "h"] \\ & \underline{B} \end{tikzcd}$$

▷ **preservation of $\mathcal{A}$-structure.**

Take any $s \in \mathcal{A}_n$ and $e_1, \ldots, e_n \in \underline{A}$. We can write

$$e_l = A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j\rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l}$$

Taking $p \coloneqq \sum_l k^l$ and $s' \coloneqq s[i \mapsto s^i]_{i=1}^n \in \mathcal{A}_p$, we have:

$$
\begin{aligned}
h\left(A_2 \llbracket s \rrbracket \langle e_l \rangle_{l=1}^n\right) &= h\left(A_2 \llbracket s \rrbracket \left\langle A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \right) \\
&= h\left(A_2 \llbracket s' \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_m \rrbracket \left\langle \eta^{A_1} \mathbf{e}_j^m \right\rangle_{j=1}^{k_m} \right\rangle_{m=1}^p \right) \\
&= B \llbracket s' \rrbracket \left\langle B \llbracket t_m \rrbracket \left\langle f\, \mathbf{e}_j^m \right\rangle_{j=1}^{k_m} \right\rangle_{m=1}^p \\
&= B \llbracket s \rrbracket \left\langle B \llbracket s^l \rrbracket \left\langle B \llbracket t_i^l \rrbracket \left\langle f\, \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \\
&= B \llbracket s \rrbracket \langle h\, e_l \rangle_{l=1}^n
\end{aligned}
$$

▷ **preservation of $\mathcal{M}$-structure.**

Take any $t \in \mathcal{M}_n$ and $e_1, \ldots, e_n \in \underline{A}$. We can write

$$
e_l = A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l}
$$

Since the operations of $\mathcal{M}$ distribute over the operations of $\mathcal{A}$, we have:

$$
\begin{aligned}
h\left(A_2 \llbracket t \rrbracket \langle e_l \rangle_{l=1}^n\right) &= h\left(A_2 \llbracket t \rrbracket \left\langle A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \right) \\
&= h\left(A_2 \llbracket s^1 \rrbracket \left\langle \cdots A_2 \llbracket s^n \rrbracket \left\langle A_2 \llbracket t \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{i_l}^l \rrbracket \left\langle \eta^{A_0} \mathbf{e}_{\langle i_l,m \rangle}^l \right\rangle_{m=1}^{k_{i_l}^l} \right\rangle_{l=1}^n \right\rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \right) \\
&= h\left(A_2 \left\llbracket s^1 \left\langle \cdots s^n \langle x_{i_1}, \ldots, x_{i_n} \rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \right\rrbracket \left\langle A_2 \llbracket t \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \left\llbracket t_{\langle q \rangle_l^{-1}}^l \right\rrbracket \left\langle \eta^{A_0} \mathbf{e}_{\langle \langle q \rangle_l^{-1}, m \rangle}^l \right\rangle_{m=1}^{k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{l=1}^n \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \right) \\
&= h\left(A_2 \left\llbracket s^1 \left\langle \cdots s^n \langle x_{i_1}, \ldots, x_{i_n} \rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \right\rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t'_q \rrbracket \left\langle \eta^{A_0} \mathbf{e}^m \right\rangle_{m=1}^{\sum_l k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \right) \\
&= B \left\llbracket s^1 \left\langle \cdots s^n \langle x_{i_1}, \ldots, x_{i_n} \rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \right\rrbracket \left\langle B \llbracket t'_q \rrbracket \left\langle f\, \mathbf{e}^m \right\rangle_{m=1}^{\sum_l k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
&= B \left\llbracket s^1 \left\langle \cdots s^n \langle x_{i_1}, \ldots, x_{i_n} \rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \right\rrbracket \left\langle B \llbracket t \rrbracket \left\langle B \left\llbracket t_{\langle q \rangle_l^{-1}}^l \right\rrbracket \left\langle f\, \mathbf{e}_{\langle \langle q \rangle_l^{-1}, m \rangle}^l \right\rangle_{m=1}^{k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{l=1}^n \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
&= B \llbracket s^1 \rrbracket \left\langle \cdots B \llbracket s^n \rrbracket \left\langle B \llbracket t \rrbracket \left\langle B \llbracket t_{i_l}^l \rrbracket \left\langle f\, \mathbf{e}_{\langle i_l,m \rangle}^l \right\rangle_{m=1}^{k_{i_l}^l} \right\rangle_{l=1}^n \right\rangle_{i_n=1}^{k^n} \cdots \right\rangle_{i_1=1}^{k^1} \\
&= B \llbracket t \rrbracket \left\langle B \llbracket s^l \rrbracket \left\langle B \llbracket t_i^l \rrbracket \left\langle f\, \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \\
&= B \llbracket t \rrbracket \langle h\, e_l \rangle_{l=1}^n
\end{aligned}
$$

Note to self: never again.

▷ **commutation of the diagram.**

We need to show that for all $x \in X$, we have $h(\eta\, x) = f\, x$. We have $\eta x = \eta_{A_1}^{A_2} \eta^{A_0}\, x$. We notice that

$$\begin{aligned}\eta\, x &= \eta^{A_2}_{A_1}\eta^{A_0}\, x\\ &= \eta^{A_2}_{A_1} A_0\, [\![\mathrm{id}]\!]\, \langle \eta^{A_1}\, x\rangle\\ &= A_2\, [\![\mathrm{id}]\!]\, \langle \eta^{A_2}_{A_1} A_0\, [\![\mathrm{id}]\!]\, \langle \eta^{A_1}\, x\rangle\rangle\end{aligned}$$

Therefore

$$\begin{aligned}h(\eta\, x) &= h\left(A_2\, [\![\mathrm{id}]\!]\, \langle \eta^{A_2}_{A_1} A_0\, [\![\mathrm{id}]\!]\, \langle \eta^{A_1}\, x\rangle\rangle\right)\\ &= B\, [\![\mathrm{id}]\!]\, \langle B\, [\![\mathrm{id}]\!]\, \langle f\, x\rangle\rangle\\ &= f\, x\end{aligned}$$

▷ **uniqueness.**

Let $h' : A \to B$ be another morphism of $\mathfrak{T} \lhd \mathcal{A}$-algebras such that $h' \circ \eta = f$. We need to show that for all $e \in \underline{A}$, we have $h\, e = h'\, e$. We can write

$$e = A_2\, [\![s]\!]\, \left\langle \eta^{A_2}_{A_1} A_0\, [\![t_i]\!]\, \langle \eta^{A_1} \mathbf{e}_{\langle i,j\rangle}\rangle_{j=1}^{k_i}\right\rangle_{i=1}^{k}$$

Therefore

$$\begin{aligned}h\, e &= B\, [\![s]\!]\, \left\langle B\, [\![t_i]\!]\, \langle f\, \mathbf{e}_{\langle i,j\rangle}\rangle_{j=1}^{k_i}\right\rangle_{i=1}^{k} && \text{by definition}\\ &= B\, [\![s]\!]\, \left\langle B\, [\![t_i]\!]\, \langle h'\eta\, \mathbf{e}_{\langle i,j\rangle}\rangle_{j=1}^{k_i}\right\rangle_{i=1}^{k} && \text{since the diagram commutes}\\ &= h'\left(A_2\, [\![s]\!]\, \left\langle \eta^{A_2}_{A_1} A_0\, [\![t_i]\!]\, \langle \eta^{A_1} \mathbf{e}_{\langle i,j\rangle}\rangle_{j=1}^{k_i}\right\rangle_{i=1}^{k}\right) && \text{since } h' \text{ is an homomorphism}\\ &= h'\, e\end{aligned}$$

and $h = h'$.

Therefore h is the unique morphism of $\mathfrak{T} \lhd \mathcal{A}$-algebras such that $h \circ \eta = f$.

2.8 Link with distributive laws

There seem to be a strong link between the existence of a distributive law between the two monads underlying the theories we are trying to combine and the success of the construction we described.

[ZM18] gives a bunch of theorems about the non-existence of distributive laws, and it would be useful to check that our construction indeed does not apply in the cases mentioned.

2.8.1 Plotkin-type no-go

One of the condition for the non-existence of a distributive law between two theories given in the Plotkin-like no-go theorems 3.5 and 3.12 of [ZM18] is the existence of an operator that is idempotent in both theories. Our construction forbids such axioms as the presentation would not be $\mathfrak{T}$-affine for any translation $\mathfrak{T}$.

2.8.2 Too many constants

A theory $\mathcal{A}$ is commutative iff for every n-ary operation f and every m-ary operation g,

$$f\big(g(x_{11},\ldots,x_{1m}),\ldots,g(x_{n1},\ldots,x_{nm})\big) = g\big(f(x_{11},\ldots,x_{n1}),\ldots,f(x_{1m},\ldots,x_{nm})\big) \qquad (1)$$

This equation still holds for constants, therefore if

$$f = c \in \mathcal{A}_0$$
$$g = d \in \mathcal{A}_0$$

then the commutativity of $\mathcal{A}$ states that $c = d$.

Therefore is the no-go theorem 4.7 of [ZM18] applies then there are two distinct constants and the theory can't be commutative, meaning our construction doesn't apply.

2.8.3 Lacking abides

For an operator $*$ of arity 2, abides hold if

$$(a * b) * (c * d) = (a * c) * (b * d) \tag{2}$$

Theorem 4.17 of [ZM18] states that if some conditions apply on the theories $\mathcal{A}$ and $\mathcal{M}$ and that additionaly equation (2) does not hold then there are no distributive law of $\mathcal{M}$ over $\mathcal{A}$.

However if $\mathcal{A}$ is commutative, then (2) holds by taking $f = g = *$ in (1). There are no contradiction here either.

2.8.4 Idempotence and units

Theorem 4.24 of [ZM18] states that if some conditions apply on the theories $\mathcal{A}$ and $\mathcal{M}$ and that moreover $\mathcal{M}$ has a binary term that is idempotent then there are no distributive law of $\mathcal{M}$ over $\mathcal{A}$.

Our construction forbids idempotent terms, therefore there are no contradiction.

3 Constructing free extensions

3.1 Semigroups

3.1.1 Semigroup

Let's construct the free extension of an **SG**-model $A = (\underline{A}, (\cdot_A))$ by a set of variable X.

We define a **SG**-algebra $\mathrm{Frex}_A(X)$ as follows:

- The carrier set of $\mathrm{Frex}_A(X)$ is the set of non-empty alternating lists of elements of A and X^+.
- The operator $(\cdot)$ on $\mathrm{Frex}_A(X)$ is defined by concatenation of lists, and by using the operator $(\cdot_A)$ of A to combine adjacent elements of A in the resulting list and list concatenation to combine adjacent elements of X^+.

This is indeed an **SG**-aglebra, as list concatenation is associative, and the operator $(\cdot_A)$ of A is associative.

We define embed : $A \to \mathrm{Frex}_A(X)$ by $\mathrm{embed}(n) = [n]$ for all $n \in A$. For all $m, n \in A$, we have $\mathrm{embed}(m \cdot_A n) = [m \cdot_A n] = [m] \cdot [n] = \mathrm{embed}(m) \cdot \mathrm{embed}(n)$, so embed is a morphism of **SG**-algebras.

We also define $\mathrm{var} : X \to \mathrm{Frex}_A(X)$ by $\mathrm{var}(x) = [x]$ for all $x \in X$.

Proposition 126

$\langle \mathrm{Frex}_A(X), \mathrm{embed}, \mathrm{var}\rangle$ is the free extension of A by X.

Proof

Suppose $\langle B, f, g\rangle$ is another extension of A by X.
We extend g to X^+ by defining $g(x_1 \cdots x_n) = g(x_1)\cdots g(x_n)$ for all $x_1, \ldots, x_n \in X$.
Now for an element $[b_1, \ldots, b_n]$ of $\mathrm{Frex}_A(X)$, we define $h([b_1, \ldots, b_n]) = [\![b_1]\!] \cdot \cdots \cdot [\![b_n]\!]$ where:

- if $b_i \in A$, then $[\![b_i]\!] = f(b_i)$
- if $b_i \in X^+$, then $[\![b_i]\!] = g(b_i)$

This is well-defined and homomorphic because the only merges in the extension are:

- concrete/concrete, handled by $f(a \cdot_A b) = f(a) \cdot f(b)$
- abstract/abstract, handled by $g(uv) = g(u) \cdot g(v)$

The diagram below commutes:

$$\begin{tikzcd}
A \arrow[r, "\mathrm{embed}"] \arrow[dr, "f"'] & \mathrm{Frex}_A(X) \arrow[d, "h"] & X \arrow[l, "\mathrm{var}"'] \arrow[dl, "g"] \\
& B &
\end{tikzcd}$$

Uniqueness of h is straightforward, as h is defined by the values of f and g on the elements of A and X, and by the homomorphic property of h. $\square$

3.1.2 Commutative semigroup

Let's construct the free extension of an **CSG**-model $A = (\underline{A}, (+_A))$ by a set of variable X.

We pose $A_\bot = \underline{A} \sqcup \{\bot\}$ where $\bot$ is a new element, designating the absence of a concrete part in the polynomial. We extend the operator $+_A$ and $\cdot$ of A to $A_\bot$ by defining $\bot +_A a = a +_A \bot = a$ for all $a \in A$ and $\bot +_A \bot = \bot$.

We define a **CSG**-algebra $\mathrm{Frex}_A(X)$ as follows:

- The carrier set of $\mathrm{Frex}_A(X)$ is the set

  $$\{(c, M) \mid c \in A_\bot,\ M \in \mathrm{Bag}(X),\ c \neq \bot \text{ or } M \neq \varnothing\}$$

  where $\mathrm{Bag}(X)$ are finite-support multisets over X.

- The operator $(+)$ on $\mathrm{Frex}_A(X)$ is defined by

  $$(c, M) + (d, N) = (c + d, M \uplus N)$$

This is indeed an **CSG**-algebra, as union of multisets is associative and commutative, and so is addition in $A_\bot$.

We define $\mathrm{embed} : A \to \mathrm{Frex}_A(X)$ by $\mathrm{embed}(n) = (n, \varnothing)$ for all $n \in A$. For all $m, n \in A$, we have $\mathrm{embed}(m + n) = (m, \varnothing) + (n, \varnothing) = (m + n, \varnothing) = \mathrm{embed}(m) + \mathrm{embed}(n)$, so embed is a morphism of **CSG**-algebras.

We also define $\mathrm{var} : X \to \mathrm{Frex}_A(X)$ by $\mathrm{var}(x) = (\bot, \{\!|x|\!\})$ for all $x \in X$.

Proposition 127

$\langle \mathrm{Frex}_A(X), \mathrm{embed}, \mathrm{var}\rangle$ is the free extension of A by X.

Proof

Suppose $\langle B, f, g\rangle$ is another extension of A by X. We define $h : \mathrm{Frex}_A(X) \to B$ by

$$h((c, M)) = \begin{cases} (f(c), \{g(m) : n \mid (m : n) \in M\}) & \text{if } c \neq \bot \\ \{g(m) : n \mid (m : n) \in M\} & \text{if } c = \bot \end{cases}$$

This is well-defined and homomorphic because the only merges in the extension are:

- concrete/concrete, handled by $f(a + b) = f(a) + f(b)$

The diagram below commutes:

$$\begin{tikzcd} A \arrow[r, "\mathrm{embed}"] \arrow[dr, "f"'] & \mathrm{Frex}_A(X) \arrow[d, "h"] & X \arrow[l, "\mathrm{var}"'] \arrow[dl, "g"] \\ & B & \end{tikzcd}$$

Uniqueness of h is straightforward, as h is defined by the values of f and g on the elements of A and X, and by the homomorphic property of h. □

3.2 Sem-rgs

Let $A = (\underline{A}, (+), (\cdot))$ be a $\mathbf{SG} \lhd \mathbf{CSG}$-model and X be a set.

We define

$$\mathbf{Mon}_A(X) = \{[b_1, \ldots, b_n] \mid n \geqslant 1,\ b_i \in X^+ \sqcup \underline{A},\ b_i b_{i+1} \text{ alternating kinds},\ \exists i, b_i \in X^+\}$$

We define $A_\bot$ as in the previous section, and extend the operations with $\bot \cdot a = a \cdot \bot = \bot$ for all $a \in A$, and $\bot \cdot \bot = \bot$.

We define

$$\mathrm{Frex}_A(X) = \{(c, M) \mid c \in A_\bot,\ M \in \mathrm{Bag}(\mathbf{Mon}_A(X)),\ c \neq \bot \text{ or } M \neq \varnothing\}$$

Defining the addition is straightforward:

$$(c, M) + (d, N) = (c + d, M \uplus N)$$

We define the monomial multiplication $m \otimes n$ for $m, n \in \mathbf{Mon}_A(X)$ by concatenation of lists, and by using the operator $\cdot$ of A to combine adjacent elements of A in the resulting list and list concatenation to combine adjacent elements of X^+ (as in Section 3.1).

We also define multiplication of a concrete element with a monomial by treating the concrete element as a monomial of length 1:

- $r \rhd m = [r] \otimes m$
- $m \lhd r = m \otimes [r]$

This is well-defined because m already contains an element of X^+.

We also pose

$$\bot \rhd N = \varnothing$$
$$M \lhd \bot = \varnothing$$

We naturaly extend the multiplication to bags, which then gives us a definition for the multiplication of two elements of $\mathrm{Frex}_A(X)$:

$$(c, M) \cdot (d, N) = (c \cdot d, (c \triangleright N) \uplus (M \otimes N) \uplus (M \triangleleft d))$$

We define $\mathrm{embed} : A \to \mathrm{Frex}_A(X)$ by $\mathrm{embed}(a) = (a, \varnothing)$ for all $a \in A$. For all $a, b \in A$, we have $\mathrm{embed}(a + b) = (a + b, \varnothing) = (a, \varnothing) + (b, \varnothing) = \mathrm{embed}(a) + \mathrm{embed}(b)$ and $\mathrm{embed}(a \cdot b) = (a \cdot b, \varnothing) = (a, \varnothing) \cdot (b, \varnothing) = \mathrm{embed}(a) \cdot \mathrm{embed}(b)$, so embed is a morphism of **SG** $\triangleleft$ **CSG**-algebras.

We also define $\mathrm{var} : X \to \mathrm{Frex}_A(X)$ by $\mathrm{var}(x) = (\bot, \{[x]\})$ for all $x \in X$.

Proposition 128

$\langle \mathrm{Frex}_A(X), \mathrm{embed}, \mathrm{var} \rangle$ is the free extension of A by X.

Proof

- **$\mathrm{Frex}_A(X)$ is a SG $\triangleleft$ CSG-model.** All axioms of **SG** $\triangleleft$ **CSG** but the distributivity axioms are straightforward to check. For the distributivity axioms, we have:

$$\begin{aligned}
(c, M) \cdot ((d, N) + (e, P)) &= (c, M) \cdot (d + e, N \uplus P) \\
&= (c \cdot (d + e), (c \triangleright (N \uplus P)) \uplus (M \otimes (N \uplus P)) \uplus (M \triangleleft (d + e))) \\
&= (c \cdot d + c \cdot e, (c \triangleright N) \uplus (c \triangleright P) \uplus (M \otimes N) \uplus (M \otimes P) \uplus (M \triangleleft d) \uplus (M \triangleleft e)) \\
&= (c, M) \cdot (d, N) + (c, M) \cdot (e, P)
\end{aligned}$$

  and

$$\begin{aligned}
((c, M) + (d, N)) \cdot (e, P) &= (c + d, M \uplus N) \cdot (e, P) \\
&= ((c + d) \cdot e, ((c + d) \triangleright P) \uplus ((M \uplus N) \otimes P \uplus ((M \uplus N) \triangleleft e))) \\
&= (c \cdot e + d \cdot e, (c \triangleright P) \uplus (d \triangleright P) \uplus (M \otimes P) \uplus (N \otimes P) \uplus (M \triangleleft e) \uplus (N \triangleleft e)) \\
&= (c, M) \cdot (e, P) + (d, N) \cdot (e, P)
\end{aligned}$$

- **Universality.** Let B be any **SG** $\triangleleft$ **CSG**-model and let $f : A \to B$ be a morphism of **SG** $\triangleleft$ **CSG**-algebras and $g : X \to B$ be a homomorphism.

  We extend g to X^+ by defining $g(x_1 \cdots x_n) = g(x_1) \cdots g(x_n)$ for all $x_1, \ldots, x_n \in X$.

  We define $h([b_1, \ldots, b_n]) = [\![b_1]\!] \cdot \cdots \cdot [\![b_n]\!]$ where:

  - if $b_i \in A$, then $[\![b_i]\!] = f(b_i)$
  - if $b_i \in X^+$, then $[\![b_i]\!] = g(b_i)$

  We extend h to $\mathrm{Frex}_A(X)$ by defining

$$h(c, M) = \begin{cases} f(c) + \sum_{m \in M} h(m) & \text{if } c \neq \bot \text{ and } M \neq \varnothing \\ \sum_{m \in M} h(m) & \text{if } c = \bot \\ f(c) & \text{if } M = \varnothing \end{cases}$$

  This is well-defined and homomorphic[a] because the only merges in the extension are:

  - concrete/concrete, handled by $f(a + b) = f(a) + f(b)$ and $f(a \cdot b) = f(a) \cdot f(b)$
  - abstract/abstract, handled by $g(uv) = g(u) \cdot g(v)$

The diagram below commutes:

$$\begin{tikzcd}
A \arrow[r, "\text{embed}"] \arrow[dr, "f"'] & \mathrm{Frex}_A(X) \arrow[d, "h"] & X \arrow[l, "\text{var}"'] \arrow[dl, "g"] \\
& B &
\end{tikzcd}$$

Uniqueness of h is straightforward, as h is defined by the values of f and g on the elements of A and X, and by the homomorphic property of h.

□

[a]it is less straightforward than in the examples for the semigroups, so doing it may be a good idea

3.3 Distributive combination of free extensions

We'll try to adapt the construction of Section 2.5 to the case of free extensions.

Let B be a $\mathfrak{T} \lhd \mathcal{A}$ model, and X be a set.

Step 1.
Let $B_{\mathcal{M}}$ be the underlying $\mathcal{M}$-model of B. Let $E_0 \coloneqq \mathrm{Frex}_{B_{\mathcal{M}}}(X) \vDash \mathcal{M}$ be the free extension of $B_{\mathcal{M}}$ by X. From the free extension construction, we also have the canonical inclusion $i_{\mathcal{M}} : B_{\mathcal{M}} \to E_0$.

Step 2.
We apply the functor

$$\underline{}_{\mathfrak{T}} \coloneqq \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \to \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$$

to E_0. This gives $E_1 \coloneqq \underline{E_0}_{\mathfrak{T}} \vDash \mathcal{O}$, and the canonical inclusion $i_{\mathcal{O}} : B_{\mathcal{O}} \to E_1$.

Step 3.
Now the next part is tricky. Let's first look at this abstractly, and then find a way to actually construct the required elements.

From [All+25], we know that we can construct free extensions based on free algebras. This means that there exists E_1^{new} such that $E_1 \cong B_{\mathcal{O}} \amalg E_1^{\text{new}}$. The model of $\mathcal{O} \lhd \mathcal{A}$ that we want is then

$$E_2 = B_{\mathcal{O} \lhd \mathcal{A}} \amalg F_{\mathcal{O} \to \mathcal{A}} E_1^{\text{new}}$$

with

$$F_{\mathcal{O} \to \mathcal{A}} : \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) : \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle) \to \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$$

obtained as in step 3 of Section 2.5. The problem with this is that we can't get E_1^{new} from E_1, so we'll have to find another approach.

Let's note $F_{\mathcal{O} \to \mathcal{A}} \dashv U_{\mathcal{O} \to \mathcal{A}}$ the adjunction between $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$ and $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$. We have $B_{\mathcal{O}} = U_{\mathcal{O} \to \mathcal{A}} B_{\mathcal{O} \lhd \mathcal{A}}$. Therefore the identity $\mathrm{id}_{B_{\mathcal{O}}} : B_{\mathcal{O}} \to U_{\mathcal{O} \to \mathcal{A}} B_{\mathcal{O} \lhd \mathcal{A}}$ has an adjunct $\varepsilon_B : F_{\mathcal{O} \to \mathcal{A}} B_{\mathcal{O}} \to B_{\mathcal{O} \lhd \mathcal{A}}$. We can then define E_2 as the pushout of ε_B and $F_{\mathcal{O} \to \mathcal{A}} i_{\mathcal{O}}$:

$$\begin{tikzcd}
F_{\mathcal{O} \to \mathcal{A}} B_{\mathcal{O}} \arrow[r, "F_{\mathcal{O} \to \mathcal{A}} i_{\mathcal{O}}"] \arrow[d, "\varepsilon_B"'] & F_{\mathcal{O} \to \mathcal{A}} E_1 \arrow[d] \\
B_{\mathcal{O} \lhd \mathcal{A}} \arrow[r] & E_2
\end{tikzcd}$$

Lemma 129
Let $\mathcal{C}$ be a category, X, Y, Z be objects of $\mathcal{C}$, and $f : X \to Z$.
The pushout of $\iota_X : X \to X \amalg Y$ and f is isomorphic to $Y \amalg Z$:

$$(X \amalg Y) \amalg_X Z \cong Y \amalg Z$$

Proof

□

Lemma 130

$$E_2 \coloneqq B_{\mathcal{O}\lhd\mathcal{A}} \amalg_{F_{\mathcal{O}\to\mathcal{A}}B_{\mathcal{O}}} F_{\mathcal{O}\to\mathcal{A}}E_1 \cong B_{\mathcal{O}\lhd\mathcal{A}} \amalg F_{\mathcal{O}\to\mathcal{A}}E_1^{\text{new}}$$

Proof
Because $F_{\mathcal{O}\to\mathcal{A}}$ is a left adjoint, it preserves coproducts, so

$$F_{\mathcal{O}\to\mathcal{A}}E_1 \cong F_{\mathcal{O}\to\mathcal{A}}B_{\mathcal{O}} \amalg F_{\mathcal{O}\to\mathcal{A}}E_1^{\text{new}}$$

$F_{\mathcal{O}\to\mathcal{A}}i_{\mathcal{O}} : F_{\mathcal{O}\to\mathcal{A}}B_{\mathcal{O}} \to F_{\mathcal{O}\to\mathcal{A}}E_1$ is then the first coproduct injection. Therefore the pushout span for E_2 is

$$B_{\mathcal{O}\to\mathcal{A}} \xleftarrow{\varepsilon_B} F_{\mathcal{O}\to\mathcal{A}}B_{\mathcal{O}} \xrightarrow{\iota_1} F_{\mathcal{O}\to\mathcal{A}}B_{\mathcal{O}} \amalg F_{\mathcal{O}\to\mathcal{A}}E_1^{\text{new}}$$

By Lemma 129, the pushout of this span is isomorphic to the coproduct $B_{\mathcal{O}\lhd\mathcal{A}} \amalg F_{\mathcal{O}\to\mathcal{A}}E_1^{\text{new}}$.
□

Since the pushout is taken in $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$, E_2 is a model of $\mathcal{O}$ in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ and we have the canonical inclusion $B_{\mathcal{O}\lhd\mathcal{A}} \to E_2$.

Intuition: The intuition behind this construction is that we can't directly apply $F_{\mathcal{O}\to\mathcal{A}}$ to E_1 because the concrete part is then duplicated. By doing the pushout, we separate the concrete part (with ε_B) from the new part (with $F_{\mathcal{O}\to\mathcal{A}}i_{\mathcal{O}}$), and we get the desired result.

Step 4.
Let $E_3 \coloneqq \mathbf{Mod}(\mathcal{O}, \_)E_2 \vDash \mathcal{O}$ be the underlying operadic algebra. We use surjectivity as in step 4 of Section 2.5 to get an $\mathcal{M}$-model $E_{\mathcal{M}} \vDash \mathcal{M}$.

Step 5.
We can the define E as the pullback of $E_{\mathcal{M}}$ and E_2. This should give us the free extension of B by X.

3.4 Practical combination

While the construction of Section 3.3 works in thoery and has showed that we can indeed construct the free extension for the distributive combination using both of the free extensions, it is not practical as it uses a pushout, which is hard to compute.

The goal of this section is to give a practical way of construction the free extension for the combination that could be implemented in Idris2.

We call $B \vDash \mathfrak{T} \lhd \mathcal{A}$ the model we are trying to extend. We denote by $B_{\mathcal{M}}[-]$ (resp. $B_{\mathcal{A}}[-]$) the functor that takes a set X and returns the free extension of B by X as an $\mathcal{M}$ (resp. $\mathcal{A}$) model. We write $B[X]$ for the free extension of B by X as a $\mathfrak{T} \lhd \mathcal{A}$-model.

Hypothesis: We would need to have the following structure with the following property:

- a model $C \vDash \mathcal{O}$
- a function $X \xrightarrow{\nu} C$
- a function $C \xrightarrow{u} B[X]$

such that

- for all $D \vDash \mathcal{O}$

- for all function $X \xrightarrow{f} \underline{D}$
- for all homomorphism $B_{\mathcal{O}} \xrightarrow{p} D$

there exists a unique homomorphism $q : C \to D$ such that the following diagram commutes:

$$\begin{tikzcd} & X \arrow[dl, "\nu"'] \arrow[dr, "f"] & \\ C \arrow[rr, dashed, "q"] \arrow[dr, "u"'] & & D \\ & B_{\mathcal{O}}[X] \arrow[ur, "{[f,p]}"'] & \end{tikzcd}$$

Then we would define

$$A_2 \coloneqq \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])\ C \vDash \mathcal{O} * \mathcal{A}$$

and continue the construction as for the case of the free algebra.

3.4.1 Example of the semigroups

Remark : In the semigroup example, let

$$\mathcal{M} = \mathbf{SG}, \qquad \mathcal{A} = \mathbf{CSG},$$

let $\mathcal{O}$ be the semigroup operad, with one operation $\mu_n \in \mathcal{O}_n$ for each $n \geq 1$, and let

$$\mathfrak{T} : \mathcal{O} \to \mathcal{M}$$

be the surjective translation defined by

$$\mathfrak{T}(\mu_n) = x_1 \cdots x_n.$$

If D is an $\mathcal{O}$-model, then its underlying set carries a semigroup structure

$$x \cdot_D y \coloneqq D\, [\![\mu_2]\!]\, (x, y),$$

and the operad laws imply that this multiplication is associative. We denote the resulting semigroup by $D_{\mathcal{M}}$. By construction,

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(D_{\mathcal{M}}) = D.$$

Proposition 131 (The translated monomial object)
Let $B \vDash \mathcal{M} \lhd \mathcal{A}$, and write

$$B_{\mathcal{O}} \coloneqq \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(B_{\mathcal{M}}).$$

Let

$$\mathbf{Mon}_B(X) = \{[b_1, \ldots, b_n] \mid n \geq 1,\ b_i \in X^+ \sqcup \underline{B},\ b_i, b_{i+1} \text{ alternate kinds},\ \exists i,\ b_i \in X^+\}.$$

Equip $\mathbf{Mon}_B(X)$ with the semigroup structure of Section 3.2, and write

$$C_{\mathcal{M}} \coloneqq \mathbf{Mon}_B(X).$$

Set

$$C \coloneqq \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(C_{\mathcal{M}}).$$

Let

$$\nu : X \to \underline{C}, \qquad \nu(x) \coloneqq [x],$$

and let

$$u_{\mathcal{M}} : C_{\mathcal{M}} \to B_{\mathcal{M}}[X]$$

be the inclusion of variable-containing alternating lists into the alternating-list representation of the multiplicative free extension $B_{\mathcal{M}}[X]$. Define

$$u \coloneqq \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(u_{\mathcal{M}}) : C \to B_{\mathcal{O}}[X],$$

where

$$B_{\mathcal{O}}[X] \coloneqq \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(B_{\mathcal{M}}[X]).$$

Then the triple (C, ν, u) satisfies the hypothesis of Section 3.4.

Proof
By definition, $C_{\mathcal{M}} = \mathbf{Mon}_B(X)$ is a semigroup: multiplication is given by concatenation followed by seam normalisation, and associativity follows from associativity of concatenation, associativity of multiplication in $B_{\mathcal{M}}$, and associativity of concatenation in X^+. Applying the translation functor

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \to \mathbf{Mod}(\mathcal{O}, \mathbf{Set})$$

therefore yields an $\mathcal{O}$-model

$$C = \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(C_{\mathcal{M}}).$$

Now let $D \vDash \mathcal{O}$, let

$$f : X \to \underline{D},$$

and let

$$p : B_{\mathcal{O}} \to D$$

be an $\mathcal{O}$-homomorphism. By the preceding remark, the carrier of D carries a semigroup structure

$$x \cdot_D y \coloneqq D [\![\mu_2]\!] (x, y),$$

which we denote by $D_{\mathcal{M}}$, and we have

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(D_{\mathcal{M}}) = D.$$

Since p is an $\mathcal{O}$-homomorphism, it is also a semigroup homomorphism

$$p_{\mathcal{M}} : B_{\mathcal{M}} \to D_{\mathcal{M}}.$$

Indeed, for all $b, b' \in \underline{B}$,

$$\begin{aligned} p_{\mathcal{M}}(b \cdot b') &= p\big(B_{\mathcal{O}} [\![\mu_2]\!] (b, b')\big) \\ &= D [\![\mu_2]\!] (p(b), p(b')) \\ &= p_{\mathcal{M}}(b) \cdot_D p_{\mathcal{M}}(b'). \end{aligned}$$

By freeness of the multiplicative extension $B_{\mathcal{M}}[X]$, there exists a unique semigroup homomorphism

$$[f, p]_{\mathcal{M}} : B_{\mathcal{M}}[X] \to D_{\mathcal{M}}$$

such that

$$[f, p]_{\mathcal{M}} \circ \iota_{B_{\mathcal{M}}} = p_{\mathcal{M}} \qquad \text{and} \qquad [f, p]_{\mathcal{M}} \circ \eta_X = f.$$

Applying $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$ gives an $\mathcal{O}$-homomorphism

$$[f, p]_{\mathcal{O}} : B_{\mathcal{O}}[X] \to D.$$

Now define

$$q := [f, p]_{\mathcal{O}} \circ u : C \to D.$$

Since u and $[f, p]_{\mathcal{O}}$ are $\mathcal{O}$-homomorphisms, q is an $\mathcal{O}$-homomorphism. Moreover,

$$q \circ \nu = [f, p]_{\mathcal{O}} \circ u \circ \nu = f,$$

because $u \circ \nu$ is the variable map into $B_{\mathcal{O}}[X]$.
Finally, if

$$q' : C \to D$$

is another $\mathcal{O}$-homomorphism satisfying the same condition from Section 3.4, namely

$$q' \circ \nu = f \qquad \text{and} \qquad q' = [f, p]_{\mathcal{O}} \circ u,$$

then clearly $q' = q$. Thus (C, ν, u) satisfies the required hypothesis. $\square$

Theorem 132 (Applying the Section 3.4 construction in the semigroup case)
Let C be the $\mathcal{O}$-model defined in the previous proposition, and let

$$E_2 := \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])(C).$$

Let

$$E_3 := \mathbf{Mod}(\mathcal{O}, -)(E_2).$$

Using the surjectivity of

$$\mathfrak{T} : \mathcal{O} \to \mathcal{M},$$

reconstruct from E_3 a semigroup model $E_{\mathcal{M}}$ on the same carrier, exactly as in Step 4 of the construction, and let E be the resulting model of $\mathcal{M} \triangleleft \mathcal{A}$ obtained by Step 5.

Then E is the free extension of B by X for the distributive combination $\mathbf{SG} \triangleleft \mathbf{CSG}$.

Proof
We compare the object obtained from Section 3.4 with the explicit free extension constructed in Section 3.2, and explain where the argument stops.

Underlying carrier of E_2.
By definition,

$$E_2 = \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])(C),$$

so the underlying $\mathcal{A}$-model of E_2 is the free extension of $B_{\mathcal{A}}$ by the carrier of C. Since

$$C = \mathbf{Mon}_B(X),$$

the underlying carrier of E_2 is

$$\{(c, M) \mid c \in B^{\perp},\ M \in \mathrm{Bag}(\mathbf{Mon}_B(X)),\ c \neq \perp \text{ or } M \neq \varnothing\}.$$

This is exactly the carrier of the explicit model of Section 3.2.

The $\mathcal{A}$-structure on E_2.
Since the underlying $\mathcal{A}$-model of E_2 is the free extension of $B_{\mathcal{A}}$ by C, its addition is the one from the free commutative semigroup extension of Section 3.1.2. Therefore, for all $(c, M), (d, N) \in E_2$, we have

$$(c, M) + (d, N) = (c + d,\ M \uplus N).$$

This is exactly the addition defined in Section 3.2.

The $\mathcal{O}$-structure on E_2.
Since $\mathcal{O}$ is the semigroup operad, it is generated by the binary operation μ_2. It therefore suffices to determine the interpretation of μ_2 in E_2.

By definition of the functor $\mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])$, we have

$$E_2 [\![\mu_2]\!] = B_{\mathcal{A}}[-](C [\![\mu_2]\!]).$$

Now C is the translated monomial object, so

$$C [\![\mu_2]\!] (m, n) = m \otimes n$$

for all $m, n \in \mathbf{Mon}_B(X)$.

Let

$$\mathrm{embed} : B \to E_2 \qquad \text{and} \qquad \mathrm{var} : \mathbf{Mon}_B(X) \to E_2$$

be the canonical maps of the free $\mathcal{A}$- variables. One must also know its values on the three other kinds of generator pairs

$$(\mathrm{embed}\, b, \mathrm{embed}\, d), \qquad (\mathrm{embed}\, b, \mathrm{var}\, y), \qquad (\mathrm{var}\, x, \mathrm{embed}\, d).$$

The data of the binary operation

$$C [\![\mu_2]\!] : \mathbf{Mon}_B(X) \times \mathbf{Mon}_B(X) \to \mathbf{Mon}_B(X)$$

only determines the variable/variable case. It does not determine the concrete/concrete, concrete/variable, or variable/concrete cases. Therefore the definition of $\mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])$ does not justify the formulas

$$\begin{aligned} E_2 [\![\mu_2]\!] (\mathrm{embed}\, b, \mathrm{embed}\, d) &= \mathrm{embed}(b \cdot d) \\ E_2 [\![\mu_2]\!] (\mathrm{embed}\, b, \mathrm{var}\, m) &= \mathrm{var}(b \rhd m) \\ E_2 [\![\mu_2]\!] (\mathrm{var}\, m, \mathrm{embed}\, d) &= \mathrm{var}(m \lhd d) \end{aligned}$$

Consequently, the proof does not show that the $\mathcal{O}$-structure on E_2 is the one induced by the explicit multiplication of Section 3.2. Since this identification is missing, the passage from E_2 to E_3, the reconstruction of the semigroup structure on E_M, and the final comparison with the explicit extension from Section 3.2 cannot be completed. □

Why this argument does not work. The problem is that the notation $B_{\mathcal{A}}[-]$ currently only records the object part of the free $\mathcal{A}$-extension construction. To make the practical construction work, one would need additional structure describing how the concrete part of B interacts with the monomial object. In the semigroup case, this means that one would need, in addition to the multiplication

$$\mu_2^C : C \times C \to C,$$

compatible left and right actions

$$\lambda_L : B_{\mathcal{O}} \times C \to C, \qquad \lambda_R : C \times B_{\mathcal{O}} \to C,$$

corresponding to the operations $b \triangleright m$ and $m \triangleleft d$, together with the concrete multiplication on $B_{\mathcal{O}}$.

4 Idris2 implementation

4.1 Recalling some important definitions

```
record Signature where
  constructor MkSignature
  ||| Type family of operators indexed by their arity.
  OpWithArity : Nat -> Type

record Op (sig : Signature) where
  constructor MkOp
  {arity : Nat}
  snd : sig.OpWithArity arity

(^) : Type -> Nat -> Type
(^) a n = Vect n a

algebraOver : (sig : Signature)
  -> (a : Type) -> Type
sig `algebraOver` a =
  (f : Op sig) -> a ^ (arity f) -> a

record Algebra (Sig : Signature) where
  constructor MakeAlgebra
  0 U    : Type
  Semantics : Sig `algebraOver` U

data Term : (0 sig : Signature) -> Type -> Type where
  ||| A variable with the given index
  Done : {0 sig : Signature} -> a -> Term sig a
  ||| An operator,applied to a vector of sub-terms
  Call : {0 sig : Signature} -> (f : Op sig) ->
         Vect (arity f) (Term sig a) -> Term sig a

  CongruenceWRT : {n : Nat} -> (a : Setoid) ->
    (f : (U a) ^ n -> U a) -> Type
  CongruenceWRT a f = SetoidHomomorphism (VectSetoid n a) a f

record SetoidAlgebra (Sig : Signature) where
  constructor MkSetoidAlgebra
```

```
  ||| Underlying algebra
  algebra : Algebra Sig
  ||| Equivalence relation making the carrier a setoid
  equivalence : Equivalence (U algebra)
  ||| All algebraic operations respect the equivalence relation
  congruence : (f : Op Sig) -> (MkSetoid (U algebra) equivalence)
                                `CongruenceWRT` (algebra.Sem f)


record Equation (Sig : Signature) where
  constructor MkEq
  support : Nat
  lhs, rhs : Term Sig (Fin support)

record Presentation where
  constructor MkPresentation
  signature : Signature
  0 Axiom   : Type
  axiom     : (ax : Axiom) -> Equation signature

models : {sig : Signature} ->
  (a : SetoidAlgebra sig) -> (eq : Equation sig) ->
  (env : Fin eq.support -> U a.algebra) -> Type
models a eq env = a.equivalence.relation
    (a.Sem eq.lhs env)
    (a.Sem eq.rhs env)

(=|) : {sig : Signature} -> (eq : Equation sig) ->
  (a : SetoidAlgebra sig
     ** Fin eq.support -> U a.algebra) -> Type
eq =| (a ** env) = models a eq env

ValidatesEquation : (eq : Equation sig) ->
  (a : SetoidAlgebra sig) -> Type
ValidatesEquation eq a =
  (env : Fin eq.support -> U a.algebra) ->
  eq =| (a ** env)

Validates : (pres : Presentation) ->
  (a : SetoidAlgebra pres.signature) -> Type
Validates pres a = (ax : pres.Axiom) ->
  ValidatesEquation (pres.axiom ax) a

record Model (Pres : Presentation) where
  constructor MkModel
  ||| Algebra structure for the operations
  Algebra  : SetoidAlgebra (Pres).signature
  ||| The algebra validates all the equations
  Validate : Validates Pres Algebra
```

4.2 Distributive combination for free algebras

We first start by defining what the coproduct signature and the coproduct presentation are.

We use the built-in type `Either` to model the alternative between the two underlying signatures:

```
coproductSignature : Signature -> Signature -> Signature
coproductSignature sigA sigM =
  MkSignature (\n => Either (sigA.OpWithArity n) (sigM.OpWithArity n))
```

The axioms are composed of both the axioms for the additive and multiplicative part, to which we add the distributivity axioms:

```
data DistributivityAxiom : Signature -> Signature -> Type where
  DistAxiom :
       {addSig, mulSig : Signature}
    -> {m, n : Nat}
    -> (addOp : addSig.OpWithArity m)
    -> (mulOp : mulSig.OpWithArity n)
    -> (i : Fin n)
    -> DistributivityAxiom addSig mulSig

data CombinationAxiom : (additive, multiplicative : Presentation) -> Type where
  AddAx : additive.Axiom -> CombinationAxiom additive multiplicative
  MulAx : multiplicative.Axiom -> CombinationAxiom additive multiplicative
  DistAx : DistributivityAxiom additive.signature multiplicative.signature
         -> CombinationAxiom additive multiplicative
```

The distributive combination of the two presentations is then defined as:

```
distributiveCombination : Presentation -> Presentation -> Presentation
distributiveCombination additive multiplicative =
  MkPresentation
    (coproductSignature additive.signature multiplicative.signature)
    (CombinationAxiom additive multiplicative)
    (\case
       AddAx ax                         => injectAddEquation (additive.axiom ax)
       MulAx ax                         => injectMulEquation (multiplicative.axiom ax)
       DistAx (DistAxiom addOp mulOp i) => distributivityEquation addOp mulOp i
    )
```

If `addOp` is an operator of arity `n` and `mulOp` is an operator of arity `m`, then `distributivityEquation addOp mulOp i` is the equation

$$x_1, \ldots, x_m, y_1, \ldots, y_n \vdash \texttt{mulOp}\big(x_1, \ldots, x_{i-1}, \texttt{addOp}(y_1, \ldots, y_n), x_{i+1}, \ldots, x_m\big)$$
$$= \texttt{addOp}\big\langle \texttt{mulOp}(x_1, \ldots, x_{i-1}, y_j, x_{i+1}, \ldots, x_m)\big\rangle_{j=1}^{n}$$

5 Evaluatation

5.1 Performance

5.2 Comparison with [GM05]

Some thoughts while reading their article:

- user need to choose their own coefficient set in some cases, otherwise some equalities may not be provable

- presence of commutativity of the multiplication in the almost-ring axioms → different definitions of rings and semiring
- reference to article about setoids in type theory [BCP03]: may be good to cite in the report
- potentially interesting example ring to test: boolean ring

The current `ring` tactic implementation in Rocq has evolved from this article. Some basic ring structure are pre-defined, such as the natural numbers semiring or the integer ring, and Rocq provides a dedicated syntax to add ring structures. The user can still chose their coefficient set, with the default being the set of integers. The equality relation can now be any relation as long as it is an equivalence relation of some setoid. The operations need to be setoid homomorphisms.

Pros of the fral approach

- Can solve equalities for theories more restricting than rings or semirings, such as non-commutative semirings: `forall x y :  N, (x + y) * (x + y) = x*x + 2*y*x + y*y` is solvable in Rocq using the `ring` tactic, but we may encounter a case where we don't want the multiplication to be commutative → this is not possible with the `ring` tactic.
- We can have more operations that just `+` and `*`, such as involutions, and this with no additional cost.
- Automatic proof extraction: can give a proof understandable by humans (might not be the smallest proof)

Cons of the fral approach

- The `ring` tactic in Rocq let's the user give additional know equations to solve equalities:

  ```
  Goal forall a b:Z, 2 * a * b = 30 -> (a + b) ^ 2 = a ^ 2 + b ^ 2 + 30.
  Proof.
    intros a b H; ring [H].
  Qed.
  ```

  Though there seems to be limitations to adding additional axioms.

  For instance, define an abstract ring and add an involution operator `~~`. By giving the axioms

  ```
  Rinv_inv : forall x, ~~ (~~ x) == x.
  Rinv_antidistr : forall x y, ~~ (x * y) == ~~ y * ~~ x.
  ```

  one would expect the `ring` tactic to solve the basic goal

  ```
  Lemma invInv : forall x, ~~ (~~ x) == x.
  Proof.
    intros.
    ring [Rinv_inv Rinv_antidistr]. (* fails *)
  Abort.
  ```

  but it is unable to do so.

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A Daily recap

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad
- 06/05/2026: reading Ohad's notes, starting to write my own notes. lots of definitions to try and understand how things work.

- 07/05/2026: read the paper on frexes and frals, wrote down the relevant definitions. Meeting with Ohad: 3 equivalent definitions of adjunctions ⇒ universal arrows, examples where/when they are used + equivalence of definitions of free algebras and how to construct a free algebra for (commutative) semigroups.

  Gave examples of uses of universal arrows: defining adjunctions, defining products and coproducts, and defining free algebras.

  Defined and proved the free algebra for commutative semigroups, started to work on the free algebra for semigroups.

- 08/05/2026: Defined and proved the free algebra for semigroups and commutative monoids. Proved the details that were missing for the free algebra for commutative semirings (in the commutative monoid part).

  Started to work on the free algebra for the combination of semigroups and commutative semigroups, but still need to define the product operator and check that the distributivity axioms hold, and that the resulting algebra is indeed a free algebra for the combination.

- 11/05/2026: finished the definition of the product operator for the free algebra for the combination of semigroups and commutative semigroups, and proved that it is indeed a free algebra for the combination.

  Defined multiplication for the free semigroup and commutative semigroup, and proved that we get a monad structure on **Set** in both cases.

  Defined the distributive law of the monad for the free semigroup over the monad for the free commutative semigroup, and proved that it is indeed a distributive law → we get a monad structure on **Set** for the free algebra for the combination of semigroups and commutative semigroups.

- 12/05/2026: definition of Kleisli category and related objects, research and definition of algebra and distributive laws of monads in extension form, and link with the previous definition of distributive laws. Example 8.5 of [MW10] is almost the same thing as Section 2.2.

- 13/05/2026: started looking into the distributive combination of semigroups and commutative semigroups. Defined the operad $\mathcal{O}$ spanned by the multiplication operator of semigroups, and defined a translation from $\mathcal{O}$ to the theory of semigroups. Proved that $\mathcal{O}$ is minimal and that the translation is surjective.

- 14/05/2026: defined the monad on $\mathbf{SG} \lhd \mathbf{CSG}$ with the adjunction, and proved that it is the same as the one given by the distributive law.

  Copied over relevant information to start describing the pullback square for the combination of models in more details. A lot of reading Ohad's notes.

- 15/05/2026: more copying definitions to understand the pullback square. Figured out what all of the objects (both vertices and arrows) are. 2-hour meeting with Ohad to check that I got everything right, and to talk about what I have to do next.

- 18/05/2026: finished giving the full description of all objects in the pullback square.

  Jesse helped me understand how to create the pullback in the category of categories, and started working on the proof that the square is a pullback square. Proved the commutativity of the square, and started working on the pullback condition. It seems like the distributivity axioms pose a problem.

- 19/05/2026: sick day, stayed home. Solved the distributivity problem thanks to Ohad. Finished proving the pullback and started working on constructing the free algebra for the example of the semigroups.

- 20/05/2026: finished constructing the free algebra for the semigroups, giving explicit description at each step, and proved that the resulting algebra is the free algebra for the combination. Starting working on completing Ohad's notes for the general case.

- 21/05/2026: finished Ohad's proof of lemma 34 in the general case. Started proving step 5 for the general case → 2 approaches: build morphisms along the steps or constructive approach (which I think might work).

- 22/05/2026: the constructive approach cannot be used to prove that we indeed have a free algebra: we need the fact that the algebra is free so that the morphism is well-defined. Almost finished the proof using the construction with the adjunctions and the pullback, still need to tie some loose ends.

- 26/05/2026: finished the proof. Proved that the homomorphism defined constructively is indeed the right one.

  Started reading about free extensions.

- 27/05/2026: read about free extensions. I'm pretty sure I know how to adapt the construction of the free algebra to free extensions.

- 28/05/2026: wrote the definition of free extensions. Defined and proved the free extensions for the semigroups and commutative semigroups. Started looking into extending the result to free extensions.

- 29/05/2026: (almost) proved that you can indeed extend the result to free extensions with some modifications. Meeting with Ohad → what I came up with works but it uses a pushout which is not easy to calculate, so we started searching for another way to do this.

- 01/06/2026: Started learning some Idris2.

  Ohad gave a counter-example to the construction of the free algebra → I had overlooked something in the proof of Lemma 34 of Ohad's notes. Partially fixed things, but stronger hypothesis now.

- 02/06/2026: read [ZM18] and checked every no-go theorem they provide and proved that our construction doesn't apply in those cases.

  Worked on the construction of the free extension.

- 03/06/2026: finished learning the basics of Idris2, started looking into the current implementation of frex in Idris2 (`https://github.com/frex-project/idris-frex`).

- 04/06/2026: looked at the whole structure of the project and read every file, understanding of where and how things are defined and how they are used and work together.

  Started working on defining the distributive combination.

  Defined the theory of semigroups and commutative semigroups.

- 05/06/2026: defined some models for the semigroups.

  Came back to the drawing board of last friday with Ohad, and tried to make sense of things. Applied it to the example of the semigroups, I can't quite picture how to get the definition the the operations once you apply the lifted functor.

- 08/06/2026: did some proofreading and corrected some parts of my notes.

  Continued with the attempt at construction the free extension for the combination of the semigroups, but it indeed does not work as the lift does not give enough information to define the operations on concrete/variable and concrete/concrete cases.

  Started implemented the frex part of the semigroup frexlet.

- 09/06/2026: finished implementing the structure of the free extension for the semigroups, and proved that is is a valid semigroup model. Started working on the proof of freeness.
- 10/06/2026: properly defined the distributive combination of theories in Idris.

  SPLS in the afternoon.
- 11/06/2026: started working on the construction of the free extension for the distributive combination of theories. Defined some of the hypotheses needed, such as being a commutative theory or an affine presentation.

  Read the proof of Theorem 24 of Ohad's notes in more detail to understand how the left adjoint is defined on morphisms, but didn't manage to get a clear picture of it.

  Lots of battling with Idris. Tried proving that commutative semigroups are indeed a commutative theory.
- 12/06/2026: Managed to prove that the theory of commutative semigroups is indeed a commutative theory, and proved that the theory of semigroups is indeed an affine presentation.

  Started reading in more detail the proof of Theorem 24 of Ohad's notes to understand how the left adjoint is defined on morphisms. 2h30min long meeting with Ohad where we read the proof together and discussed it → I have to read his notes on the Yoneda lemma for 2-categories and multi-categories.
- week-end 13-14/06/2026: worked a bit on the proof of freeness for the free extension of semigroups in Idris2, and defined the free algebra based on the free extension. (This was to get some more time off during the week)
- 15/06/2026: reading Ohad's notes and completing some notions of category theory that I was still missing, such as limits, opposite categories and presheaves.
- 16/06/2026: figured out how to use Katla to embed Idris2 code inside of the LaTeX notes. Started copying over the relevant preliminary definitions, and explaining how the construction of the distributive combination works.

  Started making sense of the definition of the multifunctor $F_{\mathcal{A}}$.

  LFCS seminar.
- 17/06/2026: I finaly have the full definition of $F_{\mathcal{A}}$ and can therefore start implementing the distributive combination construction → started implementation.
- 18/06/2026: implementation of the semantics of the distributive combination → some proofs need to be done for ψ.
- 19/06/2026: finished the implementation of the semantics and started working towards proving the axioms.

  Meeting with Ohad: will compare our work with [GM05] once more free algebras have been implemented, and we are aiming for a submission in October in ??
- 22/06/2026: finished implementing the free extension for semigroups. Reading [GM05].
- 23-26/06/2026: (trying) to implement free algebras for commutative semigroups, groups and abelian groups. The structure is up but the proofs or freeness are messy.
- 26/06/2026: meeting with Ohad:
  - turns out I don't need the proof of freeness to compare with the other article, I only need to implement the semantics and the extender function

  - need to rework the constructions so that the equivalence of the additive theory is the equal equivalence so that Idris can check equivalence by itself later when testing → new form for commutative semigroups

- 29-30/06/2026: struggling with Idris that won't typecheck obvious things while trying to implement the commutative semigroup fral for ordered setoids. Reflexion about what order is necessary and how to represent it.

- 01/07/2026: managed to implement the semantics of the commutative semigroup fral.

  First successful test of the distributive combination construction.

- 02/07/2026: wrote a bunch more tests for the distributive combination.

  Implementation of the fral on ordered setoids for commutative monoids and abelian groups.

- 03/07/2026: Some more tests. Struggled to define the distributive combinations involving involutive monoids (some export problems, function application not normalising).

- 06/07/2026: fixed problem thanks to Ohad.

  Defined an abstract semiring/ring in Rocq and check that all axioms could be solved. Tried to add an involution and give the solver the axioms about involutions, but can't solve anything.

  Started writing the report.

- 07/07/2026: finished laying down the content of each section, except the introduction and conclusion.

  Finally read [YGK18] and re-read some of the source articles.

B TODO

- write the definition of free extensions with universal arrows

C Questions for Ohad

- how does the solver work?